

Resolution of the Continuum Hypothesis in the Holographic Closure

Discrete Construction of the Real Line $\mathbb{R}$ and Cardinal Classification of Internal Subsets

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Abstract

The Continuum Hypothesis is proved in the genealogical closure constructed by Triadic Modular Holography (HMT): every internal subset of its real line is countable or has the cardinality of the complete line. HMT is the formal core composing three objects. All Possible Paths (APP) organizes sum and product evaluations on a modular grid whose Cayley graph is the skeleton of a two-dimensional toroidal cell complex. TRIT is the oriented ternary signature $\{-1, 0, +1\}$. The Topological Phase Kernel (TPK) selects and transports transitions, preserves their memory, and prolongs compatible histories. The inverse limit X_∞^{enr} preserves value, sheet, orientation, residue, quotient, carry, boundary, route, memory, and survival. The joint discrete structure of the continuum and its Archimedean real line are obtained as outputs of this action.

Let h be the effective code of the graphs of the APP, TRIT, and TPK operators, their prolongations, their restrictions, and the two projections. For each realized history m_∞ , the genealogical closure $\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ is obtained by transfinite iteration of finite definitions with parameters from earlier levels. The main theorem establishes

$$\mathfrak{G}_{\text{HMT}}(h, m_\infty) \models 2^{\aleph_0} = \aleph_1$$

and, for every $A \in \mathcal{P}(\mathfrak{G}_{\text{HMT}})$, proves the dichotomy

$$|A| \leq \aleph_0 \quad \text{or} \quad |A| = |\mathfrak{G}_{\text{HMT}}|.$$

Consequently, there is no intermediate cardinal between the natural numbers and the real line within the integral continuum generated by HMT.

The proof links four steps. Exhaustive cylinder coverage establishes the equinumerosity of the real line obtained through cylinder reduction with $\mathcal{P}(\omega)$. Genealogical condensation constructs an injection $j_{\text{HMT}} : \mathfrak{G}_{\text{HMT}} \hookrightarrow \omega_1^\omega$. The coding of countable well-orders and its ternary lift construct the injection in the opposite direction k_{HMT} . Cantor–Bernstein yields cardinal equality, and the cofinality argument extends the conclusion to every subset of the complete closure. None of these steps presupposes the equality it proves.

The local fiber of $3^6 = 729$ refinement subcylinders does not bound the result: the coinductive law acts at every prescribed depth. The nonadic return recovers the phase while carry and memory advance; it does not reset the state. The same genealogy produces, as correlated results, the holonomy Γ_9 , the integral dodecaphase register K , the fourfold relation $(\pi, \varphi, e, \alpha)$, and the Paley–Witt–Golay–Leech– $V^\natural$ –Monster incidence realization. These outputs do not prove cardinal equality by numerical coincidence; they certify the unity of the state whose Archimedean projection determines value and whose incidence projection preserves structure.

Finally, $\mathfrak{G}_{\text{HMT}}(h, m_\infty) = L[h, m_\infty]$ is proved as a subsequent recognition of the hierarchy already constructed. It does not generate the real line, does not select the injections, and does not enter the cylinder reduction.

The description of the emitter is completed by the proof of its decimal alphabet of 42 blocks, an explicit cycle of incidences between regional half-emissions, and a contrast of ordered memory at fixed histogram. These consequences specify the information preserved by the finite readings preceding prolongation. Cardinal comparison retains the domain and proofs proper to genealogical closure.

Keywords: Triadic Modular Holography; APP; TRIT; TPK; enriched state; discrete structure of the continuum; genealogical closure; Continuum Hypothesis; cardinality; ordinality; coinduction.

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1 Introduction

The Continuum Hypothesis asks whether there exists a set of real numbers whose cardinality is strictly greater than that of the natural numbers and smaller than that of the complete real line. This work gives a negative answer to that existence question, and hence an affirmative answer to the hypothesis, in a genealogical closure defined through the operations of Triadic Modular Holography (HMT). Its contribution links the construction of the real line by compatible prolongations to the cardinal comparison of all its internal subsets.

The HMT formal core comprises three objects developed from their definitions. All Possible Paths (APP) equips the support $(\mathbb{Z}/9\mathbb{Z})^2$ with sum and product evaluations, retaining their residues and quotients. Translations by $(\pm 1, 0)$ and $(0, \pm 1)$ determine its Cayley graph; square cells and the identification of opposite boundaries give it a two-dimensional toroidal realization. TRIT denotes the oriented ternary signature $\{-1, 0, +1\}$, which distinguishes transition regimes. The Topological Phase Kernel (TPK) composes selection, transport, and memory to prolong states. These distinct functions act jointly: the support locates paths, the evaluations determine their arithmetic content, and the dynamics preserves the history of each transition.

The argument follows this composition from finite operations to the domain in which the cardinal comparison is performed:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_\infty^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow m_\infty \longrightarrow \mathfrak{G}_{\text{HMT}}(h, m_\infty).$$

The composition produces enriched states, compatible prolongations, complete histories, and only thereafter an Archimedean realization. The real line is an output of the construction, not its initial universe; the cardinal question is decided over the entire domain produced by that genealogy.

The central question is whether the line obtained by Archimedean projection admits, within its complete genealogical closure, a cardinal strictly intermediate between $\aleph_0$ and the cardinal of the line itself. The theorem is formulated in $\mathfrak{G}_{\text{HMT}}(h, m_\infty)$, the closure generated by the complete system. In that domain it quantifies over $\mathcal{P}^{\mathfrak{G}}(\mathbb{R}_{\text{HMT}}^{\mathfrak{G}})$ and establishes the complete cardinal dichotomy.

Theorem 7.12, proved in Section 7, establishes the main result. In the cardinal comparisons, the real line and the power sets are interpreted in the genealogical closure $\mathfrak{G} = \mathfrak{G}_{\text{HMT}}(h, m_\infty)$. This domain and its quantifiers are constructed explicitly before the comparison is performed. The result is as follows. For every realized history m_∞ produced by the complete system associated with

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_\infty^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}},$$

whose restrictions are compatible and whose prolongations are nonempty, exhaustive, and surviving at every horizon, the genealogical extension $\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ generates the integral real line, quantifies over its entire power set, and satisfies

$$\mathfrak{G}_{\text{HMT}}(h, m_\infty) \models 2^{\aleph_0} = \aleph_1.$$

In particular, for every $A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}_{\text{HMT}}^{\mathfrak{G}})$,

$$|A| \leq \aleph_0 \quad \text{or} \quad |A| = |\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}|.$$

Consequently, HMT proves the Continuum Hypothesis affirmatively for the integral continuum that it defines and generates. The formal statement and its proof are given once in the main theorem.

The essential difference from a purely extensional presentation lies in the identity of the mathematical object. An isolated real coordinate preserves a value; an HMT state also preserves the operations that produced it, the arithmetic sheet, the tritic orientation, the residues and quotients traversed, the carry, the boundary, the route, and the accumulated memory. The Archimedean projection neither defines nor exhausts that structure: it determines one of its coordinates. The second projection preserves the boundary incidence that, in downstream developments, reaches Paley–Witt–Golay–Leech, $V^{\natural}$, and the Monster. Both realizations arise from the same state, and neither is introduced to select the other retrospectively.

1.1 Construction of the continuum through compatible prolongations

APP arranges the sum and product sheets over the same support and preserves the difference between their additive and multiplicative evaluations. TRIT incorporates orientation and regime. TPK selects, transports, memorizes, and prolongs the admissible transitions. The composition does not terminate in a finite state: it defines a system of restrictions

$$\rho_{N+1,N} : X_{N+1} \longrightarrow X_N$$

and nonempty extensions

$$\text{Ext}_N(x) \subseteq \rho_{N+1,N}^{-1}(x).$$

Compatibility at every depth produces the inverse limit

$$X_\infty^{\text{enr}} = \varprojlim_N (X_N, \rho_{N+1,N}).$$

The quantifier is $\forall N$; no execution to one thousand, ten thousand, or one million digits constitutes the bound of the theorem.

Each successor block subdivides a cylinder into $729 = 3^6$ disjoint, exhaustive refinement subcylinders of decreasing diameter. This number describes the local branching of the refinement; it is not identified with the cardinal of the continuum. The complete block tape determines a history, and its arithmetic projection determines a unique real point. Conversely, the canonical expansion of each real reconstructs a tape, and the coverage property lifts it to a compatible HMT history. Thus the real line is a surjective image of the constructed tree, never an external set used to choose its branches.

Measurement enters as a coupling. On the domain under consideration, the fiber-separation condition is imposed first: the enriched reader must distinguish two different states that yield the same value under Archimedean evaluation. Under that condition, formalized in Proposition 5.5, the instrument distinguishes a compatible continuation, adds the result and the back-action to the genealogical register, and restricts future histories to those that prolong the new state. The realized history m_∞ is formed through these updates. The inverse transformation then reconstructs the interior from the memory transported to the boundary, so that the update does not amount to a loss of information. Proposition 5.6 shows when changes of representation also preserve the measure of histories.

1.2 Genealogical semantic closure

Let h be the effective code of the graphs of APP, TRIT, and TPK, the prolongations, the restrictions, and the two projections. The pair (h, m_∞) is primitively coded by a single real $u_{h,m} = h \oplus m_\infty$. Define

$$\begin{aligned} \mathfrak{G}_0(h, m_\infty) &:= \text{TC}\{\omega, u_{h,m}, h, m_\infty\}, \\ \mathfrak{G}_{\beta+1}(h, m_\infty) &:= \text{Def}_{\Sigma_{\text{HMT}}}(\mathfrak{G}_\beta(h, m_\infty)), \\ \mathfrak{G}_\lambda(h, m_\infty) &:= \bigcup_{\beta < \lambda} \mathfrak{G}_\beta(h, m_\infty), \\ \mathfrak{G}_{\text{HMT}}(h, m_\infty) &:= \bigcup_{\beta \in \text{Ord}} \mathfrak{G}_\beta(h, m_\infty). \end{aligned}$$

This closure is formulated through relative definability and condensation in the precise sense of constructible hierarchies [16]. The Continuum Hypothesis is not introduced among the conditions defining the closure; it is obtained by the proof developed in this article. Neither an enumeration of the real line nor a target bijection is introduced as data. Its successor operation forms the subsets definable by finite formulas of the HMT language and parameters already produced. Equivalently, every admitted object has a derivation tree, and every admissible tree determines an object.

The resulting hierarchy is transitive, contains its ordinals, satisfies the axioms employed in the cardinal comparison, and has a global well-order determined by the first level of birth and the least derivation name. The notation

$$\mathfrak{G}_{\text{HMT}}(h, m_\infty) = L[u_{h,m}] = L[h, m_\infty]$$

is obtained later as a representation theorem. It is not the engine of APP, TRIT, or TPK, it does not select m_∞ , and it does not replace the material construction of the continuum.

1.3 Cardinal comparison between the HMT real line and the first uncountable ordinal of the closure

The cardinal proof is divided into two directions independent of the equality sought. First, genealogical condensation proves that every generated real appears before $\omega_1^{\mathfrak{G}}$. To retain the convention used in the proof, let $\iota(r)$ be the code of the rational cut of r , let $\beta(r)$ be the least ordinal such that $\iota(r) \in \mathfrak{G}_{\beta(r)+1}$, and let $n(r)$ be the least index of this cut in the enumeration $e_{\beta(r)+1}$. Define

$$j_{\text{HMT}}(r) = \omega\beta(r) + n(r), \quad j_{\text{HMT}} : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \hookrightarrow \omega_1^{\mathfrak{G}}.$$

In the opposite direction, every ordinal $\beta < \omega_1^{\mathfrak{G}}$ has a real code w_β for a countable well-order of type β . The codes are chosen by means of the genealogical well-order already constructed. The ternary embedding

$$E(w) = \sum_{n \geq 0} \frac{2\mathbf{1}_w(n)}{3^{2n+2}}$$

is injective, and the uniform cylinder lift proves that $E(w_\beta)$ belongs to the real line obtained by the HMT Archimedean projection. Therefore,

$$k_{\text{HMT}} : \omega_1^{\mathfrak{G}} \hookrightarrow \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}.$$

Cantor–Bernstein closes the cardinal equality. The proved identification

$$\mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \approx \mathcal{P}^{\mathfrak{G}}(\omega)$$

turns it exactly into $2^{\aleph_0} = \aleph_1$. Here $A \approx B$ denotes equinumerosity; it does not assert a field isomorphism between a real line and a power set.

1.4 Logical structure of the proof

The proof is organized into four logical steps. First, the domain of histories, its inverse limit, and the exhaustive coverage of cylinders are constructed. Second, the semantic closure is formed and condensation is applied to its reals. Third, without using the equality sought, the two injections between the HMT real line and $\omega_1^{\mathfrak{G}}$ are defined. Fourth, Cantor–Bernstein yields the equality, and the cofinality argument establishes the dichotomy for every internal subset. In this sequence, 729 describes local branching; coverage relates histories and values; condensation controls the level at which internal reals appear. The constants are correlated outputs of the same state: their conventional digits do not select the cardinal comparison, while their regions, routes, and memories remain in the enriched object on which the real line is constructed. The subsequent identification with $L[h, m_\infty]$ represents the closure already constructed.

The order of exposition first reproduces the formal core shared by the series, the prolongation, the joint construction of the continuum, and the coinductive generation; it then presents the integral semantic closure and the cardinal proof. Only thereafter does it develop the reduced phase-and-memory core, the integral dodecaphase seal K , the complete dodecaphase route to α_{HMT} , the finite analytic witness of order nine, the correlated analytic representation with recurrence, convergence, and common cylinders, and the exceptional corridor. These coordinates share an originating state but have different codomains and functions; jointly, they exhibit the complete dual realization and preserve the evaluation and projection routes, incidence, boundary, and memory. No target value selects the cardinal injections. The comparison with Cantor, Gödel, and Cohen is introduced at the end of the construction [11, 12, 13, 14, 15] and does not govern the definition of the HMT object.

Every arrow used by the cardinal closure is defined in the body of the article by its domain, codomain, action, and compatibility with refinement: the three oriented loops of the nonadic return, their grouping into six-symbol blocks, the 54-step block extension, the construction of cylinders, and the uniform lift of their branches. The nine transitions are constructed explicitly and provide the uniform lift used in the cardinal comparison. The reproducible calculations accompany, but do not replace, this proof.

The transition from seeds to compatible histories contains finite readings distinguished by the information they retain. The additive and multiplicative signatures determine the 42 possible blocks of one decimal window, with explicit seeds proving reachability. Six-window emissions establish a junction cycle between regional halves. Temporal transport retains phase, orientation, and memory in addition to those incidences. A pair of dodecaphasic records with the same histogram and different correlation proves why the archive requires sample order. These proofs specify the finite antecedent of prolongation; coverage, semantic closure, and cardinal injections are developed in their own domains.

The finite foundational part includes the emission multigraph, its phase quotient, and five minimal eight-edge closures. Each closure jointly covers APP, propagation, autoscale, and one closure region. The oriented witnesses, finite search, and parity of memory readers specify the data retained before the prolongations and cardinal readings developed in this work.

2 Formal core of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information required to extend them. Only thereafter are numerical realizations formed. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The designation *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics [1]; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory update, and extension. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks of a graduated line between zero and ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must not be confused with the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to select a path.

The cyclic identification of positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself it does not preserve the integer prior to reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i , repeated j times, produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ exchanges the accumulation directions and preserves both evaluations. It is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^-, q_{ij}^-)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

Here the term *sheet* denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:
 $S(i, j) = i + j$, $P(i, j) = ij$.

Each row is generated by accumulation:
 $P(i, j + 1) - P(i, j) = i$.

The chart represents 0 (mod 9) by 9.

The indicated block is
 $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$,
 with eigenvalues 9 and 5.

Figure 1: Multiplication table of APP in positive representatives. Nonunit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonal entries. The displayed table is an observable of the arithmetic torus; its representational edges do not constitute an intrinsic topological boundary of the torus.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The isolated residues do not determine those integer evaluations or their quotients.*

Proof. The first assertion is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, positions (5, 5) and (8, 2) give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

2.1.1 Cayley graph, toroidal topology, and winding number

In row and column coordinates, the cardinal translations are given by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southward and the second eastward. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges at each vertex. It is a Cayley graph for the *additive* structure of the support. Residue multiplication has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement prior to reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Reversal of the path changes the sign of this vector. Return to the same residue position therefore does not force the winding to vanish. This is the first instance of a distinction that will recur in the phase return: one coordinate can close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums are equal to

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The later decomposition of the calendar into phase and window will use the representatives from zero through eight.

The totals of both evaluations distinguish the visible contribution from that preserved in the quotient:

$$\begin{aligned} \sum S &= 810, & \sum P &= 2025, & \sum \Sigma &= 405, & \sum \Pi &= 459, \\ \sum q^+ &= 45, & \sum q^\times &= 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Suppressing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has group of units $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus, even before any continuous geometry, multiplicative folding distinguishes the units from the nilpotent sector.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readings: its direct reduction modulo three is 0, 0, 0, whereas extracting the factor three and then reducing its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

2.1.4 Additive evaluation by means of a Latin square

The difference between the two evaluations can also be expressed by means of operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projections $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, every translation $b \mapsto a + b$ permutes the nine residues. This yields a realization of a Latin square by rank-one projections, in the terminology of quantum Latin squares [2, 4]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different content: for $a = 3$, the vectors e_{3b} repeat residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

Multiplicity is recorded in the operator and characterizes the multiplicative radical. Consequently, the toroidal support, the Latin property, and the operator resolution of the identity are properties that require different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[3]. This reference provides a precise language for comparing APP realizations.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1), (0, 0), (0, +1)$ can exhibit the same zero residue and different carries.

The signature does not replace the path. A zero residue does not signify absence of orientation, phase, or quotient information. Nor does it introduce a physical magnitude by itself. Its function is to distinguish algebraically the transport classes to be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by the oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	-I	$\cos t I + \sin t J_\tau$
0	0	$I + tJ_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ yields the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to all three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity across realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projections and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly along two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an order of parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of the lifts

Ternary reduction makes it possible to express an integer APP evaluation in a visible coordinate and an extension coordinate. Its function is not merely to abbreviate the notation of an integer: it provides a composition rule that determines how the quotient changes when evaluations are added or multiplied. The balanced choice also makes explicit the compatibility of that rule with sign reversal.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and lies in the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When the reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined below uses those classes to determine the operative mode. Reading an evaluation and selecting a mode thus have the same ternary reduction, but different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). *Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define*

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

The integer operations are transported to the pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. The sum of quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the extension coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity guarantees that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses that accounting: $1 + 1 = -1 + 3$, whereas the residue reading preserves only -1 . In this case, the information eliminated is determined by $\chi(1, 1) = 1$.

The relation with modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3 and then $\bar{c}$. The sum transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus, the two ternary levels do not compose by an independent sum of coordinates: normalization at the first level modifies the second.

In particular, the classes 3, 6, and 0 in $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving that coordinate prevents the zero-residue sector from being artificially collapsed into a single position. Modulus 1000, used later to organize decimal blocks, serves a different positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas coincidence of residues in different bases does not by itself identify their transport states.

2.2.3 Reversal, residue class, and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share the zero signature but represent distinct evaluations. In a transition restricted to a quotient of maximum amplitude 1, they are the three possibilities in the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the path that produced it. Two APP paths may share evaluation, residue, and quotient, while differing in orientation, mode sequence, or winding. The decomposition $\mathcal{B}$ is therefore integrated into the transport state rather than replacing it. The arithmetic memory of a division and the memory of a path are related coordinates, not synonyms.

For a sequence of signatures of length n , the oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of the positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it exchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while retaining the threshold 0. Transport of the remaining path data is specified by the corresponding TPK involution.

This reversal of signatures must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within the type already fixed. The distinction prevents attributing to sign reversal an algebraic equivalence between different regimes.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and product (2.9) does not replace (2.8); it expresses a different operation in the stated quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, and in that case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the stated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero even though its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the

nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. Here the algebraic norm is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types can be faithfully realized by

$$aI + bJ_\tau \longmapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. Thus, by a single matrix expression, it represents the complex structure, the dual-number structure, and the split-real structure. Each preserves its law and its zero divisors. The signature- $(1, 1)$ form in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of elements of norm one:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation is maintained under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau) \exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparison of the two components yields the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution in the selected regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential path traverses the unit circle. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between both coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotence eliminates terms of order at least two but preserves the linear term that transports the parameter. The visible symbol 0 , the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with absence of state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio between components, not the divided-out scalar factor. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law remains part of the complete algebra; singularity of a coordinate does not automatically imply singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Therefore, the signature change that organizes a route cannot be interpreted as an undifferentiated isomorphism among the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

Within the electronic chapter, the pair of projections from the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution with square I and two nilpotent directions. The joint realization will be proved on that specific pair of projections; the preceding classification explains why the three quadratic relations can appear in the same matrix algebra without being identified with one another.

In the exceptional extension, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its link with a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of that same relation. Each preserves its characteristic, dimension, and incidence information. Development of the respective operators establishes the link, not equality of their cardinalities.

Preservation of norm one consequently describes the quadratic realizations of the evolutions. Preservation of quotients, orientation, and path belongs to the transported arithmetic state. Both properties enter the construction with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and extend the operations that originate them.

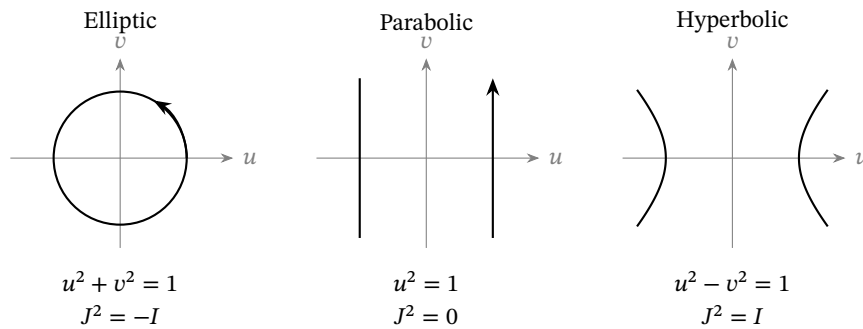


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. The parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore distinct functions: a ternary sign is not a displacement on the torus.

We denote by $\hat{X}$ the space of arithmetic states with path memory. An element contains the positions and directions of both cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation prevents replacement of the object by a reduced version that preserves only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the update of the records. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each map preserves the fields it does not modify and updates the remaining ones by Euclidean division and path concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The two-cursor recurrence of Section 4 completely specifies this finite realization. Thereafter, extension will not be obtained by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. In this way, APP provides the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamical state, observables, and dependence of the operations

The topological phase kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make extensions compatible. Equation (2.12) acquires operational content by specifying which coordinates each of its factors receives and modifies. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; the quotients record the arithmetic evaluation prior to its reduction. These coordinates are related through the transition and retain distinct mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. In the fibers of complete states over q , the ordered path, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary records are preserved. A precision extension adds the prefix and the compatible cylinder. Thus, two states with the same projection q may preserve different records.

The dependence of the operations has three levels. At the local level, the selector activates a sheet, transport modifies its position, and the update calculates the new arithmetic coordinates. At the path level, these transitions compose and preserve the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the prefix, signature, and boundary of the antecedent. The finite calendar describes the phase of those operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operative phase and the dodecaphase sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The selector already defined can be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector, in the ordered chart of D_9 , is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

The cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

At phases 3, 6, 9, both positions remain fixed; the neutral event forms part of the chronology and its record. The cardinal directions determine the vectors v_d , while the phase signature determines which of those vectors is applied.

The integer lift preserves the boundary crossing. If $s : X_9 \rightarrow D_9^2$ selects the positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records the preceding crossings, the update is $z' = z + b_d(x, a)$. Thus,

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport datum remains reconstructible after the position on the torus is reduced.

Proposition 2.12 (Preservation of the lifted displacement). *For a cardinal path of n steps, whose activation indicators are a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved when two compatible paths are concatenated.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Dividing the summation range into two segments produces exactly the concatenation law. For a closed path, the difference $z_n - z_0$ is the previously defined winding number. $\square$

The same principle governs the APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

The spatial-crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint preservation maintains the geometric and arithmetic provenance of each transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j exchanges N with S and E with O . After applying (2.15), both directions are reversed when a block of 27 steps ends. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The reading that forms the emission uses the positions prior to transport, according to the explicit recurrence of Section 4. The observable update and the emission therefore compose in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. In a block of 54 steps, positions and directions are restored; the Θ_{108} coordinate advances by 54.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108} . That phase determines whether the directions were reversed and which cursor was displaced. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

Set aside

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projection of the history to the calendar permits this finite verification. The path record, its evaluations, and the new prefixes are preserved in the dynamical state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine transport of memory and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves the orientation and the sheet; h , the residues and quotients; c , the additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded path. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q , and S_q denote, respectively, the sets of residue–quotient data, the abelian group of transport memory, the recorded paths, and the boundary signatures. Realizations of this signature are specified componentwise in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data that produce it have been fixed.

The elementary residue-and-quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives of D_9 congruent modulo 9 are equal. This chart provides the local integer reconstruction. The precision towers also incorporate the truncation maps and the cylinders developed in 4 and 6.

Let $\gamma : q \rightarrow q'$ be an observable path. Transport of the fibers is expressed by maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is a homomorphism of abelian groups. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

Here the symbol k_{tr} denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the chapter on dodecaphase closure.

The transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports the preceding memory, and the cocycle adds the increment produced by the path. In the spatial-crossing realization of each cursor separately, $C(\gamma) = I$, and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance links the abstract law with the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Update (2.19) by γ_1 , followed by update by γ_2 , coincides with update by $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from composition of H , and that for λ from associativity of paths. The final signature coincides because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible extension. The states, together with the triples (γ, x, x') admitted by those relations, form a category over the observable paths:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition simultaneously preserves arithmetic transport and boundary extendability. This is the structure in which returns with memory and successive refinements are effected.

2.3.5 Dodecaphasic record, phase, and accumulated memory

The 108 transitions are distributed among the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the subroute in the window, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart to be developed in 11.2; its window indices correspond to $m = \beta + 1$. The domain comprises the compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (2.19). The group

C_{12} , by contrast, describes translation of the window index; Θ_{108} preserves the step calendar. This distinction specifies what is transformed and what is observed upon completion of a cycle.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancing 9 steps preserves r and increases b by one. Addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a record of circuits preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.6). Its function is to describe the relation between these coordinates before using them in the construction of the signed state.

The signed record receives the phase and memory data through the composition developed in Section 11; on the compatible sublattice, the notations $\mathcal{S}_{12}$ and R_{sgn} denote the same reader:

$$\mathcal{S}_{12} \equiv R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between that signed state and K , as well as the rational reading of K , are proved in the corresponding chapter. The recording operations precede those transformations. Reconstruction of a record from its cyclic differences verifies its recoverability; the dynamical provenance must be preserved through the events that produced it. Equation (2.20) fixes the type and effective composition. The terminal state is produced by the one hundred eight updates $\mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$; $\mathcal{R}_{12}$ preserves the twelve windows with orientation, carry, boundary, and ledger, and $\mathcal{E}_{108}^{90,120}$ expresses the channels $(b^{90}, b^{120}, Q_{\text{TPK}})$. Section 11 incorporates the exact evaluation and proves that Π_H produces U_{term} before the Hadamard inversion that produces K . The focal verifier deliberately begins at the channel coordinate to check the subsequent chart and its return focally and exactly; the scope of that auxiliary program is not confused with the scope of the mathematical construction.

For the coded trace (d_t) preserved by the record, the event reader counts codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols in $\mathcal{D}$ are distinct coordinates of the record. Chapter 11 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences that characterize its image, and its exact inverse. The inversion proof establishes which data that chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and let $O_{\text{ph}} = D_9 \setminus \{9\}$ be the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. The internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; the subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and the marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Multiplication by 6, 8, 9 gives 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and takes the marked point to its image. Products and disjoint unions thus preserve incidence. $\square$

Here the cardinality 271 has a realization by phase incidence. The cubic completion of Section 4 has its own construction of the shell $1000 - 729$. The correspondence between the two realizations must preserve the maps and marks that identify them, in addition to the cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ preserve the domains and operations just distinguished.

2.3.7 Path reversal and conjugate extensions

Bidirectionality begins with an operation on paths, before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The origin of the reversed path is the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and the integer displacement changes sign. These identities are obtained by reversing the order of the steps and replacing each cardinal vector by its opposite. For a loop it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On the records, reversal also requires transporting the order of their components and the orientation of their events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a particular record. In particular, the regional involution that exchanges the closure and propagation components preserves the self-scaling axis; its formula is presented in Section 6.14. This regional operation acts on a domain different from reversal of a cardinal path.

Extensions along a route $r : q \rightarrow q'$ are expressed by relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, the orientation, the quotients, the prefix, and the boundary. Reversal of the route does not by itself imply that every extension relation is a bijection. The conjugate extension is defined on the domain where transport of those data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

One phase circuit composes the transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \cdots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain, $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$ is preserved, with admissible extensions developed below. The cyclic base has nine phases, while the fiber preserves the path, quotients, and boundary. Monodromy is the action of one circuit of that base on the fibers; the holonomy $\Gamma_{9,k}$ expresses its realization by the effective TPK transitions.

If g observes the phase and M the number of preserved circuits,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure, and the second identifies the new temporal depth. Coinductive extension also preserves its cylinders and prefixes. The restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them by the latest observation. The development of this arithmetic of histories and its identity occupies 4.8–4.10; construction of the regions and their readers is carried out in Section 6.

Phase transfer on already constructed emissions has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, of cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed by means of the recurrence in Section 4. Once obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. Transport of the histories remains given by (2.22), with its memory and boundary.

The resulting hierarchy determines the causal order of subsequent constructions. The finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection extends them; the dodecaphase register determines the data entering K and the closure of α . The numerical and incidence realizations receive that preceding structure. This order makes it possible to present each consequence in its chapter while preserving, from the formal core onward, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

The return of the calendar does not eliminate the increments produced during the traversal. To express this distinction in operator terms, we first construct an accumulated record of path events and then its generating series. Memory variables will be reduced on that transport, while also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer codes for direction preserved by a TPK path. The code d_t and the cursor's cardinal direction are distinct coordinates; no identification between their alphabets is assumed. Number the windows from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase register. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For an extended history, $\sigma_m \in \{-1, 1\}$ is the sign preserved in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts events in the trace; it does not select directions by means of a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and let $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix without accumulated history begins at $z_0 = 0$; at another origin, z_0 preserves the preceding record. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = R y_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplication of the z_m -update by $S^{-(m+1)}$ yields $y_{m+1} = S^{-1} y_m + a_m S^{-1} \mathbf{e}_0$, because $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$.

Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.26). $\square$

Thus, twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of every step within each window; that order remains in the recorded path.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Balance produced by the event). *Update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \tag{2.27}$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The sum of the coordinates is invariant under the permutation R , which yields the second equality. $\square$

2.4.2 Generating series and uniform control of memory

The series preserves every increment, not merely their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)). \tag{2.28}$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \tag{2.29}$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 -norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with the variable of another reader requires preserving that clock and the map relating the two readings.

2.4.3 Elimination of memory: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C)\mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1 w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.30); applying ℓ_0 and ℓ_1 to both components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2 B_1 D_u C = \sum_{k \geq 0} u^{k+2} B_1 D^k C$ describes an exit into memory, k steps within it, and a return. By contrast, $u\ell_1 D_u C$ reads the memory before that return. Consequently, reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n \mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example pertains to the twelve-window record, not to the entirety of the TPK state.

2.4.4 Composition of segments and coherence of reduction

Let three composable affine transports be $F_i(y) = R_i y + b_i$, with compatible domains and codomains. The memory produced by the complete traversal is

$$F_3 F_2 F_1(y) = R_3 R_2 R_1 y + b_3 + R_3 b_2 + R_3 R_2 b_1. \quad (2.33)$$

The proof is the successive substitution of the three affine formulas. Grouping either the first two segments or the last two first produces the same expression; the factors transport each increment from the point at which it originated. In (2.24), $R_i = R$, and b_i is the event of its window.

Elimination of several layers preserves analogous coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

After eliminating the third component, the four blocks are

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories jointly:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.34). Subsequently eliminating the second component gives the left-hand side of (2.35). Solving the two memory equations simultaneously gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . In both procedures, the source and reader are also transformed according to (2.30)–(2.31). $\square$

Balance (2.27) describes the increments of the record. The preceding identities allow them to be transported when memory is reused in the dodecaphase closure; by themselves, they do not determine additional analytic coefficients.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t - 1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b - 1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i + 1, j + 1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = \lfloor [\nu/4] \rfloor_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \tag{3.9}$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \tag{3.10}$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Decimal alphabet and check-digit congruence of the emitter

Retain the emission law (3.9), with the APP readings and observable ℓ_9 of (3.8), the cursors of Section 3.1, and the six-window calendar of Section 3.4. The accumulators are S_m, E_m, Z_m , with $Z_m = 3$, and the output digits are denoted $d_{m,1}, d_{m,2}, d_{m,3}$. Proposition 3.4 already recovers both residual signatures. Combining this recovery with the actual accumulator images determines the following alphabet.

Proposition 3.5 (Check-digit congruence for the third digit). *Every emitted block satisfies*

$$d_{m,3} = [5d_{m,2} - 2d_{m,1} + 1]_{10}.$$

Proof. The inverse in Proposition 3.4 gives $[E_m]_{10} = [d_{m,2} - d_{m,1}]_{10}$. Substituting $E_m \equiv d_{m,2} - d_{m,1} \pmod{10}$ and $Z_m = 3$ into the third digit gives $3d_{m,1} + 5(d_{m,2} - d_{m,1}) + 21 \equiv 5d_{m,2} - 2d_{m,1} + 1 \pmod{10}$. $\square$

Proposition 3.6 (Decimal alphabet of forty-two blocks). *The per-window image of the emitter consists exactly of the blocks*

$[E_m]_{10}$	U_m						
0	114	227	443	556	669	885	998
1	129	232	458	561	674	890	903
3	149	252	478	581	694	810	923
5	169	272	498	501	614	830	943
7	189	292	418	521	634	850	963
9	109	212	438	541	654	870	983

Proof. A cardinal trajectory reads three consecutive terms of the additive cycle

$$(2, 3, 4, 5, 6, 7, 8, 9, 1).$$

Their sums are (9, 12, 15, 18, 21, 24, 18, 12, 6); reversing direction preserves the set of sums. Hence

$$\text{Im } S_m = \{6, 9, 12, 15, 18, 21, 24\}, \quad \text{Im } [S_m]_{10} = \{1, 2, 4, 5, 6, 8, 9\}.$$

In the multiplicative channel, a fixed factor divisible by three contributes zero. For invertible factors, the cyclic sums of three readings are

fixed factor	possible sums
1, 8	$\{1, 3, 7, 9\}$
2, 7	$\{3, 5, 9\}$
4, 5	$\{1, 5, 7\}$

Thus $\text{Im } E_m = \{0, 1, 3, 5, 7, 9\}$. Attainability can already be verified in the first window: every cursor in the following two tables has direction E .

S_1	6	9	12	15	18	21	24
(i_+, j_+)	(0, 8)	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)
E_1	0	1	3	5	7	9	
$(i_\times, j_\times)$	(2, 0)	(0, 0)	(0, 1)	(1, 1)	(0, 2)	(0, 4)	

Independence of the two cursors realizes every one of the $7 \cdot 6$ pairs. The first two digits recover the residual pair, so their 42 images are distinct. The check-digit congruence fixes the third digit and yields exactly the stated table. $\square$

The decimal block therefore preserves both residual signatures through an injective encoding and a check digit. This result describes the finite emitter alphabet; continuation of the regional readings additionally retains the phase, orientation and memory conditions of the enriched state.

The per-window image therefore has 42 blocks. Compatible concatenation of six windows preserves the census of 468 emissions in Proposition 3.8, whose ternary reduction yields 243 words within the 729-word ambient space. Each cardinality counts the object of its stage and retains the provenance of the same emitter.

3.4.4 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.7 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left\lfloor \left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right\rfloor_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 1: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

Table 2: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2), \quad \mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 3: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 4: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.8 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$$\begin{array}{c|cccccc} |\Psi^{-1}(w)| & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \text{number of words } w & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (3.14)$$

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.9 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.10 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.11 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 5; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 5: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 5.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

3.11 Cyclic correlation and preservation of path order

The record (3.17) preserves the ordered concatenation of samples. The following example isolates order information extracted by two distinct readers from a twelve-position word, retaining the domain of each reading.

Let $\tau = (\tau_1, \dots, \tau_{12}) \in \{-1, 0, 1\}^{12}$ be an ordered word, with indices taken modulo 12. Define

$$\begin{aligned} G_a &= \{a, a + 4, a + 8\} \quad (1 \leq a \leq 4), \\ v_{120}(\tau) &= \sum_{a=1}^4 \text{sgn} \left(\sum_{m \in G_a} \tau_m \right), \\ v_{270}(\tau) &= \sum_{m=1}^{12} \tau_m \tau_{m+3}. \end{aligned}$$

The marginal histogram is $p_j(\tau) = \#\{m : \tau_m = j\}/12$ for $j \in \{-1, 0, 1\}$, and its entropy is $H(\tau) = -\sum_j p_j(\tau) \log p_j(\tau)$, with $0 \log 0 = 0$.

Proposition 3.12 (Order information beyond the marginal distribution). *There exist words with identical histograms and distinct correlations ν_{270} , even when ν_{120} has the same value.*

Proof. Consider

$$\tau^{(a)} = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0), \quad \tau^{(b)} = (1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0).$$

Both have $p_1 = 1/6$, $p_0 = 5/6$ and $p_{-1} = 0$, so their marginal entropies coincide. In each case, two of the sets G_a contain a unit and the other two contain only zeros: $\nu_{120} = 2$. In $\tau^{(a)}$, no pair of units has cyclic separation three, whereas in $\tau^{(b)}$ precisely the term $m = 1$ contributes. Thus $\nu_{270}(\tau^{(a)}) = 0$ and $\nu_{270}(\tau^{(b)}) = 1$. $\square$

This separation identifies the information retained by the ordered record: incidences between positions in addition to symbol frequencies. A reduced state retaining only the histogram assigns the same output to both words and loses this correlation. Marginal entropy, uncertainty between bases and entanglement entropy are defined on different objects; relating them requires retaining the distribution, the change of basis, and the state together with its partition, respectively.

Proposition 3.13 (Parity of the readers under oriented reversal). *With cyclic indices in $\mathbb{Z}/12\mathbb{Z}$, let $(C_M \tau)_m = -\tau_{-m}$ and $\text{sgn}(0) = 0$. The two readers satisfy*

$$\nu_{120}(C_M \tau) = -\nu_{120}(\tau), \quad \nu_{270}(C_M \tau) = \nu_{270}(\tau).$$

Proof. The reflection $m \mapsto -m$ permutes the four position classes modulo four. Reversal changes the sign of each class sum and hence, by the oddness of sgn , the sign of ν_{120} . In the second reader, the signs of the two factors cancel:

$$\sum_m (C_M \tau)_m (C_M \tau)_{m+3} = \sum_m \tau_{-m} \tau_{-m-3} = \sum_k \tau_{k+3} \tau_k = \nu_{270}(\tau).$$

$\square$

Orientation thus distinguishes an odd reading from an even correlation. Both depend on the arrangement of samples in the record; their transformation laws specify which information persists when the route is reversed.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal core: initial conditions, geometry, and refinement

The support of eighty-one positions does not coincide with the domain of initial conditions of the dynamics. An initial condition must also specify an orientation, and the additive and multiplicative sheets have independent cursors. This distinction permits exhaustive reconstruction of the emission without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For the enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to the positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, and hence

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The preceding parametrization is already an internal enumeration: each of the 81 pairs of positions appears with all four directions, and the Cartesian product preserves the 324^2 combinations of the two cursors. The rows of an emission table are fibers of this domain of initial conditions; no external catalog is added to define them.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t - 1) \bmod 9))$. If its value is $+1$, the positive residue $\Sigma(i + 1, j + 1) = \rho_9(i + j + 2)$ of the first cursor is read and that cursor is then displaced; if its value is -1 , the positive residue $\Pi(i + 1, j + 1) = \rho_9((i + 1)(j + 1))$ of the second cursor is read and the second cursor is then displaced. The zero value increments the neutral counter. At the end of transitions twenty-seven and fifty-four, the cardinal orientations are reversed. The integer evaluations and their quotients are preserved in the record, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm on $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For the finite emission it is extended with value zero to the nonunit sector. This extension is an observable, not an identification of elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the path record.

Let S_m be the sum of the three positive additive residues Σ in window m , let E_m be the sum of the three discrete exponents of the multiplicative residues Π , and let $Z_m = 3$ be the number of neutral events. Thus, S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and the calendar of this recurrence are an explicit part of the stated emission law. No values of π, φ, e , or α are received. Distinguishing an emission rule from a constant recognized from its output prevents the genealogy from being concealed in a table of decimals.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule can be written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transporting subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (4.4).

Lemma 4.1 (Complete census of cursor signatures). *The 324 initial conditions of each cursor are distributed among eighteen additive signatures of cardinality 18 and twenty-six multiplicative signatures: one of cardinality 108, one of cardinality 24, and twenty-four of cardinality 8. These cardinalities and signatures are obtained from the emitter rules, without consulting real values or tables of constants.*

Proof. Identify the directions with

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1) \quad \text{in } (\mathbb{Z}/9\mathbb{Z})^2.$$

On the additive sheet, the signature depends only on $c = i + j \pmod{9}$ and on the sense $\epsilon(d) = +$ for $d \in \{E, S\}$, $\epsilon(d) = -$ for $d \in \{N, O\}$. For each pair (c, ϵ) , there are nine choices of i , which fix $j = c - i$, and two directions of the indicated sense. Hence each fiber contains $9 \cdot 2 = 18$ seeds. Substitution into the six windows gives the complete table

c	$\epsilon = -$	$\epsilon = +$
0	212988	988212
1	645212	212645
2	988546	546988
3	221889	889221
4	564122	122564
5	898465	465898
6	122898	898122
7	456221	221456
8	889654	654889

(A)

and its fibers are, explicitly,

$$\mathcal{P}_{c,+} = \{(i, c - i, d) : i \in \mathbb{Z}/9\mathbb{Z}, d \in \{E, S\}\}, \quad \mathcal{P}_{c,-} = \{(i, c - i, d) : i \in \mathbb{Z}/9\mathbb{Z}, d \in \{N, O\}\}.$$

For the multiplicative sheet, write $M(a, b) = \rho_9((a + 1)(b + 1))$ and, for $m = 0, \dots, 5, r = 0, 1, 2$,

$$a_{m,r} = \begin{cases} 3m + r, & 0 \leq m \leq 2, \\ -(3(m - 3) + r), & 3 \leq m \leq 5. \end{cases}$$

The signature of $T = (i, j, d)$ is determined componentwise by

$$e_m(T) = \left[\sum_{r=0}^2 \ell_9(M((i, j) + a_{m,r}v_d)) \right]_{10}. \quad (4.5)$$

This expression contains all $81 \cdot 4 = 324$ possible substitutions. The exhaustive classification of its values is

signatures	number of signatures	cardinality of each fiber
000000	1	108
555555	1	24
177393, 177555, 177771, 339555, 339771, 339933, 393177, 393393, 393555, 555177, 555339, 555393, 555717, 555771, 555933, 717555, 717717, 717933, 771177, 771339, 771555, 933339, 933555, 933717	24	8

(B)

Each entry in (B) is verified by substituting $i, j \in \{0, \dots, 8\}$ and the four directions into (4.5); no seed is omitted because $108 + 24 + 24 \cdot 8 = 324$. The fibers are disjoint because they are preimages of distinct signatures. Together with the calculation of the additive fibers, this proves the complete census. $\square$

Proposition 4.2 (Injective factorization of the emission). *The word $U = (U_1, \dots, U_6)$ determines the two signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The image of the emission has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient space of 729 words.*

Proof. The hundreds digits of U_m recover s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit verifies the third congruence in (4.3). The coupling is therefore injective on the product of signatures.

Lemma 4.1 gives eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures: twenty-four with eight preimages, one with twenty-four, and one with one hundred eight. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} \underbrace{432}_{18 \cdot 24} \text{ fibers of } 144, & 18 \text{ fibers of } 432, & 18 \text{ fibers of } 1944. \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. To close the second quotient without an external reference, apply (4.3) and then (4.4) to the $18 \cdot 26$ pairs formed by the rows of (A) and (B). The count by fiber cardinality is

$$\begin{array}{c|cccccc} k = \#\Psi^{-1}(w) & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \#\{w : \#\Psi^{-1}(w) = k\} & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (C)$$

The two global checks

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2(66) + 3(18) + 4(3) + 5(6) + 6(18) = 468$$

prove both the cardinality of the image and the exhaustion of the emissions. The number 243 is the cardinality of that image, not that of the ambient space $\mathbb{F}_3^6$, whose cardinality is 729. $\square$

This factorization explains the difference between finite exhaustiveness and unbounded depth. The domain (4.1) is complete for the stated support and emitter; it does not contain in advance every prefix of every extension. Depth belongs to the system of compatible extensions constructed on those conditions.

4.2 Closure of the calendar and advance of memory

The calendar of one hundred eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.6)$$

If $t = 9b + r$ with $0 \leq r < 9$, the sum of two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. The composition preserves the term that links the two coordinates.

Proposition 4.3 (Nonsplitting of the calendar). *The sequence (4.6) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would imply $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred eight restore the finite calendar. Neither fact requires erasing the path or restoring the prefix of an extension. The nonadic holonomy Γ_9 denotes transport through one circuit on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.7)$$

Equality of phase and the increment of memory are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by the TPK operations on $\hat{X}$. A path parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order makes it possible to express its transitions as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and the condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also specified. This determines the state space, the temporal order, and the information being transported.

The signature τ selects a quadratic regime. Relative to that parametrization, it also admits a temporal reading. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening along two eigendirections. In the

temporal terminology of the model, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: preserved antecedent, update, and admissible extension.

This interpretation acquires content through three distinct operations. Restriction of a path recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording at the observed transition. Extension considers the continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deletion of the final step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unbounded continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the constraints that each depth imposes on the preceding ones. The existence of a compatible collection is established in the corresponding extension domain; merely defining the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the sequence of signatures, the involution C_{rev} reverses the order and changes the sign of every component. The position, sheet, and boundary transformations specified by the transport also act on the path. The reversed signature is an observation of the conjugate path, not a replacement for its remaining coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation to exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. The phase return of Γ_9 , by contrast, acts on the path with memory, in accordance with (4.7). A path can recover its phase while preserving a new antecedent: the historical datum has changed although the phase observation is the same. The parabolic threshold retains a linear displacement; the hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, while the TPK determines their effective selection and concatenation.

The aperiodic suspension of Section 6 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on paths and by their quadratic representations. Its content goes beyond

a classification of signs: it preserves the regime, the direction of transport, and the relation among antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 6.11. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension of Section 6.13 defines the aperiodic time-crystal model, and Section 6.15 composes that temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization permits a quantitative expression of what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one circuit, $U^*U = I$, and let $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ be an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.8)$$

The final equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.4 (Conservation at a finite horizon). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r}D^*DT^r + T^{*n}T^n. \quad (4.9)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r and summing produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n}T^n$. $\square$

The final summand preserves the terminal state. Suppressing it before justifying a limit would alter the equality. This distinction is the operational content of preservation of unity under compression: the norm is distributed among the visible component, memory, and terminal, without being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonal entries modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. Unity under stochastic normalization and preservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. A local realization in $SO^+(1, 1)$ follows; proving it does not require introducing four-dimensional spacetime in advance.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

and hence its quadratic form has signature $(2, 1)$. The local spectral jump $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe distinct levels of the same arithmetic comparison. They are not interchangeable indices of a single operator.

Discrete curvature can be described directly before this spectral realization. On residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and let Δ_x, Δ_y denote the forward differences. The link variables are potential differences, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.10)$$

Sum and product, taken separately, induce flat connections; the ordered mixture changes the sign of the plaquette curvature. Upon summing over a rectangle with side lengths a, b , the interior edges cancel and the oriented boundary sum equals $\pm ab$ modulo nine. This yields an exact relation between order of composition and boundary orientation.

4.5 Cubic completion and two notions of boundary

The enlargement of the cubic support from side length nine to side length ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.11)$$

The first term corresponds to the three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The enlargement shell contains 271 positions. By contrast, the internal boundary of the discrete cube of side length nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.11) by 1000 gives a normalized partition of unity. Cubic incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent link with Steiner hexads and octads requires the corresponding combinatorial maps; the cube cardinalities do not replace them. Here this completion provides the relation between bases 729 and 1000, while the exceptional extension will be developed after numerical generation, the dodecaphase register, and the constructions of α .

4.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description that makes its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ is distinct from the zero class, already represented by 9. Consequently, enlarging E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without modifying the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.5 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.6 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.12)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.13)$$

The punctured enlargement contains 270 elements in addition to the initial cube; the complete enlargement contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates that contain $*$, in $\binom{3}{r}$ ways, and assigning to each remaining coordinate one of the nine classes of E_9 . Classification by the number of additional coordinates is exhaustive, and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *extreme and mean holographic ratio*, abbreviated EMRH in the corpus, here denotes this articulation among the initial body, the enlargement strata, and unitary restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a determined object, the punctured completion; 1000 represents its closure by restoration of the apex. The added unit therefore has an explicit incidence residence.

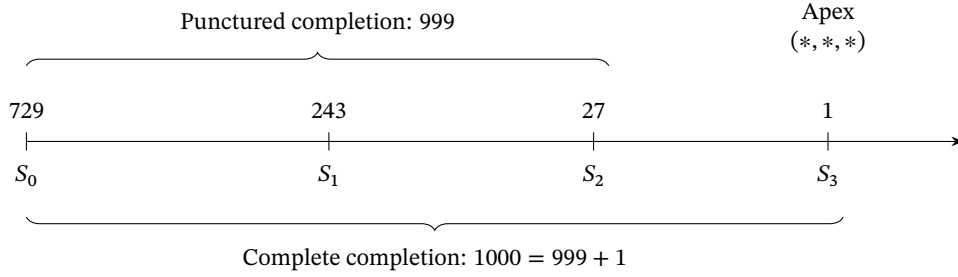


Figure 3: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 are three coordinate strata of 81 elements; the 27 elements of S_2 are three strata of 9. The apex completes the partition.

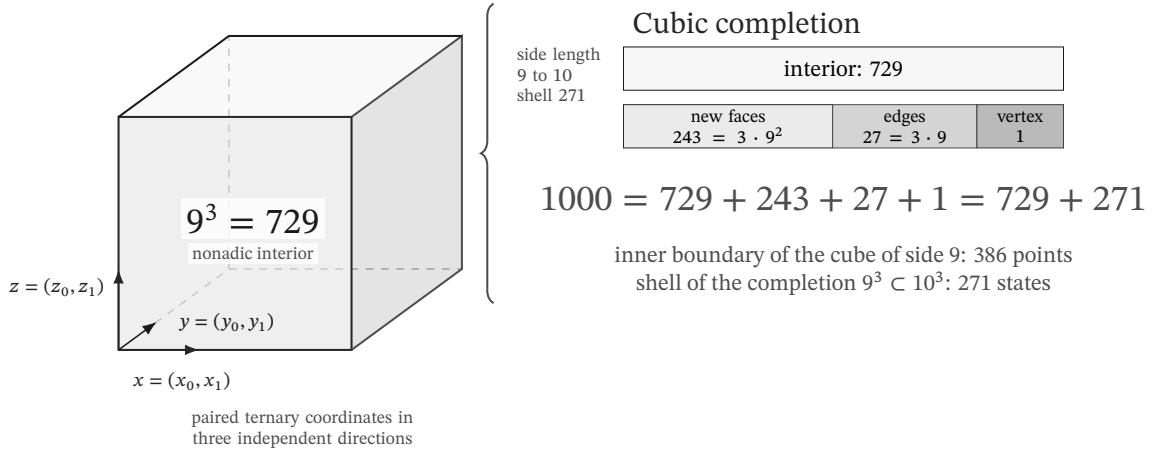


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

4.5.2 Preservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\widehat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \widehat{E}^{d+1} \rightarrow \widehat{E}^d$ deletes the final coordinate.

Proposition 4.7 (Projective compatibility of the normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\widehat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the sum of the masses of all strata is one.

Proof. Every point of $\widehat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset is obtained by summing over its points. The stratified formula follows from the same count as (4.12); the binomial $(9 + 1)^d$ gives the total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, preservation of probability mass, preservation of transport records, and thermodynamic dissipation are assertions concerning different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamical and energetic realization.

The internal geometric boundary likewise remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\widehat{C} \setminus C$ belongs to its enlargement. The cubic completion, the base-one-thousand positional expansion, and the exceptional-incidence reading share support data; their respective maps determine which information from those data is used by each representation.

4.6 Cubic incidence and Lorentzian realization

The cubic completion provides a relation between interior and boundary. Its geometry can be developed through the incidence of the six faces with the eight vertices. Label the faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and the vertices by $\nu \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),\nu} = \mathbf{1}_{\{\nu_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^\top \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^\top/6$, $P_- = (I - R_F)/2$.

Proposition 4.8 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^\top = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_\square = \mathbb{R}^6 / \ker(MM^\top) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct nonopposite faces share two vertices. The matrix $12P_u + 4P_-$ has precisely those entries. The uniform space has dimension one; the differences between opposite faces generate a three-dimensional space orthogonal to the uniform one. Its complement has dimension two and lies in the kernel. $\square$

The 1 + 3 decomposition arises from incidence. The temporal choice is specified by the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projections annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. This makes explicit the domain, the decomposition, and the sign convention that produce the Minkowski realization. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_{\square} = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_{\square}e_{i,\varepsilon} = t + \varepsilon d_i$$

maps the six face labels to null directions, since $B_{\square}(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

4.6.1 Gluing, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, which preserves the Lorentzian form, between the fibers at the endpoints of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the gluing map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibers at two consecutive levels by means of the specified refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve the label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.14)$$

Each transported term equals $\ell_m/9$ times the same null direction, which proves the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, instead of reducing the geometric correspondence to a dimension count.

Once the orientation and Lorentzian form have also been fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and specifying its coupling.

4.7 Compatibility between modular and positional reductions

The block-decimal base enters through Euclidean division, which simultaneously preserves residue and quotient:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Therefore, reduction of a block requires the contribution of the quotient in order to recover the class of the integer. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and modulo 3. This is an operational reason to preserve memory during a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated by means of a single integer residue. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

Upon adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, then

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.15)$$

Both identities are proved by substituting the definitions. Comparison of endpoints is thus reduced to integer operations that retain the contribution of each extension. This compatibility is what permits a region to be refined before its real coordinate is evaluated.

4.8 Number as a compatible section and value as a reading

The distinction between state and observation makes it possible to specify which object is extended as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: the arithmetic evaluations and preceding transports impose their relations. An extension preserves those relations in every antecedent, even as it adds new steps and new memory data. This is the meaning of the arithmetic of histories developed self-sufficiently in this article.

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be the depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.9 (Compatible numerical section). An HMT number is a compatible history of the system of states:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.16)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair formed by the section and its reader constitutes a read presentation, not a redefinition of the identity of the section.

On the Archimedean domain X_∞^{Arch} , the reader associates with each prefix a closed cylinder, preserves their nesting, and makes their diameter tend to zero. It thereby determines

$$\text{ev}_{\mathbb{R}} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_{\mathbb{R}}(x)\} = \bigcap_n I_n(x). \quad (4.17)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 6. Definition (4.16) does not by itself assert that the image of (4.17) is the entire real line: a coverage assertion requires its own theorem. Nor does equality of values establish equality of their histories without an injectivity result. This separation makes it possible to speak of precision as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first u is traversed and then v . The elementary steps arise from selection, transport, and the TPK update. A local tritic fiber can be written $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. The regime and orientation belong to the state being transported; a transition between regimes is not equivalent to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of that composition. For each object x , let A_x be a unital associative real algebra, which may be noncommutative. For each $u : x \rightarrow y$, let $\rho_u : A_x \rightarrow A_y$ be a unital homomorphism. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.18)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.19)$$

These are explicit hypotheses on the transports and their coefficients, not a consequence of calling the dynamics holonomic. When applying this presentation to a TPK sector, its ρ_u and $\Omega(v, u)$ must be identified. Transitions expressed through bimodules require preservation of that structure and are not replaced by (4.18).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.20)$$

if the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. The sum gathers histories with coefficients; the product concatenates histories while preserving the record multiplier.

Proposition 4.10 (Sectorial associativity with a central multiplier). *Under (4.18) and (4.19), the product (4.20) is associative. If the set of objects is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ permits moving it to the right of that factor; (4.19) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. The local identities and the normalization of Ω verify the stated identity element. $\square$

The calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two sums of carries of a triple are $\lfloor (a+b+c)/9 \rfloor$; hence (4.19) holds. The formal variable z records the circuit without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase register of calendar (4.6); retaining z without that relation distinguishes successive circuits. The example realizes the memory of this calendar without identifying it with the entirety of TPK memory.

4.10 Dual variance of refinement and preservation of unity

When states are restricted, observables are extended in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.21)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers the finite-depth observations. This algebra is not identified here with the entire path algebra: extending the transport operators also requires their compatibilities with refinement.

Lemma 4.11 (Unity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.22)$$

and, for every $f \in D_n$, $y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ form a partition of X_m . Their indicators are orthogonal and sum to 1_m , which proves (4.22). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant-one function on all possibilities; the individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement can increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

4.11 Phase return, Weyl representation, and two readings

Law (4.7) acquires an operator realization in the bilateral suspension to be constructed in Section 6. There T advances one step, V_9 observes the nonadic phase, W_δ the rotational phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for one circuit in this representation, the Weyl relations (6.30) imply, on vectors of finite support,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \quad (4.23)$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit increment of memory. The finite phase returns, while the irrational rotation and memory distinguish the succeeding state. The characters and parameter δ are realized subsequently from the numerical construction specified there: they do not retrospectively select the seeds. This representation describes observables and translations of the suspension; it does not replace all coordinates or all TPK operators.

The dual reading extends the same distinction between history and observable. On its Archimedean domain, (4.17) applies; on the enriched incidence domain, the maps Q_n of Section 14.1 preserve the words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines its extension Q_∞ . On the common domain of both readings, $\text{ev}_{\mathbb{R}}(x)$ and $Q_\infty(x)$ may be considered jointly: the first retains the value; the second, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and the exceptional reading without turning their realizations into new initial data.

5 Joint construction of the continuum through compatible histories and prolongations

The arithmetic expressed by a normal form preserves a precise reading of the state that produces it. To situate that reading within HMT, it is also necessary to retain the histories, the sheets, and the operations acting upon them. This chapter brings together that common substrate. Its different constructions are obtained from the same tree of APP-TRIT-TPK executions; the expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, the TRIT orientation, the change of sheet, the residue and the quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the source object. Two executions with an identical final reading are not identified for that reason alone.

5.1 Survival by pruning prolongations

Retrospective memory distinguishes histories recorded by the APP-TRIT-TPK operations; prospective compatibility organizes their admissible prolongations. Let S be the space of complete states reachable through these operations. Each state retains sheets, residues, quotients, orientation, carry, memory, and boundary data; whenever a transition depends on depth, depth forms part of the state. Let $E \subseteq S \times S$ be the effective prolongation relation and $A \subseteq S$ the admissible set determined by the compatibility of these records. Admissibility is thus decided through the operations and boundary conditions that generate the history.

For $X \subseteq S$, define

$$\text{Pre}_E(X) = \{s \in S : \exists t \in X, (s, t) \in E\}, \quad F_{\text{surv}}(X) = A \cap \text{Pre}_E(X). \quad (5.1)$$

The pruning sequence is given by

$$V_0 = A, \quad V_{n+1} = F_{\text{surv}}(V_n), \quad V_\infty = \bigcap_{n \geq 0} V_n. \quad (5.2)$$

Proposition 5.1 (Survival and the greatest fixed point). *The set V_n consists exactly of the states in A admitting an n -step prolongation all of whose states belong to A . The sequence (V_n) is decreasing. If each state has finitely many admissible successors, then*

$$V_\infty = F_{\text{surv}}(V_\infty) = \nu F_{\text{surv}},$$

where νF_{surv} denotes the greatest fixed point under inclusion. Its elements are precisely the states admitting an infinite compatible prolongation.

Proof. For $n = 0$, a prolongation of length zero consists of the state itself in A . If the characterization holds for n , a state belongs to V_{n+1} exactly when it belongs to A and has a successor in V_n ; prefixing that edge produces an admissible prolongation of $n + 1$ steps. This equivalence proves the assertion by induction. Moreover, $V_1 \subseteq V_0$, and monotonicity of F_{surv} propagates $V_{n+1} \subseteq V_n$.

Let $s \in V_\infty$, and write $E(s) = \{t \in S : (s, t) \in E\}$. Since $s \in V_{n+1}$ for every n , the sets $E(s) \cap V_n$ are nonempty, finite, and decreasing. Their intersection contains an element t , which belongs to V_∞ and satisfies $(s, t) \in E$. Hence $V_\infty \subseteq F_{\text{surv}}(V_\infty)$. Conversely, monotonicity gives $F_{\text{surv}}(V_\infty) \subseteq V_{n+1}$ for every n ; the image also lies in $A = V_0$, and therefore belongs to V_∞ . This proves the fixed-point equality.

If $W = F_{\text{surv}}(W)$, then $W \subseteq A = V_0$. From $W \subseteq V_n$ it follows that $W = F_{\text{surv}}(W) \subseteq F_{\text{surv}}(V_n) = V_{n+1}$. Induction yields $W \subseteq V_\infty$, proving maximality. Finally, the prolongation tree of a state in V_∞ is finitely branching and has a nonempty level at every depth. König's lemma provides an infinite branch. Conversely, the prefixes of any infinite prolongation place its initial state in every V_n . $\square$

Branching and stabilization. The finiteness hypothesis has an explicit control. Consider a root with a distinct terminal branch of each positive integer length, all branches being admissible and disjoint after the root. The root admits prolongations of every finite length and belongs to all the V_n ; each of its successors belongs to a branch of bounded length. Consequently, the root has neither an infinite prolongation nor a successor in V_∞ . This tree has infinite branching at the root. In contrast, if A is finite, each strict inclusion $V_{n+1} \subsetneq V_n$ removes at least one state. Pruning reaches a fixed point after at most $|A|$ strict layer removals. This bound pertains to the finite domain A .

Uniqueness of continuation. For $s \in V_\infty$, the successors retaining survival form

$$E(s) \cap V_\infty.$$

The next state is uniquely selected exactly when

$$\#(E(s) \cap V_\infty) = 1. \quad (5.3)$$

An infinite prolongation is unique when this condition holds at each state of the path. Indeed, each surviving successor has an infinite prolongation by the preceding proposition; two distinct successors produce two distinct paths. When the successor is unique at each step, the prefixes are determined successively. Several viable successors express multiplicity of prolongations within the same survival domain.

Proposition 5.2 (Conjugation of the two directions of prolongation). *Let $C : S \rightarrow S$ be an involution such that $C(A) = A$ and*

$$(x, y) \in E \iff (Cy, Cx) \in E. \quad (5.4)$$

Let V_n^+ denote pruning for E , and V_n^- pruning for E^{-1} , both with initial set A . Then

$$C(V_n^+) = V_n^- \quad (n \geq 0), \quad C(V_\infty^+) = V_\infty^-.$$

Proof. If $s_0, \dots, s_n$ is an admissible path for E , the sequence $Cs_0, \dots, Cs_n$ is an admissible path for E^{-1} : for each j , the hypothesis converts $(s_j, s_{j+1}) \in E$ into $(Cs_{j+1}, Cs_j) \in E$. All its states belong to A . Applying the same operation again recovers the initial path because $C^2 = I$; it therefore establishes a bijection between finite prolongations in the two directions. The characterization of $V_n^\pm$ by paths gives $C(V_n^+) = V_n^-$. Since C is bijective, it commutes with the intersection of these sets, yielding the equality for $V_\infty^\pm$. $\square$

The involution transports the surviving set in one direction to the surviving set in the conjugate direction. Infinite prolongability in both directions corresponds to $V_\infty^+ \cap V_\infty^-$; uniqueness of a path retains the criterion in (5.3). Retaining a genealogy and restricting its continuation tree are two typed operations on the same records. Assigning an entropy to these readings additionally requires the corresponding measure on histories.

5.2 Admissible histories and memory by composition

Let $H_0 := \{*\}$. The level H_1 is the finite nonempty set of admissible APP-TRIT seeds, and, for $k \geq 1$, H_k is the set of enriched histories of length k admitting compatible TPK prolongations at every finite horizon. Its elements are constructed through successive emissions from the initial domain of the kernel. We write $\zeta\varepsilon$ for the prolongation of ζ by an admissible emission ε , and $\rho_k(\zeta\varepsilon) = \zeta$ for deletion of the final emission. The label produced by the emission is retained, even when different labels express the same scalar state. The space of complete histories is defined by

$$H_\infty^{\text{enr}} := \varprojlim_k (H_k, \rho_k).$$

The cofinal subsystem $X_N := H_{6N}$, with $X_0 = H_0$, coincides with the sextet clock used in the cardinal proof; by cofinality there is a canonical identification $H_\infty^{\text{enr}} \cong \varprojlim_N (X_N, \rho_{6(N+1), 6N})$.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate similarly admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a path $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of those updates. Its linear part and its translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 5.3 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β yields $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the path. $\square$

This cocycle identity is the precise content of the accumulated memory. The quantity transported by a second path depends on the chart update it performs; in general, it is not the untransported sum of two records.

The nonadic return preserves a second distinction. In the phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

Over nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that traversal.

5.3 Tree refinement and measure of histories

The initial measure is the canonical one from the census: $\mu(*) = 1$ and $\mu(c) = 1/|H_1|$ for each seed $c \in H_1$. In this subsection, each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Admissibility of the kernel provides the prolongations; when a residence uses a neutral emission, that emission explicitly provides a section of ρ_k . The existence of this section concerns the tree that includes that emission, not an arbitrary restriction of routes.

Let $d(\zeta) \geq 1$ be the number of admissible emissions at ζ . The APP order and the census of prolongations determine the canonical law $p_{\text{can}}(\varepsilon \mid \zeta) = 1/d(\zeta)$. Without additional parameters, this law defines

$$\mu(\zeta\varepsilon) = \frac{\mu(\zeta)}{d(\zeta)}.$$

A nonuniform weighting may replace p_{can} only when it has already been produced and added to the enriched record. The correlative constructor of this chapter uses p_{can} exclusively; therefore, its measure is a function of the APP-TRIT-TPK tree and not of an external choice.

Proposition 5.4 (Cylindrical compatibility and conditional expectation). *At every level,*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(H_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^$, $E_k J_k = I$, and $\|J_k f\| = \|f\|$.*

Proof. The first identity is the normalization of the local probabilities. Applying it to $|f(x)|^2$ yields the isometry. Grouping the sum by fibers of ρ_k yields the formula for E_k ; on a function constant on each fiber, that formula returns its value. $\square$

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(\zeta)$. The measure extends from the algebra of cylinders, and its uniqueness follows because they generate the σ -algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under deletion of emissions is nonempty. In the cylinder topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different reading, since it would eliminate the multiplicities that the construction has just preserved.

5.3.1 Enriched readers and fiber memory

The difference between history and value admits an operational criterion. At a finite level H_k , let $q_k : H_k \rightarrow Z_k$ be a reader and let $M_k : H_k \rightarrow \mathcal{M}_k$ be the preserved record. The enriched reader is $\hat{q}_k = (q_k, M_k)$. Memory distinguishes states within a single fiber of the reader; its sufficiency is verified on the domain under consideration. This formulation incorporates into the article itself the complete development of the fiber information used below.

Proposition 5.5 (Recoverability at a finite level). *An inverse of $\hat{q}_k$ exists on its image if and only if $\hat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k ,*

$$H(Z) = H(q_k(Z)) + H(Z \mid q_k(Z)).$$

If $\hat{q}_k$ is injective on the support of Z , then

$$H(Z \mid q_k(Z), M_k(Z)) = 0.$$

Here $H(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity makes it possible to assign to each expressed pair its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z , grouping the sum defining $H(Z)$ by fibers of q_k yields the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state, and its entropy is zero. $\square$

Not publishing a coordinate and erasing the record that permits its recovery are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader, and distribution are precisely those just specified.

5.3.2 Equivariance of the measure under tree symmetries

Proposition 5.6 (Joint transport of the tree and its measure). *Let $g = (g_k)_{k \geq 0}$ be a collection of bijections $g_k : H_k \rightarrow H_k$ that commutes with truncations, fixes the root, and bijectively transports admissible emissions while preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \quad (g_k)_* \mu_k = \mu_k, \quad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, then

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The bijection of emissions preserves their number with multiplicities. The uniform seed distribution is likewise invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at every level and on every cylinder. Compatibility with truncations defines g_∞ , and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, beginning at ζ , proves the assertion about continuations. $\square$

This establishes the naturality of the tree and its measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

5.4 Transport by sheets and exact accounting of outputs

First consider the space with orthonormal basis formed by the histories at level k . The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_\zeta = \sum_{\varepsilon \text{ admissible at } \zeta} \sqrt{p_{\text{can}}(\varepsilon \mid \zeta)} e_{\zeta \varepsilon}.$$

The successor fibers of distinct antecedents are disjoint. The normalization of p_{can} therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation by masses of histories is related to this one by the diagonal change of basis incorporating $\sqrt{\mu(\zeta)}$.

The sheet label is part of the complete record of a history. We write

$$\hat{\zeta} = ((\zeta_0, s_0), \dots, (\zeta_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . The prolongation adds the emission and the sheet s_{k+1} determined by the operational TPK selector, retaining the prefix in full, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{\zeta}} = \sum_{\varepsilon \text{ admissible at } \hat{\zeta}} \sqrt{p_{\text{can}}(\varepsilon \mid \hat{\zeta})} e_{\hat{\zeta} \varepsilon}, \quad R_k e_{\hat{\zeta}} = \mathbf{1}_{s_k = \Sigma} e_{\hat{\zeta}}, \quad Q_k = I - R_k.$$

Thus, R_k reads the current sheet, while the distinction between histories is maintained by their complete prefixes. Two initial histories that reach the same active sheet retain distinct successor fibers. The proof of isometry by disjoint collections of prolongations applies verbatim to this basis. The tritic orientation is another coordinate of the record: the value 0 preserves the threshold reading on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation and sheet exchange:

$$A_k = R_{k+1} \mathcal{U}_k R_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k R_k + R_{k+1} \mathcal{U}_k Q_k.$$

Proposition 5.7 (Gram identity by sheets). *For a linear transport U , whether or not it is isometric, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, the mixed terms containing $P_{k+1} Q_{k+1}$ vanish. Summing the remaining terms produces $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$, and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second also uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross-blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it must not be attributed automatically to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$, and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 5.8 (Finite decomposition with terminal residue). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 5.7 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a positive bounded form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; this is applied to $S = T_n - T_\infty$. $\square$

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption that all histories are exhausted.

5.5 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader expresses the mean over the prolongations, $b_c = Eb_f$ is defined. The orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 5.9 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of the opposing contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine prolongations with equal weights; conditional expectation is the rule that remains valid for general branching and weightings.

5.6 Ternary incidence of the sheets and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts 4, 6, 8 belong to distinct operations on the same sheet module.

On the six pair positions, define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already retained in H_k .

To fix the ports without losing multiplicities, at depth k the labeled readings are retained: $\lambda_\Sigma(\zeta) = (\zeta, \Sigma)$ and $\lambda_\Pi(\zeta) = (\zeta, \Pi)$. Their sets are

$$\mathcal{P}_k^\Sigma = \{\lambda_\Sigma(\zeta) : \zeta \in H_k\}, \quad \mathcal{P}_k^\Pi = \{\lambda_\Pi(\zeta) : \zeta \in H_k\}.$$

They are finite and nonempty because H_k is. Values, quotients, and residues are expressed on these ports; equality of values does not identify their labels. The product $\mathcal{P}_k^\Sigma \times \mathcal{P}_k^\Pi$ is the domain of the rectangular reading, within which the support of the admissible incidences is recorded.

With this level fixed, write $I = \mathcal{P}_k^\Sigma$, $J = \mathcal{P}_k^\Pi$, and $A_m = \mathbb{Z}/3^m\mathbb{Z}$. The lexicographic APP order determines the minimal surviving branch and, within it, the reference ports i_0, j_0 . No external choice is added: changing coordinates produces a canonically isomorphic complex. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{i j_0} - w_{i_0 j} + w_{i_0 j_0}$$

define the following sequence, valid over the finite ring A_m :

$$0 \longrightarrow A_m \longrightarrow A_m^I \oplus A_m^J \xrightarrow{\kappa} A_m^{I \times J} \xrightarrow{\delta} A_m^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 5.10 (Exactness of rectangular incidence). *The preceding sequence is exact, with its first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{i j_0}$ and $v_j = w_{i_0 j_0} - w_{i_0 j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term is prolonged with zero reference row and column; its image under δ is the initial matrix. The proof uses only additions and subtractions, and therefore does not require division by a unit of the ring. $\square$

Reduction from A_{m+1} to A_m , while keeping $k, \mathcal{P}_k^\Sigma, \mathcal{P}_k^\Pi, i_0, j_0$ fixed, commutes with both maps. The reconstruction formulas are the same at every modular depth. This is the arithmetic compatibility proved in this subsection. The prolongation of histories $k \rightarrow k+1$ preserves the prefix labels; its maps between sets of ports are data distinct from the change of ring $m \rightarrow m+1$.

Those maps are obtained from deletion of emissions:

$$\rho_k^\Sigma(\zeta\varepsilon, \Sigma) = (\zeta, \Sigma), \quad \rho_k^\Pi(\zeta\varepsilon, \Pi) = (\zeta, \Pi).$$

The minimal surviving branch of the APP order just constructed has compatible prolongations. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the last module in the sequence, the matrix is first prolonged by zero on the reference row and column, this same formula is applied, and the result is restricted to the new complement of the reference ports. This rule also defines L_k when a new port has the marked port as its antecedent.

Proposition 5.11 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{m+1} \rightarrow A_m$ and compose in accordance with prefix deletion.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when an antecedent is the marked port, the four terms cancel and agree with prolongation by zero. All formulas are sums and differences with integer coefficients, so they commute with modular reduction. The composition of precompositions is precomposition with the composite prefix, proving the final assertion. $\square$

5.7 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integral complex. Its reduction to A_m uses the same integer record before reduction; therefore, successive depths are compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y by $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a path $\alpha : y \rightarrow y'$ updates the memory, its transport in the complex sends the like-named generators to their new representatives, except for

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 5.12 (Naturality of the memory complex). *The identities $\partial \Psi_\alpha = \Psi_\alpha \partial$ and $\Psi_{\beta\alpha} = \Psi_\beta \Psi_\alpha$ hold. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. Under concatenation, the memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring when appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α in this local homotopy do not denote the dodecaphase seal K : they belong to another domain, and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 5.13 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_m , one has*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_m[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integral realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y , and ι its inclusion. Verification on each generator gives

$$\partial h + h \partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish the values of that coordinate. The complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinant reading, in addition to this local complex.

5.8 Assembly with terminal paths

For a finite horizon, the residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_m[\text{Path}(\nu, \text{Term})]$, which retains each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_m} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing the adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs give the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 pairwise. Naturality in Proposition 5.12 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross terms in the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label asserting membership in a complex.

Terminal evaluation also does not require choosing a particular history. For each continuation h , the effective composition of its updates is applied. The result retains the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 5.3 proves that direct evaluation and evaluation of those two parts produce the same reading. This yields the terminal naturality square.

The path modules, incidences, memory transport, and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be retained by definition: their relations are constructed through sums, boundaries, and effective compositions, and are verified on their generators.

5.9 Determinants, limit, and arithmetic reading

At each residence, the free local complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding subsection, in turn, has the modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At finite horizon and over the same ring, they are free of finite rank and only finitely many of them are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from the direct-sum decomposition in each degree. It distinguishes the line of the local complex from that of the assembly with all paths; their bases are ordered by means of the retained labels. Modular reduction commutes with these exterior powers because the modules and their bases arise from the same free integral modules at fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_m ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases, and the relevant homology identifications. To calculate Smith factors, one uses the integer matrix produced before reduction, not an arbitrary lift of a modular matrix. An integer square block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_m only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences makes it possible to formulate and verify the stability of the relevant invariant factors; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of residues, the histories, and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in publishing nonempty cylinder intervals, compatible under inclusion and with diameter tending to zero. Their nested compact closures have a one-point intersection: existence follows from compactness and uniqueness from the diameter bound. That point is the value of the limit reading. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional notation. The value does not replace the history that produced it. The other representation preserves the incidences of the same state. The two targets share their origin without thereby receiving the same information.

The coinductive regions of Section 6, the dodecaphase record of Section 11, and the exceptional incidence of Section 14 are expressed from that substrate with their respective maps. The Hilbert-space realization subsequently retains the phase labels and specifies its own unital inclusion; duality and the screens act on the codomains constructed there. The joint operations assembled here transmit histories, sheets, orientation, and memory to those representations. Identifying a specific functional with an external analytic form in turn requires writing its action on the generators and verifying their inner products: the conservation identities of the present chapter do not replace that composition.

5.10 Mathematical scope of the joint construction

The constructions of state, tree, and measure; the correlative operations; and the terminal assembly with naturality and limit are developed in full in this article. In this chapter their balance, incidence, and memory identities have been formulated and proved on the same generators. The five constructions thus remain operations of a single tree. Each specialization keeps its exact hypotheses visible: isometry of transport, canonical weighting law, treatment of the terminal residue, and nonsingularity of the determinant block, where applicable.

The preceding equations prove the memory cocycles, weighted cancellation, rectangular exactness, and compatibility of the memory complex at every finite horizon, as well as its compatible passage to the limit. A subsequent analytic identification must compose its operator on these generators and verify the additional properties of its codomain; it does not alter or replace the joint construction proved here.

5.11 Joint emergence of the five constructions

For $0 \leq k \leq N$, let

$$\rho_{N,k} := \rho_k \circ \rho_{k+1} \circ \cdots \circ \rho_{N-1}, \quad \rho_{k,k} := \text{id}_{H_k},$$

and, for a surviving history $\zeta \in H_k$, define

$$\text{Desc}_{k,N}(\zeta) := \{x \in H_N : \rho_{N,k}(x) = \zeta\}.$$

The finite tree $\mathcal{T}_{k,N}(\zeta)$ has as vertices

$$\prod_{j=k}^N \{y \in H_j : \rho_{j,k}(y) = \zeta\}$$

and as edges only the TPK prolongations $a : y \rightarrow z$ with $z \in \rho_j^{-1}(y)$. Therefore, neither its vertices nor its prolongations arise from an auxiliary tree.

For $x \in \text{Desc}_{k,N}(\zeta)$, the marked branch $\lambda_{\zeta,x}$ is the sequence

$$\zeta = x_k \longrightarrow x_{k+1} \longrightarrow \cdots \longrightarrow x_N = x$$

determined by the truncations of x . The common base record is

$$\begin{aligned} \mathcal{R}_{k,N}(\zeta; x) := & (\mathcal{T}_{k,N}(\zeta), \lambda_{\zeta,x}; \\ & (\varepsilon_a, s_a, o_a, (\delta_{\circ,a}, r_{\circ,a}, q_{\circ,a})_{\circ \in \{\Sigma, \Pi\}}, \text{Carr}_a, C_a, t_a, \text{Fr}_a, \text{Route}_a, \text{Mem}_a, \text{Surv}_a)_a; \\ & \text{src}, \text{tgt}, \circ, (\rho_j)_{k \leq j < N}), \end{aligned} \quad (\text{C1})$$

where

$$U_a(m_0) = C_a m_0 + t_a, \quad \delta_{\circ,a} = r_{\circ,a} + 9q_{\circ,a}.$$

The symbol m_0 in this formula is a memory coordinate; it is not the modular depth m used below. The true base space is

$$\mathfrak{R}_{k,N}(\zeta) := \{\mathcal{R}_{k,N}(\zeta; x) : x \in \text{Desc}_{k,N}(\zeta)\}.$$

The marked branch is not an added input: it is the sequence of edges actually produced that ends at x . The remainder of the record simultaneously preserves the lateral fibers of the tree; therefore, the mark does not replace the branching with an isolated trajectory.

Set $R_m := \mathbb{Z}/3^m\mathbb{Z}$, and let $V_j(\zeta) := \text{Desc}_{k,j}(\zeta)$. The transport, the sheet projectors, and the terminal residue are restricted to the subtree by

$$\begin{aligned} \mathcal{U}_j^\zeta &:= \mathcal{U}_j \upharpoonright_{\ell^2(V_j(\zeta), \mu_j)}, & P_j^\zeta &:= P_j \upharpoonright_{V_j(\zeta)}, & Q_j^\zeta &:= I - P_j^\zeta, \\ A_j^\zeta &:= P_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + Q_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta, & \eta_j^\zeta &:= Q_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + P_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta. \end{aligned}$$

For $k \leq n \leq N$, let

$$F_{k,k}^\zeta := I, \quad F_{k,n}^\zeta := A_{n-1}^\zeta \cdots A_k^\zeta, \quad T_{k,n}^\zeta := (F_{k,n}^\zeta)^* F_{k,n}^\zeta.$$

The proper ports of the same subtree are

$$\mathcal{P}_j^\Sigma(\zeta) := \{(y, \Sigma) : y \in V_j(\zeta)\}, \quad \mathcal{P}_j^\Pi(\zeta) := \{(y, \Pi) : y \in V_j(\zeta)\};$$

we denote by $B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta$ the restrictions of the incidence and of the rectangular complex to those ports, and by L_j^ζ the precomposition induced by prefix deletion.

To define the path reader without abbreviations, let

$$W_\ell^\zeta(y) := R_m[\text{Path}_{\mathcal{T}_{k,\ell}(\zeta)}(y, \text{Term}_\ell)]$$

and set

$$\mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}] := \bigoplus_{y_0 \rightarrow \cdots \rightarrow y_q \text{ in } \mathcal{T}_{k,\ell}(\zeta)} W_\ell^\zeta(y_q) \otimes_{R_m} (\Gamma_r(y_0) \otimes_{\mathbb{Z}} R_m), \quad \text{Tot}_d \mathcal{D} := \bigoplus_{q+r=d} \mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}].$$

The formulas already proved produce, on each $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five objects

$$\begin{aligned} \mathcal{F}_{\text{sp}}[\mathcal{R}] &:= (\ell^2(V_j(\zeta), \mu_j), \mathcal{U}_j^\zeta, A_j^\zeta, \eta_j^\zeta, T_{k,j}^\zeta)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{sol}}[\mathcal{R}] &:= (b, \kappa_{\text{can}}, C_\alpha, \kappa_\alpha)_{\alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}] &:= (\mathcal{P}_j^\Sigma(\zeta), \mathcal{P}_j^\Pi(\zeta), B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta; R_m)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}] &:= (\Gamma_y \otimes R_m, \partial, \Psi_\alpha, H_\bullet)_{y, \alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}] &:= (\text{Det}(\text{Tot } \mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]))_{k \leq \ell \leq N}. \end{aligned} \tag{C2}$$

Here $\mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]$ is the normalized bicomplex of paths and memory from the preceding sections, restricted to the subtree of horizon ℓ and reduced over R_m . The last reader preserves the ordered collection of determinant lines up to horizon N ; in this way, its truncation is a canonical projection and does not presuppose a nonexistent determinant factorization. The law μ is the canonical law of the census, and the ports are fixed by the lexicographically minimal branch of the APP order. Therefore, none of the five readers receives an external weighting or an external port.

For $T \in \mathfrak{T} := \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}$, define the typed total space

$$\tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) := \coprod_{\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)} \{\mathcal{F}_T^{(m)}[\mathcal{R}]\}_{\mathcal{R}}, \quad p_T : \tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) \longrightarrow \mathfrak{R}_{k,N}(\zeta),$$

where $\mathcal{F}_{\text{sp}}^{(m)} := \mathcal{F}_{\text{sp}}$ and $\mathcal{F}_{\text{sol}}^{(m)} := \mathcal{F}_{\text{sol}}$; that is, the superscript m is inert for those two readers. The projection p_T forgets only the reader structure and preserves its source record. The correct fiber product is

$$\begin{aligned} \mathcal{J}_{k,N}^{(m)}(\zeta) &:= \tilde{\mathcal{F}}_{\text{sp};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{sol};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{gau};k,N}^{(m)}(\zeta) \\ &\quad \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{coh};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{det};k,N}^{(m)}(\zeta). \end{aligned} \tag{C3}$$

The equality of the five projections forces preservation of the same tree, the same edges, the same produced branch, and the same record. For each $\mathcal{R}$, the five constructions determine the element

$$s_{k,N}^{(m)}(\mathcal{R}) := (\mathcal{F}_T^{(m)}[\mathcal{R}])_{T \in \mathfrak{T}} \in \mathcal{J}_{k,N}^{(m)}(\zeta).$$

The existence of this element arises from the operations of (C2), not from the fiber-product notation.

Deletion of the last layer induces

$$\text{res}_{N+1,N}^{\mathfrak{R}}(\mathcal{R}_{k,N+1}(\zeta; x)) := \mathcal{R}_{k,N}(\zeta; \rho_N(x)).$$

The restriction of each operator to the retained generators defines

$$r_{N+1,N}^m : \mathcal{J}_{k,N+1}^{(m)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

In the determinantal reader, $r_{N+1,N}^m$ deletes only the last coordinate of the collection of lines in (C2). The reduction $R_{m+1} \rightarrow R_m$ defines, by base change,

$$q_{m+1,m}^N : \mathcal{J}_{k,N}^{(m+1)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

On each generator, the following identities hold:

$$\begin{aligned} r_{N+1,N}^m q_{m+1,m}^{N+1} &= q_{m+1,m}^N r_{N+1,N}^{m+1}, \\ \partial \Psi_\alpha &= \Psi_\alpha \partial, \quad L_j^\zeta \kappa_j^\zeta = \kappa_{j+1}^\zeta L_j^\zeta, \quad L_j^\zeta \delta_j^\zeta = \delta_{j+1}^\zeta L_j^\zeta, \end{aligned} \tag{C4}$$

and the compositions of the maps r and q satisfy the functorial identities. The first equality holds because deleting the last layer and reducing the coefficients act on distinct coordinates. The other three are the naturality identities proved above. A single pro-system in (N, m) is obtained, not five juxtaposed limits.

At the root, define

$$\mathfrak{F}_{N,\bullet}^{(m)} := \mathcal{J}_{0,N}^{(m)}(*). \tag{C5}$$

The joint discrete structure of the continuum is the limit of those correlative objects:

$$\mathcal{C}_{\text{cont}}^{\text{disc}} := \lim_{\substack{\leftarrow \\ N \geq 1 \\ m \geq 1}} (\mathfrak{F}_{N,\bullet}^{(m)}, r_{N+1,N}^m, q_{m+1,m}^N). \tag{C6}$$

This is a definition by the pro-system generated by the full tree; it is not the image or the graph of a map that retains the input state as its first coordinate.

The records possess a produced terminal mark. Their compatible projections induce

$$\Pi : \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow H_\infty^{\text{enr}}.$$

Conversely, a complete history determines, through its already emitted prefixes, the compatible sections $s_{0,N}^{(m)}$. Thus, after having defined the continuum, one obtains

$$\text{Gen}_{\text{cont}} : H_\infty^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}, \quad \Pi \circ \text{Gen}_{\text{cont}} = \text{id}_{H_\infty^{\text{enr}}}. \tag{C7}$$

This section proves recoverability; it is not used to define $\mathcal{C}_{\text{cont}}^{\text{disc}}$.

Theorem 5.14 (Correlative emergence of the five constructions). *For every surviving history $\zeta \in H_k$, every $N > k$, every $m \geq 1$, and every $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five constructions of (C2) are defined on the same record, and (C3) contains $s_{k,N}^{(m)}(\mathcal{R})$. Their truncations, modular reductions, and transports satisfy (C4). Furthermore, $\mathcal{C}_{\text{cont}}^{\text{disc}} \neq \emptyset$, the map Π is surjective and is split by Gen_{cont} . The five components are inseparable aspects of the same continuum, not five domains or five independent maps.*

Proof. The survival that defines H_k and the TPK prolongation law give $\rho_k^{-1}(\zeta) \neq \emptyset$ for each vertex. Because the fibers are finite, König's lemma yields $H_\infty^{\text{enr}} \neq \emptyset$.

For a fixed record $\mathcal{R}$, the transport of histories and sheets produces $\mathcal{F}_{\text{sp}}[\mathcal{R}]$; affine composition and the exact cancellation defect produce $\mathcal{F}_{\text{sol}}[\mathcal{R}]$; the incidences and the rectangular sequence produce $\mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}]$; the memory complex and its transport produce $\mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}]$; and the lines of the normalized path

assembly produce $\mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}]$. Each construction acts on the same generated edges, not on an assumed subdomain or on the abstract ambient space of 729 words.

The affine, transport, incidence, and memory compatibilities prove (C4). Truncating the determinant collection of (C2) commutes by definition with r ; its base change commutes with q . The sections $s_{0,N}^{(m)}$ are compatible and thus produce Gen_{cont} . The identity (C7) proves simultaneously the nonemptiness of the limit, the surjectivity of Π , and the recoverability of each history without introducing it as a defining coordinate.

The fiber, relation, number, orientation, carry, memory, boundary, and route remain in the common record; the multisection, the terminals, and the linear and probabilistic readers remain in the five correlative constructions. Therefore, passage to the limit preserves the closure of dependencies and does not disaggregate the continuum. $\square$

The subsequent recognition of the single continuum constructed through spectral, solenoidal, gauge, cohomological, or determinant language belongs to the subsequent conventional comparison. It neither selects its states nor intervenes in its APP-TRIT-TPK generation.

6 Joint coinductive prolongation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. The prolongation must preserve those data and determine a compatible collection of prefixes of arbitrary length. The word *generation* designates this operation; *genealogy* designates the reconstruction of its dependencies. Both notions enter the construction, but they perform distinct functions.

6.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

One has $r^3 = s^2 = I$ and $srs = r^{-1}$; therefore these permutations realize the dihedral group D_3 . The action preserves the catalog and the multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fiber predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is exactly one orbit with these properties in the catalog. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ for indices from zero to eight; its additive signature preserves this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of those classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, selects propagation alone; the census $\nu = (2, 2, 2)$ selects self-scaling alone. The diagonal support cannot be omitted: without it there are four balanced single-point orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and leads to

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (6.1)$$

These symbols first designate regions of the catalog. The scalar identification will be proved through their readers; they are not chosen because they coincide with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and the constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position of the support would eliminate the distinction between numerical region and electronic structure.

The microfiber also has a precise bipartite morphology. Each emission is a product of an additive-cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions taken together, another 18×24 ; and the return, a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition by connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing the form to a cardinality.

6.2 Incidences between halves of regional emissions

Let $\mathcal{U}_6$ be the image of six consecutive windows of the emitter of Section 3.4. For $u = (u_1, \dots, u_6)$, define $A(u) = (u_1, u_2, u_3)$ and $B(u) = (u_4, u_5, u_6)$. The half-emission graph has these triples as vertices and an undirected edge between $A(u)$ and $B(u)$ for each emission.

Proposition 6.1 (Cycle of regional junctions). *The vertices*

$$\begin{aligned} A &= (830, 943, 830), & B &= (109, 252, 252), \\ C &= (810, 923, 870), & D &= (169, 272, 272), \\ E &= (850, 963, 890), & F &= (149, 252, 212) \end{aligned}$$

form the cycle $A - B - C - D - E - F - A$.

Proof. Write each seed as (i, j, d) , with residual positions between 0 and 8 and directions $N = (-1, 0)$, $E = (0, 1)$ and $S = (1, 0)$. Under the 54-step recurrence, the following seed pairs produce the six emissions shown:

emission	s_+	$s_\times$
$A \mid B$	$(0, 6, E)$	$(0, 6, N)$
$B \mid C$	$(0, 6, N)$	$(1, 5, E)$
$C \mid D$	$(0, 6, E)$	$(0, 1, S)$
$D \mid E$	$(0, 6, N)$	$(0, 4, S)$
$E \mid F$	$(0, 6, E)$	$(0, 3, E)$
$F \mid A$	$(0, 6, N)$	$(0, 1, S)$

Each row witnesses membership in $\mathcal{U}_6$ by successive evaluation of the two accumulators. Splitting each emission into its two halves yields exactly the six edges of the cycle. $\square$

In the classification (6.1), $A \mid B$, $C \mid D$ and $E \mid F$ respectively produce the words w^{au} , w^{cl} and w^{pr} . The regional readers subsequently identify these outputs with φ , one region of π and e . The other three emissions exhibit their junctions in the graph. The result describes incidences between emitter outputs. Temporal realization of an incidence additionally uses the orientation, phase and memory of the enriched state. Including the APP anchor and all five regions of π imposes further path conditions; the cycle above retains its role as a local junction witness.

The six-edge cycle belongs to this undirected half-emission graph. An enriched temporal path additionally retains the anchoring, orientation and closure data of its construction. Incidence between two halves and transport of a state thus have explicit domains and composition conditions.

6.2.1 Block multigraph and five minimal regional closures

The 468 emissions in $\mathcal{U}_6$, produced by the APP-TRIT-TPK emitter, determine a block multigraph. Identifying its regional anchors makes it possible to solve a joint edge-covering problem: to traverse the APP, propagation, autoscale, and each closure region's incidences within a single closed walk. The local six-junction cycle of Section 6.2 and these covers retain their respective traversal conditions.

Block graph and exact phase quotient. For $U = (U_1, \dots, U_6) \in \mathcal{U}_6$, define

$$A(U) = (U_1, U_2, U_3), \quad B(U) = (U_4, U_5, U_6). \quad (6.2)$$

The undirected multigraph $\tilde{G}$ has as vertices the triples occurring as $A(U)$ or $B(U)$, and one edge labelled by each U , incident at those two triples. All 468 labels are retained: distinct emissions are not identified merely because they share endpoints or because one interchanges the halves of another.

Internal phase acts on each triple by cyclic rotation:

$$(a, b, c) \sim (b, c, a) \sim (c, a, b), \quad \text{can}(a, b, c) = \min_{\text{lex}}\{(a, b, c), (b, c, a), (c, a, b)\}. \quad (6.3)$$

The graph G replaces the endpoints of edge U by $\text{can } A(U)$ and $\text{can } B(U)$, retaining its label. The equivalence acts precisely by the three cyclic rotations of each vertex; every emission retains its identity as an edge.

Enumeration from the same emitter and component searches give

	$ V $	component sizes
$\bar{G}$	96	(28, 28, 28, 4, 4, 4)
G	32	(28, 4)

(6.4)

These data are obtained by collecting distinct endpoints, adding each incidence in both directions, and traversing each unvisited component by breadth-first search. Adjacency may be deduplicated when counting components, but parallel labels are retained when searching for closures with prescribed edges.

APP anchor and regional numbering. The three common edges are

$$\begin{aligned}
 U_{\text{APP}} &= 903|850|850|252|109|252, \\
 U_e &= 850|963|890|149|252|212, \\
 U_\varphi &= 830|943|830|109|252|252.
 \end{aligned}
 \tag{6.5}$$

Fix the endpoint $v_{\text{APP}} = \text{can } A(U_{\text{APP}}) = 850|850|903$. For the five already constructed regions, use the following order, which places the transverse region in the fifth row:

i	$U_\pi^{(i)}$	ρ_i	m_i
1	810 923 870 169 272 272	++	144
2	810 983 810 169 272 272	++	144
3	870 923 810 169 272 272	++	144
4	870 923 810 418 674 521	--	144
5	501 614 498 169 272 272	$\perp$	432

(6.6)

This numbering distinguishes the three ++ regions, the mutated return --, and the transverse region $\perp$. Accordingly, $\rho = (++, ++, ++, --, \perp)$ and $m = (144, 144, 144, 144, 432)$ in this order; their sum remains 1008.

Five complete witnesses with traversal directions. In each table, row j means traversing the edge labelled e_j from v_{j-1} to v_j . Endpoints are canonical phase representatives; since G is undirected, the pair (v_{j-1}, v_j) specifies the traversal direction of the labelled edge. Each table contains exactly eight edges and satisfies $v_0 = v_8 = v_{\text{APP}}$.

The breadth-first search described in the proof yields the following witnesses, one for each set of four target edges. Minimality concerns length; several walks may attain that length.

Witness $i = 1$.

j	v_{j-1}	e_j	v_j
1	850 850 903	109 252 252 850 903 850	109 252 252
2	109 252 252	109 252 252 810 923 870	810 923 870
3	810 923 870	810 923 870 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 2$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 810 983 810	810 810 983
3	810 810 983	810 983 810 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 3$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 850 963 890	850 963 890
3	850 963 890	850 963 890 149 252 212	149 252 212
4	149 252 212	149 252 212 870 923 810	810 870 923
5	810 870 923	870 923 810 169 272 272	169 272 272
6	169 272 272	169 272 272 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 4$.

j	v_{j-1}	e_j	v_j
1	850 850 903	169 272 272 850 903 850	169 272 272
2	169 272 272	169 272 272 850 963 890	850 963 890
3	850 963 890	850 963 890 149 252 212	149 252 212
4	149 252 212	149 252 212 870 923 810	810 870 923
5	810 870 923	870 923 810 418 674 521	418 674 521
6	418 674 521	418 674 521 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Witness $i = 5$.

j	v_{j-1}	e_j	v_j
1	850 850 903	109 252 252 850 903 850	109 252 252
2	109 252 252	109 252 252 501 614 498	498 501 614
3	498 501 614	501 614 498 169 272 272	169 272 272
4	169 272 272	169 272 272 850 963 890	850 963 890
5	850 963 890	850 963 890 149 252 212	149 252 212
6	149 252 212	149 252 212 830 943 830	830 830 943
7	830 830 943	830 943 830 109 252 252	109 252 252
8	109 252 252	903 850 850 252 109 252	850 850 903

Theorem 6.2 (Minimality of the five regional edge covers). *For each $i \in \{1, \dots, 5\}$, among closed walks in G traversing every edge of*

$$E_i^* = \{U_{\text{APP}}, U_e, U_\varphi, U_\pi^{(i)}\}$$

at least once, the minimum length, counting traversals with multiplicity, is eight. Vertices and edges may be repeated.

Proof. Every closed walk traversing U_{APP} passes through v_{APP} . Cyclic rotation of its step list allows it to start at that vertex without changing length or coverage. It therefore suffices to solve the problem with a fixed starting vertex.

The search space is the finite set

$$\mathcal{S}_i = V(G) \times \mathcal{P}(E_i^*), \quad |\mathcal{S}_i| \leq 32 \cdot 16 = 512.$$

The initial state is $s_0 = (v_{\text{APP}}, \emptyset)$, and the terminal state is $s_* = (v_{\text{APP}}, E_i^*)$. Traversing an edge e incident at v towards its other endpoint v' updates the state by

$$(v, M) \mapsto (v', M \cup (E_i^* \cap \{e\})). \quad (6.7)$$

The mask records edge labels, not just endpoints; distinct parallel edges must therefore not be collapsed.

Every transition has unit cost. A FIFO queue initialized at s_0 extracts states in increasing distance, marks each state on its first visit, and stores its predecessor and traversed label. Equivalently, with $\mathcal{F}_0 = \{s_0\}$, the new layers are

$$\mathcal{F}_{k+1} = \text{Succ}(\mathcal{F}_k) \setminus \bigcup_{j=0}^k \mathcal{F}_j.$$

Induction shows that $\mathcal{F}_k$ contains exactly the states whose minimum distance from s_0 is k . Finite enumeration over the specified emitter gives, for each of the five regions,

$$s_* \notin \bigcup_{k=0}^7 \mathcal{F}_k, \quad s_* \in \mathcal{F}_8, \quad (\ell_1, \ell_2, \ell_3, \ell_4, \ell_5) = (8, 8, 8, 8, 8).$$

The tables exhibit the five upper bounds, including all four required labels in each case. Exhaustive exploration of the preceding layers proves the lower bound. Together they establish minimality for the declared problem. $\square$

Length eight characterizes the joint coverage of four incidences in each closure region. The quotient retains emission labels and their junctions between phase classes; enriched temporal evolution also preserves orientation and memory. This description links all five regions to the same propagation, autoscale, and APP anchors and complements the local six-junction cycle. Coinductive prolongation retains its operators and cylinders together with these orientation and memory data.

6.3 Reconstruction of the finite transports and the prolongation calendar

The endomorphisms L_0, L_1 of the space of ternary words act before the Archimedean realization. Their determination begins with the transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite step of selection, transport, and updating, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In that row-vector chart, one obtains

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \quad (6.8)$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \quad (6.9)$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (6.9) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

6.4 Five closure regions and angular compensation

The closure region contains one transverse component and four accompanying components. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^T, \quad \mathbf{1} = (1, 1, 1, 1, 1)^T.$$

The invariance $P_5 \lambda = \lambda$ and the preservation $\mathbf{1}^T \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes unity among the reader positions; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The elliptic chart of the reader realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. The rule that passes from a normalized coordinate to a slope is thus made explicit, rather than identifying the two notions without a map. The reader composes the four accompanying positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (6.10)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. The composition of slopes is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 6.3 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to unit slope determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, whence $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{cl} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (6.11)$$

Machin's angular identity recognizes $\Pi_{cl} = \pi$. The direction of the construction is precise: the slope and the number of compositions are declared by the penta-fiber reader; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not reselect the seed orbit.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes of (6.11), they provide rational endpoints with explicit error. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

6.5 Propagation reader and normalization of the unit

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and the oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not merely its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

the normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (6.12)$$

The character at one unit of parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

The normalization of the generator matters. Without that datum, the semigroup law would admit $P_t = e^{ct}$ for distinct c ; hence the condition that fixes the unit of the parameter in the reader is not omitted. This condition precedes decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 6.4 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric majorant of the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

The strict inequality follows from the decrease of the successive ratios. $\square$

6.6 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. In modes Σ, Π , the rule is $+1 : m \mapsto \Sigma, -1 : m \mapsto \Pi, 0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Therefore its incidence, in that order of modes, is

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (6.13)$$

The four entries are obtained from the calendar, not from an equation that has previously received the value of φ . Repetition of the calendar gives counts $3F_{\text{av}}, 9F_{\text{av}}, 36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred eight. These counts are not powers of the matrix.

Theorem 6.5 (Positive self-scaling character). *The incidence (6.13) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and is $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Of its two roots, only one is positive. Since every entry of F_{av}^2 is positive, the positive eigenray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in the production of F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive quotients F_{n+1}/F_n alternate about φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive quotients. Their growth makes this rational bracket arbitrarily small.

The measure on all compatible histories does not coincide with the mode frequency of the three-event calendar. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal–volumetric reading; it is not obtained by simply replacing one counting frequency with another.

6.6.1 Path space and invariant measure between sheets

The toroidal topology of APP fixes the support of the cardinal translations and their winding classes. The correspondence between sheets arises from another operation on that same support: the temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \eta_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

The first sheet corresponds to the additive boundary reading; the second, to the multiplicative interior reading. This map transports the mode labels and preserves the incidence (6.13). The square of the self-scaling ratio is determined through the paths of that incidence.

Definition 6.6 (One-memory symbolic envelope). Let

$$X_{\text{av}} = \{(s_k)_{k \in \mathbb{Z}} \in \{\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}}\}^{\mathbb{Z}} : (F_{\text{av}})_{s_k, s_{k+1}} = 1 \text{ for every } k\}.$$

The index shift acts on this space. It is the smallest one-memory symbolic system that contains the modal sequences and preserves their consecutive pairs: it admits the two cross transitions and volumetric self-retention. The phase, cardinal orientation, and path records remain in the TPK state from which this envelope is obtained.

Proposition 6.7 (Weighting of paths between sheets). *In areal–volumetric order, the invariant measure of maximal entropy of X_{av} has transition matrix and stationary distribution*

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad (\mu_{\text{ar}}, \mu_{\text{vol}}) = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (6.14)$$

In particular, $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$.

Proof. The incidence already constructed has positive eigenvector $r = (1, \varphi)^T$ and eigenvalue φ . The normalization

$$(P_{\text{av}})_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i}$$

gives the stated matrix. Its rows sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. The symmetry of F_{av} implies that the weights proportional to r_i^2 satisfy detailed balance; hence $\mu P_{\text{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\text{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If v_i are its vertex weights and $q_i(j) = \nu(s_1 = j \mid s_0 = i)$, for $v_i > 0$, then

$$h_\nu \leq H_\nu(s_1 \mid s_0) = \log \varphi - \sum_{i: v_i > 0} v_i D(q_i \parallel R) \leq \log \varphi,$$

where $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on the allowed pairs, with the convention $0 \log 0 = 0$. The Markov measure of (6.14) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = R$. Stationarity and irreducibility then fix μ , thereby establishing uniqueness. This is the Parry measure of the defined envelope. $\square$

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the paths of the envelope. Its quadratic ratio expresses the product of the incoming and outgoing incidence amplitudes. The memory and phase constraints of the TPK preserve their own domain; the entropy of the envelope corresponds to the symbolic system specified in this definition.

6.6.2 Closure operator and limit of the marked returns

The closure coordinate π , generated by the regional reader, enters after the construction of the modal incidence. Let $e_{\text{ar}} = (1, 0)^T$, $e_{\text{vol}} = (0, 1)^T$ and $P_i = e_i e_i^T$ be the sheet projectors. The conditions of the reader are preservation of both sheets, unit volumetric normalization, and an areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \quad (6.15)$$

Diagonality expresses sheet preservation, and the two entries express their normalizations; the uniqueness of the operator for the fixed reader is thereby made explicit.

Proposition 6.8 (Ratio of boundary returns). *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \quad (6.16)$$

satisfies

$$\lim_{n \rightarrow \infty} \Theta_n = \pi \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \frac{\pi}{\varphi^2}. \quad (6.17)$$

Proof. The diagonal entries of F_{av}^n count the paths of length n that return to their initial sheet. The power formula already proved yields F_{n-1} and F_{n+1} . The operator D_π multiplies the first count by π . Finally, F_{n-1}/F_{n+1} is the product of two consecutive quotients that converge to φ^{-1} ; its limit is φ^{-2} . The identity with the ratio of weights follows from (6.14). $\square$

The closure operator appears only once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. The information from two readers of the same state is thus composed: the modal incidence determines the quadratic weighting, and the closure region contributes its coordinate at the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image of the reader $\mathcal{R}(x, y) = x/y^2$ is the interval

$$\mathcal{Q}(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that the nested closure and self-scaling cylinders produce nested intervals for the ratio; the positivity of their limits and the contraction of their diameters determine π/φ^2 . The scalar expression thus retains a chain of operations compatible with depth, which is subsequently applied in its two orientations.

6.6.3 Multiplicity and recovery of modal itineraries

The incidence F_{av} , constructed by the tritic calendar in the preceding development, acts here on its envelope of modal walks. Write

$$F = F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad 0 = \Sigma, \quad 1 = \Pi.$$

The self-scaling coordinate φ and the positive ray $r = (1, \varphi)$ are the outputs already obtained from this incidence. They determine the multiplicity of the scalar reading and support a reversible representation of each itinerary.

Consider the set of finite directed walks $\gamma = (x_0, \dots, x_n)$ of F . Their endpoints and lengths are distinguished, repetitions are allowed, and trivial walks are included. This domain is the modal envelope; the complete TPK state additionally retains the records of its enriched history. The multiplicative character takes the form

$$\chi(\gamma) = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \varphi^{k(\gamma)}, \quad r = (1, \varphi), \quad k(\gamma) = n + x_0 - x_n. \quad (6.18)$$

The transitions $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$ increment k by 0, 1, and 2, respectively. The first step modifies the path while preserving its character. In particular, both $(1, 1)$ and $(0, 1, 1)$ produce φ under this reading.

Proposition 6.9 (Multiplicity of the self-scaling reading). *Let $\mathcal{F}_k = \{\gamma : k(\gamma) = k\}$ and $N_k = |\mathcal{F}_k|$. Then*

$$N_0 = 3, \quad N_k = 2 \operatorname{tr}(F^k) = 2(\varphi^k + (-\varphi^{-1})^k) \quad (k \geq 1), \quad (6.19)$$

and

$$\lim_{k \rightarrow \infty} N_k^{1/k} = \varphi.$$

Proof. Given endpoints $i, j \in \{0, 1\}$, the length is determined by $n = k - i + j$. The number of walks with those endpoints is $(F^n)_{ij}$. For $k \geq 1$, summing the four sectors gives

$$N_k = \operatorname{tr}(F^k) + (F^{k+1})_{01} + (F^{k-1})_{10}.$$

The identity $F^2 = F + I$ implies by induction that, for $n \geq 1$,

$$F^n = \begin{pmatrix} f_{n-1} & f_n \\ f_n & f_{n+1} \end{pmatrix}, \quad f_0 = 0, \quad f_1 = 1, \quad f_{n+1} = f_n + f_{n-1}.$$

For $k \geq 2$, the two off-diagonal terms are $f_{k+1} + f_{k-1} = \operatorname{tr}(F^k)$; for $k = 1$, their sum is $1 + 0 = 1 = \operatorname{tr}(F)$, using $F^0 = I$. For $k = 0$, the only walks are (0) , (1) , and $(0, 1)$. The eigenvalues of F are φ and $-\varphi^{-1}$, so its trace gives the closed formula. Finally,

$$N_k = 2\varphi^k(1 + (-\varphi^{-2})^k),$$

and the factor in parentheses tends to one; taking k -th roots yields the limit. $\square$

Self-scaling thus coincides with the exponential growth rate of the finite modal genealogies that the scalar character makes indistinguishable.

Refinement by the composition of returns. Let $b(\gamma)$ count the edges $\Pi \rightarrow \Sigma$, and let $c(\gamma)$ count the edges $\Pi \rightarrow \Pi$. The increments in (6.18) give $k = 2b + c$. For formal indeterminates z, t , the graded incidence

$$M(z, t) = \begin{pmatrix} 0 & 1 \\ tz^2 & z \end{pmatrix}$$

records exponent and returns jointly. Its only degree-zero edge is $\Sigma \rightarrow \Pi$, and there is no degree-zero cycle. Each coefficient therefore counts a finite set of walks. If $N_{k,b}$ denotes their multiplicity with values k, b , the sum over all lengths satisfies

$$\sum_{k,b \geq 0} N_{k,b} z^k t^b = \mathbf{1}^\top (I - M(z, t))^{-1} \mathbf{1} = \frac{3 - z + tz^2}{1 - z - tz^2}, \quad \mathbf{1} = (1, 1)^\top. \quad (6.20)$$

Indeed, entry ij of M^n counts walks of length n with those endpoints and their weights. Coefficientwise formal summation yields $(I - M)^{-1}$. The determinant of $I - M$ is $1 - z - tz^2$, and the sum of the four entries of its adjugate is $3 - z + tz^2$.

For $k \geq 1$ and $0 \leq b \leq \lfloor k/2 \rfloor$, this gives

$$N_{k,b} = \frac{2k}{k-b} \binom{k-b}{b}. \quad (6.21)$$

To prove this, the expansion

$$(1 - z - tz^2)^{-1} = \sum_{m \geq 0} (z + tz^2)^m$$

gives coefficient $\binom{k-b}{b}$ at $z^k t^b$. Multiplication by the numerator of (6.20) produces

$$3 \binom{k-b}{b} - \binom{k-b-1}{b} + \binom{k-b-1}{b-1} = 2 \binom{k-b}{b} + 2 \binom{k-b-1}{b-1}.$$

Coefficients outside their range are taken to be zero. For $b \geq 1$, the identity $\binom{k-b-1}{b-1} = \frac{b}{k-b} \binom{k-b}{b}$ yields (6.21); for $b = 0$, both sides equal two. The constant term is treated separately: $N_{0,0} = 3$. For $k = 4$, there are fourteen walks, distributed as 2, 8, and 4 according to whether they contain zero, one, or two returns. The same character therefore admits distinct internal compositions.

Proposition 6.10 (Reversible recovery and transport of codes). *Each walk in the modal envelope admits a code*

$$\mathcal{C}(\gamma) = (k, a), \quad 0 \leq a < N_k,$$

with a decoding map $\mathcal{D}$ such that

$$\mathcal{D}\mathcal{C} = \text{id}, \quad \mathcal{C}\mathcal{D} = \text{id}.$$

If $A_v(\gamma) = \gamma v$ prolongs a walk by an admissible edge and T removes the last state of a positive-length walk, their representations

$$\mathcal{E}_v = \mathcal{C}A_v\mathcal{D}, \quad \hat{T} = \mathcal{C}T\mathcal{D}$$

satisfy, on their respective domains,

$$\hat{T}\mathcal{E}_v = \text{id}, \quad k(A_v\gamma) - k(\gamma) = 1 + x_n - v. \quad (6.22)$$

Proof. For each k , the nonempty sectors are ordered by (n, i, j) and, within each sector, walks are ordered lexicographically by their states. The number of length- m continuations from v to j is $(F^m)_{vj}$. Traversing a walk, one sums the sizes of preceding sectors and, at each position, the sizes of the continuations lexicographically preceding the chosen one. The result is its zero-based index a .

To invert this process, the size of each preceding sector is subtracted from the index until the unique containing sector is identified. At each position, the same procedure is applied to continuation sizes. These sets form a disjoint and exhaustive partition of the starting sector. At length zero, the only walk of a nonempty sector is its vertex. Assuming recovery of continuations of length $m-1$, the index intervals of each first successor uniquely identify that successor and the remaining index of its continuation. Induction on m proves that the procedures are inverse, both on walks and on the indices $0, \dots, N_k - 1$.

Lexicographic order serializes the existing path; the operations generating the path precede this serialization. Since $TA_v = \text{id}$, the two inverse identities give

$$\hat{T}\mathcal{E}_v = \mathcal{C}T\mathcal{D}\mathcal{C}A_v\mathcal{D} = \mathcal{C}TA_v\mathcal{D} = \text{id}.$$

Finally, the exponent definition gives

$$k(A_v\gamma) = (n+1) + x_0 - v, \quad k(\gamma) = n + x_0 - x_n,$$

whose difference is the second identity. □

Distinguishing capacity with a known exponent. With k known, an injective fixed-length binary representation of length ℓ requires $2^\ell \geq N_k$, by the pigeonhole principle. Consequently, $\ell \geq \lceil \log_2 N_k \rceil$. The preceding index a attains this length through its binary representation, padded with leading zeros where necessary. This minimum concerns transmitting the index with k known; transmitting k and delimiting variable-length codes have their own costs. Moreover,

$$\lim_{k \rightarrow \infty} \frac{\log_2 N_k}{k} = \log_2 \varphi.$$

The equality measures the combinatorial capacity to distinguish paths in the modal envelope.

Fiber information under an explicit distribution. For fixed k , let R_k be a walk chosen uniformly from $\mathcal{F}_k$. We use natural logarithms and $H(R_k) = -\sum_{\gamma \in \mathcal{F}_k} p_\gamma \log p_\gamma$, with $p_\gamma = 1/N_k$. All these walks have character φ^k , so

$$H(R_k \mid \chi(R_k)) = \log N_k, \quad H(R_k \mid \mathcal{C}(R_k)) = 0.$$

The first equality follows because the reading is constant on the fiber; the second follows from the unique decoding already proved. In particular, $\log N_k/k \rightarrow \log \varphi$.

For $k = 4$, the character leaves an uncertainty of $\log 14$ nats. The three classes $B = b(R_4)$ have probabilities $2/14$, $8/14$, and $4/14$. Conditional on each class, the distribution remains uniform on its 2, 8, or 4 elements, respectively. By the definition of conditional entropy,

$$H(R_4 \mid B) = \frac{2}{14} \log 2 + \frac{8}{14} \log 8 + \frac{4}{14} \log 4.$$

The index within the class, together with B , uniquely recovers the walk. These values pertain to the declared uniform distribution on the modal fiber. The measure on the complete space of histories, the temporal realization of k , and any thermal reading retain their respective domains and transducers.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

give

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (6.23)$$

The second expression enters the exponent of the electronic construction articulated here. The first is also obtained as the limit of the marked reading $\pi F_{n-1}/F_{n+1}$. Interchanging the arguments relates the two, but does not turn a reparametrization of the electronic equation into a second independent derivation of all its coefficients.

6.7 Exponential realizations of the generated coordinates

The regional coordinates already determined make it possible to recognize the concrete realizations of the quadratic TRIT collection. The order is substantive: the construction of the regions precedes the evaluation of an exponential at the parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not enter into the selection of the prefixes that produce them.

6.7.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, formed by the integral combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. In the residual evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. The quotients of the integral lift are preserved in the corresponding arithmetic record.

The elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (6.13) yields

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the plane of nilpotent transport, and the plane of modal incidence. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 6.11 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so that $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 6.12 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\boxed{\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.} \quad (6.24)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotence eliminates exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Therefore, the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the full turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not the absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use distinct parameters from a common exponential collection.

In that collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , obtained previously from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

6.7.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

The use of Q keeps this operator square separate from the dodecaphase register K .

Proposition 6.13 (Polynomial regime projectors). *The mappings*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take the values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross-products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

Thus one obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \tag{6.25}$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation brings together the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation makes it possible to recognize their types without replacing that dynamics with an identification of the fibers.

6.8 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integral prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, beginning with $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; therefore, their initial values are $m = 3, 2, 1$. The subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \tag{6.26}$$

The half-open convention is used to select a representative when a value lies on a boundary. For passage to the limit, the closures of these intervals are used, and the two equivalent boundary expansions are identified. This specification prevents one from assuming that every intersection of nested half-open intervals is automatically nonempty.

Theorem 6.14 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (6.26). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending by one block preserves the preceding prefix under Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints are Cauchy and have a unique limit. For a fixed decimal length, a value separated from the partition boundaries has positive distance from them. A sufficiently narrow bracket is contained in a unique cell and determines its prefix. The uniqueness of the cell and the nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; that decision does not follow from numerical width alone. $\square$

Corollary 6.15 (Finite determination and compatibility of the regional prefixes). *For each of the realizations π, φ, e and for every finite decimal length, the constructed brackets determine the corresponding prefix by a finite computation. The prefixes obtained at distinct depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ guarantees their separation from the rational boundaries of every finite decimal partition. The preceding theorem applies to each character. $\square$

Arbitrary depth expresses the quantifier of the corollary: for every finite horizon there exists a computation that determines its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

6.9 Mirror involution and synchronization of the regions

The nonadic connection jointly transports the three characters. In the chart of the ninth phase, the self-scaling axis and the ternary sum of closure and propagation remain fixed under the mirror interchange. Helical inversion reverses the direction of transport; the mirror of the two channels changes their relative orientation. They are different involutions, although both act on the same system of paths.

The census coincidences illustrate that coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert an equality among π, e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a shift of -16 per channel. The closure that will lead to α must preserve both pieces of information: relative coincidence and the evolution of memory.

6.10 Sturmian helical suspension

The relation between bases nine and ten of the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (6.27)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic reading of the refinement has an exact consequence.

Lemma 6.16 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the first member contains the prime three, while that of the second contains only the primes two and five. The equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and the sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, the separations between events are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first and is not a second free parameter.

6.10.1 Capacity index, vacancies, and induction of the tritic regime

The sequence of vacancies is linked to refinement through an increment identity. To establish it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of the refinement under the normalization that compares a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase register to be constructed subsequently.

Proposition 6.17 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (6.28)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. The irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting two consecutive terms yields (6.28). The sum over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity is retained while a transition is incorporated into the history. Here a vacancy is a capacity-retention event with a determined position and antecedent. The preceding balance preserves exactly the length between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = \lfloor \mathcal{C}_t \rfloor_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and the chronological phase $[t]_9$ are two different observations. A local capacity retention and a full turn of the holonomy have distinct mechanisms, although both allow the phase observation to be preserved while modifying the state.

The relation with TRIT can be followed at the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = c_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j \delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and therefore

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (6.29)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one unit in the negative direction. The canonical selector is applied to the positive phase $r_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(r_j).$$

A tritic regime is thereby determined by the capacity index retained at each event, in addition to the binary coding of the vacancy.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of the plateaus of the class $[v_j]_3$, once the origin and endpoint convention have been fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the initial run of 6 is the initial restriction of a plateau. Their signatures run through 0, -1 , $+1$. Equations (6.28) and (6.29) make explicit the link among refinement, vacancies, and regime selection. The quadratic TRIT reading then realizes those types through their respective generators; the temporal event and its geometric representation thus preserve an identifiable dependency.

Inverting the monotone index yields another useful observation:

$$\mathcal{T}(k) = \min\{t : c_t = k\} = \left\lfloor \frac{k}{\rho} \right\rfloor - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks required to gain m units of capacity is

$$\mathcal{T}(k + m) - \mathcal{T}(k) = m + \lfloor (k + m)\sigma \rfloor - \lfloor k\sigma \rfloor \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas the return time measured by the other index admits two values. This distinction is part of the refinement dynamics and is not an a posteriori correction of its numerical values.

Theorem 6.18 (Complexity of the phase readings). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. When the phase $n \bmod 9$ is preserved, the complexity is $p_9(m) = 9(m + 1)$; when only its ternary class is preserved, it is $p_3(m) = 3(m + 1)$. All three topological entropies are zero.*

Proof. For factors of length m , the preimages of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: the coincidence between shifted endpoints eliminates duplications, and irrationality prevents additional coincidences. Those points determine $m + 1$ distinct intervals of constant itinerary. Every irrational orbit visits each interval, whence $p(m) = m + 1$.

Once a phase modulo nine is fixed, the positions of that phase traverse a rotation of step 9δ , which is also irrational. By density, all factors appear in every phase, and the first coordinate distinguishes their nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has frequency characters

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not log 10: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor log 10 belongs to the normalization of (6.27).

An operator presentation makes memory explicit. In $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n+1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finite-support vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (6.30)$$

The first two relations are Weyl covariances; the third expresses that one nonadic turn increases the memory degree. The base rotation is Abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward and backward advance. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

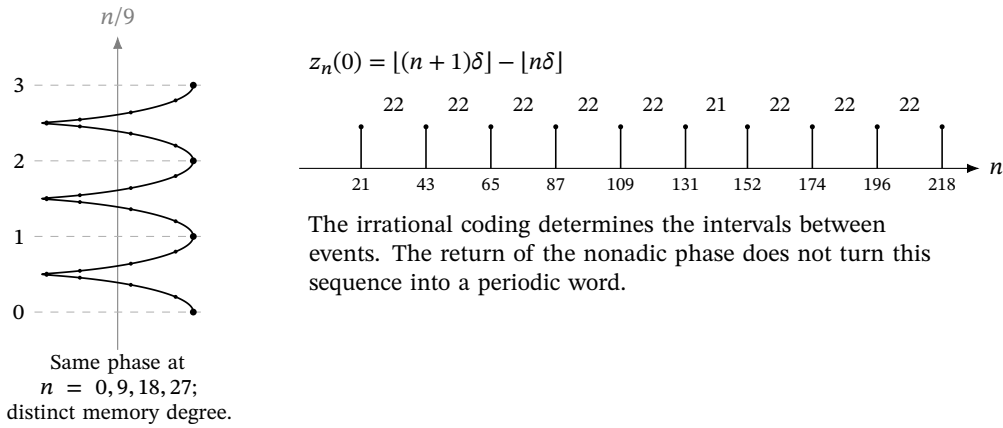


Figure 5: Two complementary representations of transport. On the left, the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$ is projected: after nine steps, the phase returns while the height increases. On the right, the first ten events of the lower word for $\delta = \log_{10}(10/9)$ are shown, with exact spacings of 21 or 22. The helical representation introduces no additional physical metric.

For this count, fix the intercept and the cumulative endpoints

$$\theta = 0, \quad (N_0, \dots, N_4) = (0, 729, 972, 999, 1000),$$

corresponding to the cubic order $(729, 243, 27, 1)$. The lower and upper counts are given by

$$c_j^- := [N_j \delta] - [N_{j-1} \delta], \quad c_j^+ := [N_j \delta] - [N_{j-1} \delta],$$

and, with the same initial zero boundary, satisfy

$$(c_1^-, \dots, c_4^-) = (33, 11, 1, 0), \quad (c_1^+, \dots, c_4^+) = (34, 11, 1, 0).$$

These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a single count prior to any unit chart; identifying them with physical exponents also requires the corresponding dimensional derivation.

The designation *aperiodic crystal* is used here for this symbolic order, with specified complexity, spectrum, and return. Zero topological entropy does not by itself amount to zero physical dissipation: a thermodynamic realization requires a defined energetic dynamics and erasure operation. Likewise, this Sturmian reading does not replace all TPK operators, nor does it by itself prove the periodicity or aperiodicity of the complete output of α . Its function is to make explicit an order structure that the calendar's simple return does not describe.

6.11 Nonadic connection, holonomy, and monodromic representation

The regional connection already constructed is now studied through three objects: transport of the complete state, representation of its turns, and suspension of its itineraries. The APP-TRIT-TPK operations and the initial catalog preserve the preceding definitions. The necessary distinction here concerns what each representation retains from the same history, with all required operations incorporated into this section.

At prolongation index k , a chart of the state preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k assembles the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts the complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphase position, while the integer m_k distinguishes its successive turns.

Definition 6.19 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible extension relations of the TPK. The realization of the holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves the admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may continue to admit more than one continuation. This difference is precisely the function of memory and of the boundary conditions in determining the path.

Proposition 6.20 (Iteration of the return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves the phase and increments the counter by one. Its application to the already transported state repeats exactly those two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integral witness to the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (4.6) expresses the composition of the calendar through its cocycle; the path also preserves the integral lift of that datum. Preservation here means transport of the preceding contributions, not immobility of all coordinates.

6.11.1 Bilateral representation of the return index

The shift T of (6.30) admits an expression in phase and turn coordinates. Number the phases from 0 to 8, by shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unit shift

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passage through the phase origin therefore incorporates the exact quotient unit.

Proposition 6.21 (Representation of the cycle monodromy). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m + 1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the group of turns of the cycle.

Proof. Nine shifts preserve the remainder upon division by 9 and increment the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m + s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose group of turns is $\mathbb{Z}$, and represents the phase together with its counter. The TPK state also contains the path coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a path begun at an initial condition uses its admissible half-axis. This provides an operator inverse for the index without turning each extension relation into a bijection of the complete state.

With $D_M|k\rangle = \lfloor k/9 \rfloor |k\rangle$, the identity $[D_M, S] = S$ expresses the increment of memory on the dense domain of finite-support vectors. If $\mathcal{I}|k\rangle = | -k\rangle$ and P_0 projects onto indices divisible by 9, one also obtains

$$\mathcal{I}T\mathcal{I} = T^{-1}, \quad \mathcal{I}D_M\mathcal{I} = -D_M - (I - P_0). \quad (6.31)$$

Indeed, for $k = 9m + r$, $\lfloor -k/9 \rfloor = -m$ if $r = 0$, and equals $-m - 1$ if $r > 0$. The inversion thus preserves the boundary term of Euclidean division.

6.12 Block clock and capacity clock

The index of a refinement mapping and the number of available decimal positions obey related but different laws. To express them, reserve n for the index of ternary blocks and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n + 1)\rho \rfloor, \quad n \geq 0.$$

Here the letter $\mathcal{C}_n$ designates capacity, not the dodecaphase record K . At the canonical phase origin, the preceding mechanical word gives

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n + 1)\delta \rfloor.$$

Proposition 6.22 (Exact balance between capacity and vacancies). *For $n \geq 0$, and in the second identity $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. The irrationality of δ implies $\lfloor (n + 1)\delta \rfloor = \lfloor (n + 1)\delta \rfloor + 1$. Then

$$\lfloor (n + 1)(1 - \delta) \rfloor = n + 1 - \lfloor (n + 1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities completes the proof. $\square$

An event $z_n = 1$ adds a block to the history while retaining capacity $\mathcal{C}_n$; an event $z_n = 0$ increments both counters. On any interval, the capacity increments and the vacancies sum to exactly its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and the capacity phase are related through the integrated memory of vacancies. In particular, a capacity retention and a complete turn of the connection have distinct mechanisms, although both may preserve a phase observation.

The monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time required to gain b units of capacity is

$$T_{\text{cap}}(a + b) - T_{\text{cap}}(a) = b + [(a + b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. Thus 9 or 10 blocks are obtained to gain 9 units of capacity, and 113 or 114 to gain 108. These are times of the inverse clock, measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

6.13 Symbolic suspension and aperiodic time crystal

The rotation coding brings together phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Preserving a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}}z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 6.23 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and the representation of the turn counter, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ : the uniform discrepancy bound below one, obtained by telescoping on any window, persists upon taking the closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension likewise has no closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. The itineraries of length m arise from a partition of the circle by $m + 1$ points; every interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases and why the already established complexities are

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

The recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

The realization through the observables V_δ, W_δ and the shift T expresses the same order through the covariances (6.30). The frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[\mathcal{C}_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when interpreting the returns. The term *time crystal* is thereby defined through a mathematical dynamical system. Its physical realization requires specification of an energetic dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

6.14 Mirror symmetry of the regions and double realization

The mirror interchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and its aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixing refers to the involution in one fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the distribution that the aggregate preserves without distinguishing. The inversion $\mathcal{I}$ of the temporal index and the channel involution $\mathfrak{c}$ therefore act on different data.

The phase return has an explicit witness:

$$\begin{aligned} B_1 &= (012222 \mid 121221 \mid 112202), \\ B_{10} &= (101222 \mid 112221 \mid 112002). \end{aligned}$$

The two positions have phase 1; the block, the axis, and the memory have changed. Likewise, synchronization of the censuses in phases 8 and 9 annihilates their relative differences, while the diagonal component passes from $(59, 59, 59)$ to $(43, 43, 43)$. Equality of a relative observable does not exhaust the state that determines it.

The double realization preserves that architecture. In Section 14.1, the same prefix determines nested cylinders on the one hand and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two mappings. The subsequent incidence path constructs codes, lattices, and the Moonshine realization in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

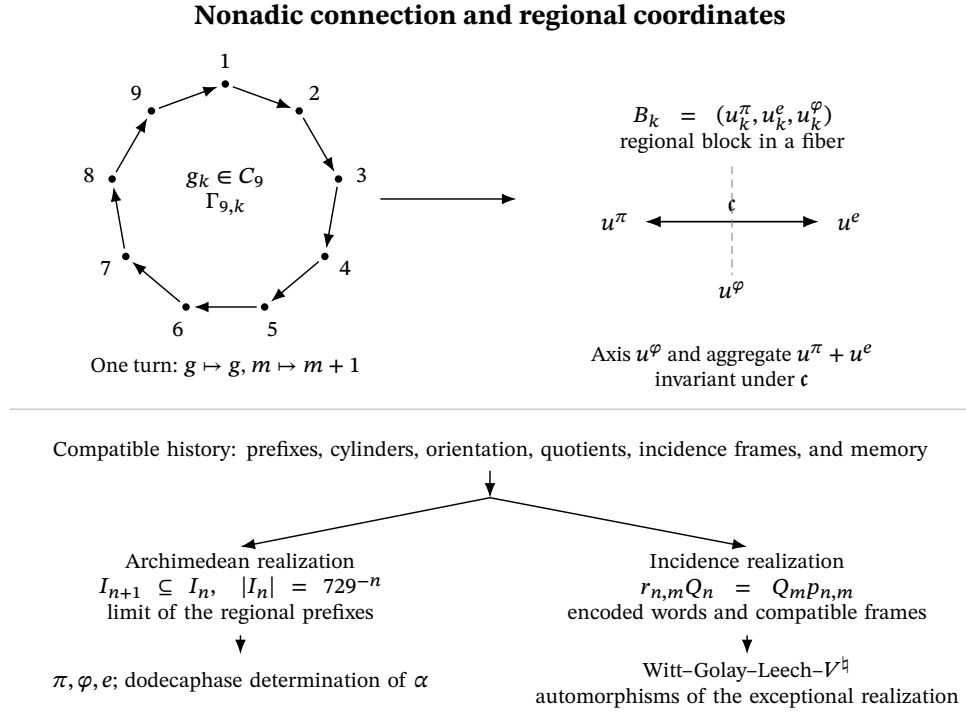


Figure 6: Nine phases of a connection and two realizations of the compatible history. The cycle represents the phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not impose that the axis remain constant during transport. The two lower mappings use the same prefix: one determines its numerical limit, and the other preserves words and incidence frames. The maps of this second realization are specified in Section 14.1; the exceptional path is developed subsequently.

6.14.1 Linear representation of the phases and the return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. The horizontal position indicates the order of the charts; the upper labels specify regional operations. Figure 7 thus brings together the initial real line, the fixing of the self-scaling axis, the mirror distribution of the two channels, and the return with memory developed in the preceding construction.

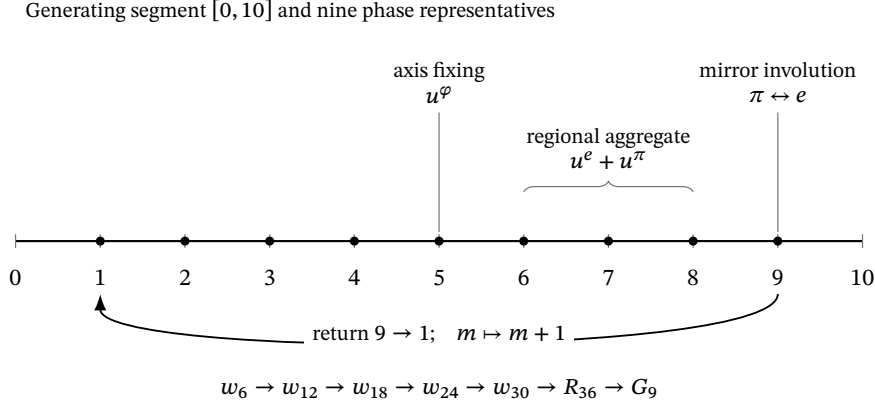


Figure 7: Linear representation of the nonadic connection. Mark 5 locates the first strong saturation and the regional self-scaling axis; the brace 6–8 identifies the ternary aggregate fixed in each fiber; phase 9 selects the closure orientation. The return preserves the phase and increments the memory. The abscissae number operative positions, while the labels u^π, u^e, u^φ designate blocks of $\mathbb{F}_3^6$.

The fixing of the axis is formulated in the fiber of a given boundary. If $q = u^\pi + u^e + u^\varphi$ and $a = u^\pi + u^e - u^\varphi$, then

$$u^\varphi = 2(q - a), \quad s = u^\pi + u^e = q - u^\varphi \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and the aggregate; the ordered distribution of s preserves the mirror freedom resolved by the prolongation. The fifth phase has a local solution and an extension, with boundary datum $c = 11111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ placed above mark 5 identifies the regional axis that enters that operation.

The intermediate phases preserve this decomposition within each fiber, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness and three prolongations; phase 7 resolves a triple mirror fiber; phase 8 synchronizes the three regional censuses. Phase 9 preserves two local distributions with the same axis and the same aggregate, one of which is prolonged. These counts express different functions within a single connection and justify the marks on the real line.

In particular, for phases 6, 7, 8, the regional sums are respectively 012100, 101001, and 112000. The preservation represented by the brace is an invariance with respect to the mirror distribution in each fiber. The subsequent real evaluation applies the readers of the compatible cylinders to the complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \rightarrow 1$ represents passage from one turn to the next. The formulas in Section 6.11 preserve the phase remainder and increment the turn counter. The linear diagram and the cyclic diagram in Figure 6 therefore express two complementary aspects of the same holonomy: the ordering of positions and the composition of the return on the state with memory.

6.15 Temporal transport and tritic exponential composition

The Euler relations in Section 6.7 describe the local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is the unit. In the elliptic type it is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear shift; the hyperbolic one separates expansion and contraction with unit determinant. At the regional parameters already generated,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unit of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transport of a local realization by an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types instead belongs to the selection of TPK sheets and regimes; it is not a conjugation that changes the value of J^2 . The temporal product therefore preserves the order of the transitions and the information at their endpoints.

This articulation distinguishes the closure of a local representation from the holonomy of a history. The equality $\exp(2\pi J_e) = I$ may hold at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the orientation of the regime; the second records another turn. The discrete structure preserves both: the local return and the accumulated provenance of the state that realizes it.

6.16 Multiplicative irreducibility and return periods

The distinction among structure, evaluation, and phase also appears in the multiplicative sector. The local APP product contains both the remainder and the quotient; its iteration makes it possible to construct normal forms and factorizations before examining their decimal periods. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, not to an isolated coincidence of periods.

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

The sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n-1$ by 9 uniquely determines the first digit and the continuation. This is the bijective nonadic numeration; it preserves the representative 9 of the boundary.

The sum $\boxplus_\#$ is obtained by applying the additive sheet of APP to the digits at each position and propagating their quotients. If there is only one digit, it constitutes the provisional remainder; a nonzero incoming quotient is normalized by a second additive operation. The sum of the quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated through

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

One begins with $c_0 = 0$, and the final quotient is incorporated when it is nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. The local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the full product is constructed by the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u), \quad u \boxtimes_{\#} v = a_0.$$

The rule composes the two local sheets. Its evaluation verifies

$$\text{ev}_9(u \boxtimes_{\#} v) = \text{ev}_9(u) \text{ev}_9(v). \quad (6.32)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the corresponding power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (6.32). The uniqueness of the normal form transports associativity and commutativity to the constructed product; its unit is (1).

Definition 6.24 (Reduced multiplicative coproduct). On the finite-support vector space generated by the nonempty normal forms, define

$$\tilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A nonunit normal form is irreducible when its image under $\tilde{\Delta}_{\#}$ is zero.

The factorization fibers are finite by (6.32). In the completion $\mathcal{H}_{\#} = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, the images of distinct forms have disjoint supports. If $d_{\#}(w)$ counts the ordered nonunit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_{\#}}) = \left\{ x \in \mathcal{H}_{\#} : \sum_w d_{\#}(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is that weighted sum. Therefore, the operator

$$A_{\#} = (\overline{\tilde{\Delta}_{\#}})^* \overline{\tilde{\Delta}_{\#}}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered nonunit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through the multiplicative evaluation, these correspond to the ordinary primes. The selection depends on the factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the permutation of remainders $r \mapsto 10r \pmod{n}$. The return of remainder 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

The composite return synchronizes the returns of the prime powers. For a prime $p \notin \{2, 3, 5\}$, one has

$$\tau_p \mid p - 1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses the nonadic closure of the decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines the increments of the capacity index; factorization determines the multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, whereas the quotient and factorization preserve information that this closure does not distinguish. The link with the spectral constructions of the treatise resides in those operators and their intertwining maps. The proof of its subsequent analytic results retains its own development; it is not replaced by the periodicity of a residual observation.

6.16.1 Vacancy discrepancy and Dirichlet readings

The link also admits an exact analytic expression. The same coding $z_n = z_n(0)$, for $n \geq 1$, is retained, and two subsequent observables are considered: the series over all indices and the product over the indices selected by irreducibility. For $\text{Re } s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in that half-plane. Their common factor is the density δ , but their discrepancies are different.

Proposition 6.25 (Separation of the vacancy density). *One has*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\text{Re } s > 0).$$

In $\text{Re } s > 1$, with logarithms defined by the absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{[x]} x^{-s-1} dx$. The integral converges locally uniformly in $\text{Re } s > 0$, which proves holomorphy. The second identity results from separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition makes visible what the reading over primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling the discrepancy over all integers gives the first holomorphic continuation; the second observable preserves a discrepancy of its own over primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation establishes its relevance to the arithmetic core without transferring to this article the proof of the terminal results of the treatise.

7 Proof of $2^{\aleph_0} = \aleph_1$ in the HMT continuum

This section brings together, in a contiguous proof, extension, double projection, and arbitrary depth; it then constructs the semantic closure, the uniform lift, the genealogical machinery, and the cardinal closure. The set-theoretic hierarchy is identified only at the end of the construction and cannot be read as its causal origin. The causal dependence is:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_\infty^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow m_\infty \longrightarrow \mathfrak{G}_{\text{HMT}}(h, m_\infty). \quad (32)$$

First, the state, its extensions, the continuum, and the genealogical ledger are constructed. Then **that produced genealogy** is extended to all its hereditary objects by an explicit semantic rule. The subsequent identification with a known set-theoretic hierarchy selects neither the seeds, the operators, the genealogical ledger, the line, nor its projections. It does, however, permit the exact logical-cardinal strength of the adopted semantic rule to be assessed with complete transparency.

7.1 Extension of states and transfinite semantic extension

Let

$$\Sigma_{\text{HMT}} = (\dot{R}_{\text{APP}}, \dot{R}_{\text{TRIT}}, \dot{R}_{\text{TPK}}, \dot{R}_U, \dot{R}_{\Gamma_0}, \dot{R}_{\text{Ext}}, \dot{R}_\rho, \dot{R}_\pi, \dot{R}_\varphi, \dot{R}_e, \dot{R}_\alpha, \dot{R}_{\text{ar}}, \dot{R}_{\text{exc}}) \quad (33)$$

be the countable **relational** signature whose effective code is h . Each $\dot{R}_S$ is the graph of the operator S on the codes where it is defined; no external function is assumed to be total within each level. The graphs $\dot{R}_\pi, \dot{R}_\varphi, \dot{R}_e, \dot{R}_\alpha$ are the output readers constructed by coinductive generation and dodecaphase closure; they do not take the conventional values of the constants as arguments. Their presence in Σ_{HMT} ensures that the closure preserves the full state: it simultaneously integrates the routes that express value and the incidence that extends toward the exceptional realization.

At each horizon, the state code records the APP–TRIT–TPK coordinates, the route, sheet, carry, boundary, and memory; at its representation stages it also records R_N, Φ_N, E_N, A_N , where A_N is the prefix of α_{HMT} produced by the same state. These coordinates do not select the branch from outside: they are part of the history being encoded. Once a primitive-recursive numbering of the finite coordinates of a state is fixed, every inverse history $x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}}$ determines the code

$$\text{code}(x_\infty) := \{(N, j, c) : (x_N)_j \text{ has code } c\} \subseteq \omega.$$

The coherence equations $\rho_{N+1, N}(x_{N+1}) = x_N$ make it possible to decode the history uniquely from that real. We write m_∞ for the code of the realized history and $m_\infty(n)$ for its characteristic function under the fixed numbering. The operator code h and this complete ledger are combined, by means of a primitive-recursive pairing function, into the real

$$u_{h, m} := \{\langle 0, \langle n, h(n) \rangle \rangle : n < \omega\} \cup \{\langle 1, \langle n, m_\infty(n) \rangle \rangle : n < \omega\}. \quad (33a)$$

The two primitive projections recover h and m_∞ from $u_{h, m}$.

The first extension is the metric and operator extension of the finite states. For $N \geq 1$, let $\hat{\mathcal{X}}_{6N}^{\text{enr, raw}}$ be the set of raw enriched states of length $6N$, and define the surviving subsystem set-theoretically by

$$\begin{aligned} X_N &:= \hat{\mathcal{X}}_{6N}^{\text{enr, surv}} \\ &:= \left\{ x \in \hat{\mathcal{X}}_{6N}^{\text{enr, raw}} : \right. \\ &\quad \forall M > N \exists z \in \hat{\mathcal{X}}_{6M}^{\text{enr, raw}} \text{ such that} \\ &\quad \left. \text{tr}_{6M, 6N}(z) = x \wedge z \text{ satisfies every intermediate TPK transition} \right\}. \end{aligned}$$

and set

$$\rho_{N+1, N} := \text{tr}_{6N+6, 6N} \upharpoonright X_{N+1},$$

and

$$\text{Ext}_N(x) := \{y \in X_{N+1} : (x, y) \in \text{Ext}_{6N+6, 6N}\}.$$

The root level is included without any implicit exception:

$$X_0 := \{*\}, \quad \rho_{1,0} : X_1 \rightarrow X_0 \text{ is constant}, \quad \text{Ext}_0(*) := X_1.$$

The superscript *surv* denotes the pruned tree of states possessing compatible TPK descendants at every finite horizon. The TPK extension law established in the preceding sections proves that the emitted seeds belong to this tree; finite branching and König's lemma turn compatibility at every horizon into an infinite branch. In particular,

$$\forall N \geq 0 \forall x \in X_N \quad \emptyset \neq \text{Ext}_N(x) \subseteq \rho_{N+1,N}^{-1}(x). \quad (34)$$

By the same finitely branching tree lemma, every surviving seed has a compatible extension

$$x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}} := \varprojlim_N (X_N, \rho_{N+1,N}). \quad (35)$$

The second extension acts **afterward** on the complete code of that realized history. It is not inferred from the mere existence of the finite operator Ext_N : at set-theoretic scale it formalizes the HMT principle that the only objects in the entire domain are those having a genealogical derivation from the state and its rules. The transfinite semantic extension is:

$$\begin{aligned} \mathfrak{G}_0(h, m_\infty) &:= \text{TC}(\{\omega, u_{h,m}, h, m_\infty\}), \\ \mathfrak{G}_{\beta+1}(h, m_\infty) &:= \text{Def}_{\Sigma_{\text{HMT}}}(\mathfrak{G}_\beta(h, m_\infty)), \\ \mathfrak{G}_\lambda(h, m_\infty) &:= \bigcup_{\beta < \lambda} \mathfrak{G}_\beta(h, m_\infty) \quad (\lambda \text{ limit}), \\ \mathfrak{G}_{\text{HMT}}(h, m_\infty) &:= \bigcup_{\beta \in \text{Ord}} \mathfrak{G}_\beta(h, m_\infty). \end{aligned} \quad (36)$$

The Continuum Hypothesis and a target bijection are not introduced among the conditions defining the successor operator; the operator is given prospectively by

$$\text{Def}_{\Sigma_{\text{HMT}}}(M) := M \cup \left\{ \{x \in M : \langle M, \in, (\dot{R}_S \cap M^{k_S})_{S \in \Sigma_{\text{HMT}}} \rangle \models \varphi(x, \bar{p})\} : \ulcorner \varphi \urcorner \in \omega, \bar{p} \in M^{<\omega} \right\}. \quad (37)$$

That is, it simultaneously expresses all subsets definable by a finite formula of the HMT relational language and parameters already produced by the genealogy. It consults neither a target real value, a conventional constant, a target cardinal, nor an external model. Formula (37) defines the transfinite closure over the previously constructed seeds, operators, enriched state, ledger, and double projection.

Lemma 7.1 (Joint coding of the HMT program and the realized ledger). *The pair formed by the operator code h and the ledger m_∞ is encoded by a single real $u_{h,m} \subseteq \omega$, and every formula of Σ_{HMT} is uniformly translated into a formula of the language $\{\in, u_{h,m}\}$.*

Proof. Both h and m_∞ are sequences of natural numbers: h encodes the signature, the finite tables, and the operator programs; m_∞ encodes one realized history, not the complete history space. Formula (33a) and its two decoders prove the first assertion. Each symbol in (33) denotes its **graph relation encoded in h** , never the external graph of all its possible evaluations. APP, TRIT, U_t , Γ_0 , the restrictions ρ , and the finite extensions have primitive-recursive graphs on integer codes and, therefore, Δ_0 formulas absolute between transitive levels. The two representations are expressed as graph relations through the formulas

$$\begin{aligned} \dot{R}_{\text{ar}}(x, y) &\iff \forall k \in \omega \exists N \forall n \geq N \quad |\mu_n(x) - y| < 2^{-k}, \\ \dot{R}_{\text{exc}}(x, c, z) &\iff \forall N \in \omega \quad z \upharpoonright N = E_N(x \upharpoonright N, c \upharpoonright N), \end{aligned} \quad (37c_0)$$

where μ_n and E_N are the rational/finite functions whose codes occur in h . Every atomic formula containing an HMT symbol is inductively replaced by its graph formula with parameter $u_{h,m}$. Connectives and quantifiers are preserved. This induction on syntactic complexity produces a translation $\varphi \mapsto \varphi^*$ such that, for every transitive level M of (36) containing the parameters,

$$\langle M, \in, (\dot{R}_S \cap M^{k_S})_S \rangle \models \varphi(\bar{a}) \iff M \models \varphi^*(\bar{a}; u_{h,m}). \quad (37c)$$

Thus, (37) cannot consult an oracle, a target decimal table, or an uncoded relation that introduces uncountably many objects in a single step. $\square$

HMT semantics is now defined **independently of identifying it with $\mathfrak{G}$** . Let $\mathcal{T}_0^{\text{HMT}}$ be the set of canonical names of the elements of $\mathfrak{G}_0(h, m_\infty) = \text{TC}(\{\omega, u_{h,m}, h, m_\infty\})$. In particular, the initial level contains the realized ledger m_∞ , not every history in X_∞^{enf} as primitive parameters. The admissible terms are the well-founded trees obtained by

$$\begin{aligned} \mathcal{T}_{\beta+1}^{\text{HMT}} &:= \mathcal{T}_\beta^{\text{HMT}} \cup \{\langle \beta, \ulcorner \varphi \urcorner, t_0, \dots, t_{k-1} \rangle : t_i \in \mathcal{T}_\beta^{\text{HMT}}\}, \\ \mathcal{T}_\lambda^{\text{HMT}} &:= \bigcup_{\beta < \lambda} \mathcal{T}_\beta^{\text{HMT}}, \quad \mathcal{T}^{\text{HMT}} := \bigcup_{\beta \in \text{Ord}} \mathcal{T}_\beta^{\text{HMT}}. \end{aligned} \quad (37a)$$

The value of a successor term is the subset of the preceding level defined by φ in the induced relational structure, with the values of $t_0, \dots, t_{k-1}$ as parameters. Before any set-theoretic comparison, define

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) := \{\llbracket t \rrbracket : t \in \mathcal{T}^{\text{HMT}}\}. \quad (37b)$$

This is the rule of **generative exhaustivity** adopted by HMT: every object in the entire domain must possess one of these derivation trees, and every admissible tree expresses an object. It is not a consequence of finite branching. The Continuum Hypothesis is not introduced into this rule; it is obtained through the proof developed in the article. Neither an enumeration of the reals nor a target bijection is introduced as data.

Theorem 7.2 (Equivalence between term semantics and the genealogical hierarchy). *The evaluation of the preceding terms satisfies*

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) = \mathfrak{G}_{\text{HMT}}(h, m_\infty). \quad (37d)$$

Proof. By transfinite induction. At the initial level, the two classes coincide. If every value of a term in $\mathcal{T}_\beta^{\text{HMT}}$ belongs to $\mathfrak{G}_\beta$, the semantics of a successor term is one of the subsets in (37). Conversely, every subset in (37) is given by a formula code and a finite tuple of parameters; by the induction hypothesis those parameters possess terms, and (37a) forms the term that denotes it. At a limit, both constructions take the same union. $\square$

The state encoder is compatible with truncations and sends every TPK extension to a successor term. Let $\mathcal{T}_N^{\text{st}} \subset \mathcal{T}^{\text{HMT}}$ be the subclass of terms encoding complete states of depth N , and let $\tau_{N+1,N} : \mathcal{T}_{N+1}^{\text{st}} \rightarrow \mathcal{T}_N^{\text{st}}$ delete the last block together with the corresponding truncation of the genealogical ledger, carry, and memory. If $J_N : X_N \rightarrow \mathcal{T}_N^{\text{st}}$ is the complete encoder, compatibility with Ext_N proves the well-typed identity

$$\forall x \in X_N \forall y \in \text{Ext}_N(x) \quad \tau_{N+1,N}(J_{N+1}(y)) = J_N(x). \quad (37e_0)$$

Define within the semantics

$$\text{Succ}_{N+1}^{\text{TPK}}(J_N(x)) := \{t \in \mathcal{T}_{N+1}^{\text{st}} : \exists y \in X_{N+1} (y \in \text{Ext}_N(x) \wedge t = J_{N+1}(y))\}.$$

Choose β such that $\mathfrak{G}_\beta$ contains x, X_N, X_{N+1} , the finite graphs of $\text{Ext}_N, J_N, J_{N+1}$, and their codes in h . Then the primitive-recursive formula for the graph of Ext_N gives

$$J_{N+1}[\text{Ext}_N(x)] = \text{Succ}_{N+1}^{\text{TPK}}(J_N(x)), \quad J_{N+1}[\text{Ext}_N(x)] \in \mathfrak{G}_{\beta+1}, \quad (37e)$$

and both sides are formed by the graph formula of the same TPK extension. Equations (37e₀)–(37e) are the explicit semantic link: the subclass of successor terms is exactly the image of the admissible extensions, and its truncation commutes. The complete fiber of ρ is not identified with Ext_N , nor is Def identified with TPK merely by nomenclature.

Lemma 7.3 (Transitivity, axioms, and well-ordering of the genealogical model). *$\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ is a transitive class, contains all ordinals, satisfies Zermelo–Fraenkel set theory (ZF), and possesses a definable global well-order; it therefore satisfies Choice and all axioms used in the subsequent cardinal comparison.*

Proof. Abbreviate $B = \mathfrak{G}_0$ and $u = u_{h,m}$. Induction on (36) proves simultaneously that every level is transitive and that $\mathfrak{G}_\alpha \in \mathfrak{G}_{\alpha+1}$; if all ordinals smaller than α are present, their set is definable at the corresponding level and expresses α . Hence the union contains all ordinals. Extensionality and Foundation are inherited from the ambient universe; Empty Set and Infinity belong to B . Once a level containing the parameters is chosen, the formulas defining $\{a, b\}$, $\bigcup a$, and subsets by Separation express them at a successor level.

For Collection–Replacement, let F be a functional relation definable in $\mathfrak{G}$ on the set a . The reflection scheme relative to the definable hierarchy $(\mathfrak{G}_\xi, \in, u_{h,m})_{\xi \in \text{Ord}}$, applied in the ambient metatheory to the finite list of formulas defining F , produces a level $\mathfrak{G}_\theta$ containing a and the parameters and correct for those formulas. The ranks of the witnesses in $F[a]$ are then bounded by θ ; Separation over $\mathfrak{G}_\theta$ forms the image in $\mathfrak{G}_{\theta+1}$. For Power Set, the internal set to be constructed is not presupposed. Given $a \in \mathfrak{G}_\gamma$, consider in the ambient metatheory the already existing set $\mathcal{P}^V(a)$. For every $b \in \mathcal{P}^V(a) \cap \mathfrak{G}$, let $r(b)$ be its first genealogical stage. Since $\mathfrak{G}$ is a definable class, Separation in the ambient metatheory first forms the set $\mathcal{P}^V(a) \cap \mathfrak{G}$. Replacement **in the ambient metatheory**, applied to that set, bounds the ordinals $r(b)$ by some θ . Consequently,

$$\mathcal{P}^\mathfrak{G}(a) = \{b \in \mathfrak{G}_\theta : b \subseteq a\},$$

and the right-hand side is expressed by Separation in $\mathfrak{G}_{\theta+1}$. It does not introduce into $\mathfrak{G}$ the subsets external to the semantics (37b). This is the noncircular rank-bounding argument.

Finally, each object is assigned its first level of birth and its least **derivation name**: a well-founded, finitely branching derivation tree (and, by (37a), finite for each individual name), labeled by a level ordinal, a formula code, and the tuple of names of its parameters. These names are ordered by transfinite recursion: first by the ordinal label, then by the formula code, and then lexicographically by the already ordered parameter names. The least name of each value defines a class relation $<_\mathfrak{G}$ that compares all objects. It is a definable global well-order and, by taking the least element of every nonempty set, proves Choice. $\square$

Reflection by witness closure and construction of the image. The Collection–Replacement step of Lemma 7.3 admits the following explicit proof. We retain the hierarchy already generated from the operator code and the realized record, and abbreviate $\mathfrak{G} = \mathfrak{G}_{\text{HMT}}(h, m_\infty)$. The argument uses Separation and Replacement in the ambient metatheory, without presupposing these schemes in $\mathfrak{G}$.

Let Φ be a finite collection of formulas. First translate their universal quantifiers using negation and existential quantification, and close the resulting collection under subformulas. For every initial level there is a later limit ordinal θ such that

$$\mathfrak{G}_\theta \models \varphi(\bar{a}) \iff \mathfrak{G} \models \varphi(\bar{a}) \quad (\varphi \in \Phi, \bar{a} \in \mathfrak{G}_\theta^{<\omega}).$$

For each fixed formula, satisfaction in the class is interpreted by relativizing its quantifiers to the definable class $\mathfrak{G}$; no uniform truth predicate is introduced.

Proof. Fix α . For each existential subformula $\exists y \psi(y, \bar{x})$ of Φ , the tuples $\bar{a} \in \mathfrak{G}_\alpha^{<\omega}$ of the corresponding arity that have a witness in $\mathfrak{G}$ form a set by Separation in the metatheory. Assign to each tuple the least ordinal β for which some $y \in \mathfrak{G}_\beta$ satisfies $\mathfrak{G} \models \psi(y, \bar{a})$. This ordinal exists because $\mathfrak{G}$ is the union of its levels. Replacement in the metatheory bounds these first stages of witnesses. Since Φ is finite, a single bound $f(\alpha) > \alpha$, valid for all its existential subformulas, can be chosen definably. Thus every tuple under consideration has *at least one witness* in $\mathfrak{G}_{f(\alpha)}$.

Take α_0 large enough to contain the initial parameters and set

$$\alpha_{n+1} = f(\alpha_n), \quad \theta = \sup_{n < \omega} \alpha_n.$$

By continuity of the hierarchy, every finite tuple from $\mathfrak{G}_\theta$ belongs to some $\mathfrak{G}_{\alpha_n}$. If an existential formula of Φ with that tuple is true in $\mathfrak{G}$, it has a witness in $\mathfrak{G}_{\alpha_{n+1}} \subseteq \mathfrak{G}_\theta$. Induction on formulas now proves the equivalence: atomic formulas are preserved in the induced structure; connectives preserve the equivalence; the existential step uses that witness in one direction and the same witness, viewed in $\mathfrak{G}$, in the other. The prior translation of universal quantifiers also includes the existential formulas expressing their possible counterexamples. This proves the required finite reflection. $\square$

Now let $F(x, y, \bar{p})$ be functional on a in $\mathfrak{G}$. Apply the construction to the formulas expressing F , the existence of its witnesses, and its image, with $a, \bar{p} \in \mathfrak{G}_\theta$. Transitivity gives $a \subseteq \mathfrak{G}_\theta$. Every $x \in a$ has its value within that level, and reflection identifies the exact image as

$$b = \{y \in \mathfrak{G}_\theta : \mathfrak{G}_\theta \models \exists x \in a F(x, y, \bar{p})\}.$$

This set is definable over $\mathfrak{G}_\theta$, with parameters from the same level. By the successor operation of the hierarchy, $b \in \text{Def}(\mathfrak{G}_\theta) \subseteq \mathfrak{G}_{\theta+1}$. This establishes Replacement in $\mathfrak{G}$. If the existence of witnesses is retained and uniqueness is omitted, the same construction provides a set of witnesses for Collection.

Arity of names and internal selection. In the well-order of Lemma 7.3, the enumeration of the base assigns to each element its least natural-number code. At each successor stage, formula codes are ordered first; once a code is fixed, the arity of its parameter tuple is fixed. Lexicographic order is then applied to a *finite product of fixed arity* of the previously constructed well-orders. The least name represents each new value, and limit levels retain the order of first appearance. This specifies the ordering of tuples without combining different arities into a single lexicographic order. Selecting the least element of every nonempty set, together with Replacement already established, forms internally the choice functions used subsequently. This construction of names precedes the cardinal injections and receives no bijection of the continuum as data.

7.2 TPK lift of the ternary fiber and complete cylinder partition

The fiber of $729 = 3^6$ refinement subcylinders does not arise from the transfer quotient $\mathcal{K}_{\text{ph}}$. That operator is a subsequent representation of reduced memory and cannot generate histories. The covering is constructed directly in the system $(H_k, \rho_k, \text{Ext}_k)$ through the three one-turn extensions of the nonadic holonomy.

7.2.1 Hensel inversion in the arithmetic tower and realization by histories

Before lifting the branches to enriched histories, it is useful to recover the finite argument that determines the capacity of the arithmetic tower. The complete proof of the matrix's unfiltered transducer is included; its domain is distinguished from the surviving tree.

Proposition 7.4 (Conjugacy of the unfiltered Hensel tower). *Let p be prime, $d \geq 1$, and $A \in \text{GL}_d(\mathbb{Z})$. For row vectors, taking residue representatives in $\{0, \dots, p-1\}^d$, define*

$$x_t A = u_t + p x_{t+1}, \quad u_t = x_t A \bmod p.$$

The map

$$\Psi_T : (\mathbb{Z}/p^T\mathbb{Z})^d \longrightarrow (\mathbb{F}_p^d)^T, \quad x_0 \longmapsto (u_0, \dots, u_{T-1})$$

is bijective, with inverse

$$x_0 \equiv \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} \pmod{p^T}.$$

The bijections commute with the reductions and yield a homeomorphism $\Psi : \mathbb{Z}_p^d \rightarrow (\mathbb{F}_p^d)^\mathbb{N}$, where $\mathbb{Z}_p := \varprojlim_T \mathbb{Z}/p^T\mathbb{Z}$. The update $x_t \mapsto x_{t+1}$ corresponds to deleting the first block.

Proof. The integral inverse of A exists because its determinant is ± 1 . Iterating $x_t = (u_t + p x_{t+1})A^{-1}$ gives

$$x_0 = \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} + p^T x_T A^{-T}.$$

Modulo p^T , a word determines a unique initial class. Given an arbitrary word, setting $x_T = 0$ and reconstructing backwards produces a class that emits it, proving surjectivity. The same formula proves compatibility under truncation. Passing to the inverse limit yields the homeomorphism and the relation with the tape shift. $\square$

For $p = 3$ and $d = 6$, the block alphabet has $3^6 = 729$ elements. This arithmetic capacity is not identified by definition with the realizability of every block after every surviving prefix. The link with the Ext edges is precisely the purpose of Lemma 7.6. The two operations distinguished by the matrix are thereby retained: representation in the unfiltered tower and realization under the APP-TRIT-TPK compatibilities.

7.2.2 Edge-by-edge realization of the three nonadic extensions

This subsection specifies the autonomous model generated by the TPK extension relation entering the cardinal proof. The existence of an edge is not inferred from a coincidence of types: its initial and final states, APP relation, fiber transport, and unary update of the ledger are constructed. The construction uses exclusively the two APP sheets, the TRIT selector, and the TPK composition Sel-Tra-Upd.

Complete cells and three balanced reductions. In the chart $D_9 = \{1, \dots, 9\}$, we always retain the complete APP cell

$$a_{p,q} := (p, q; R_{p,q}^\Sigma, q_{p,q}^\Sigma, R_{p,q}^\Pi, q_{p,q}^\Pi),$$

with

$$p + q = R_{p,q}^\Sigma + 9q_{p,q}^\Sigma, \quad pq = R_{p,q}^\Pi + 9q_{p,q}^\Pi.$$

The active sheet is recorded in a separate coordinate; $a_{p,q}$ is not split into a purported additive cell and a separate multiplicative cell. For $r \in D_9$, let

$$a(r) := \begin{cases} a_{1,9}, & r = 1, \\ a_{1,r-1}, & 2 \leq r \leq 9, \end{cases} \quad \tau(r) := M_{\text{ph}}(r), \quad m(r) := \tau(r) \in \{-1, 0, +1\}. \quad (7.1)$$

The additive residue of $a(r)$ supplies an APP representative of phase here; this does not make Σ the universally active sheet. The dynamic selector activates Σ for $r \in \{1, 4, 7\}$, activates Π for $r \in \{2, 5, 8\}$, and preserves both sheets without moving the cursor for $r \in \{3, 6, 9\}$. In all three cases, $R^\Sigma, q^\Sigma, R^\Pi, q^\Pi$ are transported jointly. We have

$$(m(1), \dots, m(9)) = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The three pairs ordered by the nonadic base are

$$\mathfrak{p}_{+1} := (1, 4), \quad \mathfrak{p}_{-1} := (2, 5), \quad \mathfrak{p}_0 := (3, 6). \quad (7.2)$$

These pairs are the normalized section taking the first two occurrences of each TRIT type under the shift $r \mapsto r + 3$; the third phase block 7, 8, 9 retains the turn-closure control. Both members belong to the same fiber of M_{ph} . Without introducing a target coefficient, the balanced reduction from the TRIT chapter gives

$$\begin{array}{c|c|c|c} \varepsilon & m(\mathfrak{p}_\varepsilon^{(1)}) + m(\mathfrak{p}_\varepsilon^{(2)}) & r_3(2\varepsilon) & \chi(\varepsilon, \varepsilon) \\ \hline +1 & 1 + 1 = (-1) + 3 \cdot 1 & -1 & +1 \\ 0 & 0 + 0 = 0 + 3 \cdot 0 & 0 & 0 \\ -1 & (-1) + (-1) = 1 + 3 \cdot (-1) & +1 & -1 \end{array} \quad (7.3)$$

Therefore,

$$\chi(\varepsilon, \varepsilon) = \frac{2\varepsilon - r_3(2\varepsilon)}{3} = \varepsilon. \quad (7.4)$$

The branch sign is thus the output of the balanced cocycle applied to two TRIT emissions produced by APP. Each pair supplies one of the three carry outputs. For clarity, we subsequently index the pair by $\varepsilon(\mathfrak{p}) := \chi(m(\mathfrak{p}^{(1)}), m(\mathfrak{p}^{(2)}))$; the symbol ε does not supply a free carry to Upd^{cal} .

The phase architecture, balanced lift, and memory transport are realized here through twenty-seven local incidences and their extension graphs. The normalized section fixes representatives of the fibers of M_{ph} ; the covering to be proved uses the existence of these representatives and remains invariant when they are replaced by another compatible section.

Typed data of the nine edges. Let $K_{\text{Carr}} := \langle \kappa_{\text{Carr}} \rangle_{\text{Ab}}$ be the carry module. The actions are typed by

$$H_{\text{Carr}} : \mathcal{P}_{\text{TPK}}^{\text{op}} \longrightarrow \text{End}(K_{\text{Carr}}), \quad Y_{\text{Carr}} : \mathcal{P}_{\text{TPK}} \longrightarrow \text{End}(K_{\text{Carr}}).$$

In this generated model, take the trivial actions $H_{\text{Carr}}(e) = Y_{\text{Carr}}(e) = \text{id}_{K_{\text{Carr}}}$: the active sheet and the route are retained as coordinates of the ledger itself and do not act on the free carry summand. Also let (M_d, J_d, d) be the graded direct system of memory. To avoid confusing two representatives identified by the colimit, the state also retains the degree as a coordinate, and the labeled carrier

$$\tilde{M} := \bigsqcup_{d \geq 1} (\{d\} \times M_d), \quad \text{gr}_M(d, \mu) = d.$$

The representation in the limit memory is given by

$$\begin{aligned} M &:= \varinjlim_d M_d, & J_d &: M_d \longrightarrow M, \\ u &:= J_d(u_d), & \iota_M &: \mathbb{Z} \longrightarrow M, & n &\longmapsto nu. \end{aligned}$$

Here J_d is the canonical colimit map, and the u_d form a compatible collection. Its free component permits the reader $\deg_u(nu) = n$. The configuration, phase, memory, carry, route, and survival projections will be typed on the calibrated carrier constructed recursively below; no ambient space will be imported for them. For every constructed edge e , the calibrated component $\text{Upd}_e^{\text{cal}} := \text{Mem}_e^{\text{cal}}$ will denote the third map of the factorization; its law is fixed field by field below and is not presupposed as the restriction of an ambient relation.

Fix the initial depth $k_0 = 1$, a complete configuration $q_\star \in \mathcal{Q}_{\text{obs}}$ of phase one, and

$$q_{n,r} := F_{\text{obs}}^{9n+r-1}(q_\star), \quad q_{n,10} := q_{n+1,1}.$$

Because the observable calendar is periodic, its value $q_{n,r}$ does not by itself distinguish occurrences separated by twelve turns. The calibrated model therefore uses the labeled lift

$$\tilde{q}_{n,r} := (q_{n,r}, n, r), \quad \tilde{q}_{n,10} := \tilde{q}_{n+1,1}, \quad \pi_{\text{obs}}(\tilde{q}_{n,r}) = q_{n,r}.$$

The lifted dynamics $\tilde{F}_{\text{obs}}(\tilde{q}_{n,r}) = \tilde{q}_{n,r+1}$ projects to F_{obs} ; the label does not alter the physical phase, but makes its levels of occurrence distinct. The copy of the cell at each level is labeled by

$$a_{n,r} := (n, r; a(r)), \quad (n, r)^+ := \begin{cases} (n, r+1), & 1 \leq r < 9, \\ (n+1, 1), & r = 9. \end{cases}$$

The label only prevents two occurrences of the same cell from being identified; the two Euclidean divisions and their four coordinates are those of $a(r)$. For turn $n \geq 0$ and phase $r \in D_9$, the symbol

$$e_{\varepsilon,n,r} : \tilde{q}_{n,r} \longrightarrow \tilde{q}_{n,r+1}, \quad r^+ := 1 + (r \bmod 9),$$

occupies level $d_{n,r} := k_n + r - 1$, where $k_n := k_0 + 9n$. Its local ledger contains the complete cell $a(r)$, the active sheet determined by $M_{\text{ph}}(r)$, the type $\tau(r)$, the membership mark in $\mathfrak{p}_\varepsilon$, and, upon reaching the second member of that pair, the certified identity (7.3). We define its operators solely from that ledger:

$$C_M(e_{\varepsilon,n,r}) = J_{d_{n,r}+1, d_{n,r}}, \quad \kappa_M(e_{\varepsilon,n,r}) = \mathbf{1}_{\{r=9\}} u_{d_{n,r}+1}, \quad (7.5)$$

$$\text{Carr}(e_{\varepsilon,n,r}) = d_{\varepsilon,r} \kappa_{\text{Carr}}, \quad d_{\varepsilon,r} := \mathbf{1}_{\{r=\mathfrak{p}_\varepsilon^{(2)}\}} \chi\left(m(\mathfrak{p}_\varepsilon^{(1)}), m(\mathfrak{p}_\varepsilon^{(2)})\right). \quad (7.6)$$

For identities and every admissible composition, also define

$$C_M(\text{id}_{\tilde{q}}; d) := \text{id}_{M_d}, \quad \kappa_M(\text{id}_{\tilde{q}}; d) := 0, \quad (7.7)$$

$$C_M(e_t \cdots e_1) := C_M(e_t) \circ \cdots \circ C_M(e_1), \quad \kappa_M(e_2 e_1) := C_M(e_2)(\kappa_M(e_1)) + \kappa_M(e_2). \quad (7.8)$$

The identity carries degree d because the same observable configuration may recur at different depths. Iterated, the final equality defines $\kappa_M(e_t \cdots e_1)$ for every length and is the cocycle law of the direct system; the inclusions satisfy $J_{d+2, d+1} J_{d+1, d} = J_{d+2, d}$. In addition, set $s(e_{\varepsilon,n,r}) = \top$. The following twenty-seven entries certify the coordinates distinguishing the three extensions; the universal transport fields are specified immediately afterward. The symbols *I* and *II* designate, respectively, the first retained emission and the second emission performing the reduction; a dash denotes a zero carry increment, not the absence of an edge.

r	$M_{\text{ph}}(r)$	$\mathfrak{p}_{+1}$ mark	$d_{+1,r}$	$\mathfrak{p}_0$ mark	$d_{0,r}$	$\mathfrak{p}_{-1}$ mark	$d_{-1,r}$	$\Delta \text{ mem}$
1	+1	I	0	–	0	–	0	0
2	–1	–	0	–	0	I	0	0
3	0	–	0	I	0	–	0	0
4	+1	II	+1	–	0	–	0	0
5	–1	–	0	–	0	II	–1	0
6	0	–	0	II	0	–	0	0
7	+1	–	0	–	0	–	0	0
8	–1	–	0	–	0	–	0	0
9	0	–	0	–	0	–	0	u

Thus, $d_{\varepsilon,r}$ is computed from two outputs of the selector and is distinct from the cell quotients q^Σ, q^Π . The ledger also preserves the marked pair and the balanced residue; in particular, the zero branch is not confused with an absence of event.

The carry composition law used by the TPK category is

$$\text{Carr}(\text{id}_{\tilde{q}}) = 0, \quad \text{Carr}(e_2 e_1) = H_{\text{Carr}}(e_1)(\text{Carr}(e_2)) + Y_{\text{Carr}}(e_2)(\text{Carr}(e_1)). \quad (7.9)$$

On identities and paths, they are extended functorially:

$$\begin{aligned} H_{\text{Carr}}(\text{id}) &= Y_{\text{Carr}}(\text{id}) = \text{id}, \\ H_{\text{Carr}}(e_t \cdots e_1) &= H_{\text{Carr}}(e_1) \circ \cdots \circ H_{\text{Carr}}(e_t), \\ Y_{\text{Carr}}(e_t \cdots e_1) &= Y_{\text{Carr}}(e_t) \circ \cdots \circ Y_{\text{Carr}}(e_1). \end{aligned}$$

In this model all these actions remain the identity, but their typing makes explicit which endomorphism acts on each summand of (7.9). On the calibrated edges, $H_{\text{Carr}} = Y_{\text{Carr}} = \text{id}$, so composition adds the increments already produced. Equation (7.9), not an identification between orientation and quotient, governs the accumulated ledger.

Independent construction of the transition tuples. A fiber state is written, in the order of its relevant fields, as

$$x = (a; q; \tau, o, h; b_{\text{ref}}, R^\Sigma, q^\Sigma, R^\Pi, q^\Pi; c; \text{Sig}; C, \partial C; p_{\text{APP}}, p_{\{-1,0,+1\}}; \lambda; \text{gr}_M; \text{mem}; s; \text{Led}).$$

The update notation $x[f \leftarrow v]$ changes only the indicated field. For $h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\}$, define

$$h_r(h) = \begin{cases} (\Sigma, \Pi), & r \in \{1, 4, 7\}, \\ (\Pi, \Sigma), & r \in \{2, 5, 8\}, \\ h, & r \in \{3, 6, 9\}. \end{cases}$$

The complete sheet-and-arithmetic coordinate is abbreviated as

$$\hat{h}(x) := (h(x), R^\Sigma(x), q^\Sigma(x), R^\Pi(x), q^\Pi(x)).$$

Let $\hat{\mathcal{H}}_{\tilde{q},r}^{\text{cal}}$ be the set of quintuples

$$(h, R_{a_{n,r}}^\Sigma, q_{a_{n,r}}^\Sigma, R_{a_{n,r}}^\Pi, q_{a_{n,r}}^\Pi), \quad h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\},$$

whose four arithmetic data are the two Euclidean divisions of the labeled APP cell $a_{n,r}$. This direct definition does not presuppose a state in a carrier not yet constructed. Let $\text{Hist}_{\tilde{q}}^{\text{cal}}$ be the set of admissible finite paths in the graph of labeled edges ending at $\tilde{q}$, including the identity, and let $B_{\tilde{q}}^{\text{cal}}$ be the set of their ordered endpoint sequences. Define

$$S_{\tilde{q}}^{\text{cal}} := \hat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times B_{\tilde{q}}^{\text{cal}} \times M.$$

The calibrated signature is not postulated: it is the typed assembly map

$$\text{Sig}_{\tilde{q}}^{\text{cal}} : \hat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times B_{\tilde{q}}^{\text{cal}} \times M \longrightarrow S_{\tilde{q}}^{\text{cal}}, \quad (\hat{h}, o, c, \lambda, \partial C, \mu) \longmapsto (\hat{h}, o, c, \lambda, \partial C, \mu).$$

Likewise, $\pi_{\mathcal{F}}$ denotes the projection that removes only the first two coordinates $(a; q)$ of the state tuple; it preserves $\tau, o, h, b_{\text{ref}}$, the four Euclidean data, carry, signature, cylinder, boundary, prefixes, route, memory degree, memory, survival, and ledger. The corresponding orientation is fixed by $o(+1) = o_{\rightarrow}$, $o(0) = o_{\circ}$, and $o(-1) = o_{\leftarrow}$. The calibrated APP fiber over $\tilde{q}_{n,r}$ contains the already defined labeled copy $a_{n,r}$. The initial state is fixed without free coordinates:

$$\begin{aligned} \hat{h}_{\star} &:= ((\Sigma, \Pi), R_{1,9}^\Sigma, q_{1,9}^\Sigma, R_{1,9}^\Pi, q_{1,9}^\Pi). \\ x_{\star} &:= (a_{0,1}; \tilde{q}_{0,1}; +1, o_{\rightarrow}, (\Sigma, \Pi); \perp; R_{1,9}^\Sigma, q_{1,9}^\Sigma, R_{1,9}^\Pi, q_{1,9}^\Pi; 0; \\ &\quad \text{Sig}_{\tilde{q}_{0,1}}^{\text{cal}}(\hat{h}_{\star}, o_{\rightarrow}, 0, \text{id}_{\tilde{q}_{0,1}}, \emptyset, J_{k_0}(0_{M_{k_0}})); \emptyset, \emptyset; (a_{0,1}), (+1); \text{id}_{\tilde{q}_{0,1}}; k_0; 0_{M_{k_0}}; \top; \emptyset). \end{aligned} \quad (7.10)$$

The tuple contains no indeterminate coordinates and satisfies $s(x_{\star}) = \top$. Let $G_0 := \{x_{\star}\}$. Assuming G_n has been constructed, fix $x \in G_n$, $\varepsilon \in \{-1, 0, +1\}$, and $x_{\varepsilon,0}(x) := x$. For $r = 1, \dots, 9$, first construct the elementary ledger

$$\begin{aligned} \ell_{\varepsilon,n,r} &:= (e_{\varepsilon,n,r}, a_{n,r}, a_{(n,r)^+}, \tilde{q}_{n,r}, \tilde{q}_{n,r+1}, M_{\text{ph}}(r), \\ &\quad h_r(h(x_{\varepsilon,r-1}(x))), o(M_{\text{ph}}(r)), \mathfrak{p}_{\varepsilon}, d_{\varepsilon,r}, \\ &\quad C_M(e_{\varepsilon,n,r}), \kappa_M(e_{\varepsilon,n,r}), \text{Carr}(e_{\varepsilon,n,r})), \end{aligned} \quad (7.11)$$

and set $\text{Led}_{\varepsilon,n,r}^{\text{new}} := \text{Led}(x_{\varepsilon,r-1}(x)) \frown \ell_{\varepsilon,n,r}$. Then construct the tuple

$$y_{\varepsilon,r}(x) := x_{\varepsilon,r-1}(x)[\tau \leftarrow M_{\text{ph}}(r), h \leftarrow h_r(h(x_{\varepsilon,r-1}(x))), o \leftarrow o(M_{\text{ph}}(r)), b_{\text{ref}} \leftarrow \mathfrak{p}_{\varepsilon}], \quad (7.12)$$

where selection leaves the cell, quotients, carry, route, memory, cylinder, boundary, survival, and ledger unchanged; the coordinate b_{ref} makes the chosen balanced reduction explicit. Membership in an ambient relation is not presupposed here.

Next, define $z_{\varepsilon,r}(x)$ field by field: its configuration is $\tilde{q}_{n,r+1}$; its complete cell is $a_{(n,r)^+}$, with the four residues and quotients recomputed by the two Euclidean divisions; it preserves the selected sheet

and orientation; and it performs the following six updates, in which only the common argument x is omitted:

$$\begin{aligned}
 C(z_{\varepsilon,r}) &:= C(y_{\varepsilon,r}) \frown e_{\varepsilon,n,r}, \\
 \partial C(z_{\varepsilon,r}) &:= \partial C(y_{\varepsilon,r}) \frown (\tilde{q}_{n,r}, \tilde{q}_{n,r+1}), \\
 p_{\text{APP}}(z_{\varepsilon,r}) &:= p_{\text{APP}}(y_{\varepsilon,r}) \frown a_{(n,r)+}, \\
 p_{\{-1,0,+1\}}(z_{\varepsilon,r}) &:= p_{\{-1,0,+1\}}(y_{\varepsilon,r}) \frown \tau(r), \\
 \text{Led}(z_{\varepsilon,r}) &:= \text{Led}_{\varepsilon,n,r}^{\text{new}}, \\
 \lambda(z_{\varepsilon,r}) &:= e_{\varepsilon,n,r} \lambda(y_{\varepsilon,r}),
 \end{aligned} \tag{7.13}$$

and still retains the fields $c, \text{gr}_M, \text{mem}, s$ of $y_{\varepsilon,r}(x)$. Its signature, by contrast, is retyped at the target configuration through the transported presignature

$$\text{Sig}(z_{\varepsilon,r}(x)) := \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\hat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(y_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)), J_{d_{n,r}}(\text{mem}_{d_{n,r}}(y_{\varepsilon,r}(x))) \right). \tag{7.14}$$

Thus, and only in Tra^{cal} , the edge, its endpoints, the mark $(p_\varepsilon, r, d_{\varepsilon,r})$, and its cocycles are added to the incidence ledger.

The initial state satisfies

$$c(x_\star) = 0 = \text{Carr}(\text{id}_{\tilde{q}_{0,1}}) = \text{Carr}(\lambda(x_\star)).$$

In the complete states $x_{\varepsilon,r-1}(x)$, and also after the selection $y_{\varepsilon,r}(x)$, the typed invariant $c(v) = \text{Carr}(\lambda(v))$ is preserved, since Sel^{cal} modifies neither field. Transport advances the route but leaves the carry pending; hence the pre-update tuple satisfies exactly

$$c(z_{\varepsilon,r}(x)) = c(y_{\varepsilon,r}(x)) = \text{Carr}(\lambda(y_{\varepsilon,r}(x))).$$

The map Upd^{cal} then restores $c(x_{\varepsilon,r}(x)) = \text{Carr}(\lambda(x_{\varepsilon,r}(x)))$ by means of the composition law, without attributing that invariant to the intermediate state z .

Finally, apply the unary function $\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} = \text{Mem}_{e_{\varepsilon,n,r}}^{\text{cal}}$ to that tuple. Its result $x_{\varepsilon,r}(x)$ is given by

$$\begin{aligned}
 c(x_{\varepsilon,r}(x)) &= \text{Carr}(\lambda(z_{\varepsilon,r}(x))) \\
 &= H_{\text{Carr}}(\lambda(y_{\varepsilon,r}(x))) (\text{Carr}(e_{\varepsilon,n,r})) + Y_{\text{Carr}}(e_{\varepsilon,n,r})(c(y_{\varepsilon,r}(x))), \\
 \lambda(x_{\varepsilon,r}(x)) &= \lambda(z_{\varepsilon,r}(x)), \\
 \text{gr}_M(x_{\varepsilon,r}(x)) &= d_{n,r} + 1, \\
 \text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x)) &= C_M(e_{\varepsilon,n,r}) \text{mem}_{d_{n,r}}(z_{\varepsilon,r}(x)) + \kappa_M(e_{\varepsilon,n,r}), \\
 s(x_{\varepsilon,r}(x)) &= s(e_{\varepsilon,n,r}) \wedge s(z_{\varepsilon,r}(x)), \\
 \text{Sig}(x_{\varepsilon,r}(x)) &= \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\hat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(x_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)), J_{d_{n,r}+1}(\text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x))) \right).
 \end{aligned} \tag{7.15}$$

All fields not enumerated in (7.15) coincide with those of $z_{\varepsilon,r}(x)$. In particular, the signature is computed from the sheet, orientation, boundary, and route already fixed by Sel^{cal} and Tra^{cal} , and from the carry and memory computed in the preceding lines; it is not circularly defined from the final state. Neither ε , nor $d_{\varepsilon,r}$, nor u is an additional argument of Upd^{cal} : the function reads all three data from the already constructed edge.

Simultaneous induction of the levels. Once the preceding nine stages have been completed for an already constructed G_n , define immediately

$$\gamma_{\varepsilon,k_n}(x) := x_{\varepsilon,9}(x), \quad G_{n+1} := \{\gamma_{\varepsilon,k_n}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}. \tag{7.16}$$

The base $G_0 = \{x_\star\}$, the intermediate tuples, and (7.16) constitute a single definition by recursion on n . Therefore, no subsequent carrier is used to assume the existence of G_n : the next level is introduced only after constructing all its edges from the preceding level.

TPK extension relation and effective membership. Let $\mathcal{E}_{\tilde{q}}^{\text{cal}}$ be the least labeled collection containing $x_\star$ and, at each step of the preceding recursion, the four tuples $x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x), z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)$ in their source and target configurations. Also let $\mathcal{A}_{\tilde{q},r}^{\text{cal}} := \{a_{n,r}\}$, and set

$$\tilde{\mathcal{Q}}_{\text{obs}} := \{\tilde{q}_{n,r} : n \geq 0, 1 \leq r \leq 9\}, \quad \mathcal{E}^{\text{cal}} := \bigsqcup_{\tilde{q}} \mathcal{E}_{\tilde{q}}^{\text{cal}}, \quad \mathcal{F}_{\tilde{q}}^{\text{cal}} := \pi_{\mathcal{F}}(\mathcal{E}_{\tilde{q}}^{\text{cal}}).$$

For each depth d , let $\mathcal{E}_d^{\text{cal}} := \text{gr}_M^{-1}(d) \subset \mathcal{E}^{\text{cal}}$. The explicit degree coordinate, rather than the mere membership of a representative in some image of the direct system, makes these carriers disjoint. On them, the projections are typed as

$$\begin{aligned} \text{cfg} : \mathcal{E}^{\text{cal}} &\longrightarrow \tilde{\mathcal{Q}}_{\text{obs}}, \\ g(x) &:= \text{phase}(\pi_{\text{obs}}(\text{cfg}(x))) \in C_9, \\ c : \mathcal{E}^{\text{cal}} &\longrightarrow K_{\text{Carr}}, \\ s : \mathcal{E}^{\text{cal}} &\longrightarrow \{\perp, \top\}, \\ \lambda : \mathcal{E}_{\tilde{q}}^{\text{cal}} &\longrightarrow \text{Hist}_{\tilde{q}}^{\text{cal}}, \\ \text{gr}_M : \mathcal{E}^{\text{cal}} &\longrightarrow \mathbb{N}_{\geq 1}, \\ \text{mem}_d : \mathcal{E}_d^{\text{cal}} &\longrightarrow M_d, \\ \text{mem}(x) &:= J_d(\text{mem}_d(x)) \in M \quad (x \text{ of depth } d). \end{aligned}$$

Here $\text{cfg}(x)$ is the second coordinate of the tuple, and phase reads the phase of the observable calendar. For the three carrier classes, fix the diagonals

$$\begin{aligned} \Delta_{\text{APP},\tilde{q}} &:= \{(a, a) : a \in \mathcal{A}_{\tilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{fib},\tilde{q}} &:= \{(f, f) : f \in \mathcal{F}_{\tilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{enr},\tilde{q}} &:= \{(x, x) : x \in \mathcal{E}_{\tilde{q}}^{\text{cal}}\}. \end{aligned}$$

In particular,

$$a_{0,1} \in \mathcal{A}_{\tilde{q}_{0,1}}^{\text{cal}}, \quad x_\star \in \mathcal{E}_{\tilde{q}_{0,1}}^{\text{cal}}.$$

On these carriers, the following relations are defined by their complete graphs, and not through necessary implications:

$$\text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} := \{(x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x)) : x \in G_n\}, \quad (7.17)$$

$$\text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} := \{(y_{\varepsilon,r}(x), z_{\varepsilon,r}(x)) : x \in G_n\}, \quad (7.18)$$

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{APP,cal}} := \{(a_{n,r}, a_{(n,r)^+})\}, \quad (7.19)$$

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{fib,cal}} := \{(\pi_{\mathcal{F}}x_{\varepsilon,r-1}(x), \pi_{\mathcal{F}}x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (7.20)$$

The synchronized enriched extension is

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{enr,cal}} := \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (7.21)$$

The third map is defined by

$$\text{graph}(\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}}) := \{(z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)) : x \in G_n\}, \quad (7.22)$$

where the second component is exactly the one in (7.15). Consequently,

$$\mathcal{U}_{e_{\varepsilon,n,r}}^{(1),\text{cal}} := \text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} = \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (7.23)$$

Each operator is indexed by the edge $e_{\varepsilon,n,r}$ produced by the reduction $\mathfrak{p}_\varepsilon$, rather than by a trit supplied from outside. The coordinate b_{ref} also distinguishes their codomains. Therefore, $\text{Sel}_e^{\text{cal}}$, Tra^{cal} and Upd^{cal} are functions on their respective domains, whereas the union $\bigcup_{\varepsilon \in \{-1,0,+1\}} \mathcal{U}_{e_{\varepsilon,n,r}}^{(1),\text{cal}}$ deliberately preserves the three outputs of the trisection.

For $\bullet \in \{\text{APP}, \text{fib}, \text{enr}\}$, identities and compositions are not left implicit. Define

$$\text{Ext}_{\text{id}_{\tilde{q}}}^{\bullet, \text{cal}} := \Delta_{\bullet, \tilde{q}}, \quad (7.24)$$

$$\text{Ext}_{e_i \dots e_1}^{\bullet, \text{cal}} := \text{Ext}_{e_i}^{\bullet, \text{cal}} \circ \dots \circ \text{Ext}_{e_1}^{\bullet, \text{cal}} \quad (7.25)$$

for every admissible edge word. The two APP and fiber path closures are formed separately,

$$\text{Path}(\text{Ext}^{\text{APP}, \text{cal}}), \quad \text{Path}(\text{Ext}^{\text{fib}, \text{cal}}), \quad (7.26)$$

and Tra^{cal} synchronizes them edge by edge.

Finally, the levels of complete states and their truncations are fixed by the recursion itself:

$$H_{d_{n,r}}^{\text{cal}} := \{x_{\varepsilon, r-1}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}, \quad (7.27)$$

$$H_{d_{n,r}+1}^{\text{cal}} := \{x_{\varepsilon, r}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}, \quad (7.28)$$

$$\rho_{d_{n,r}+1, d_{n,r}}^{\text{cal}}(x_{\varepsilon, r}(x)) := x_{\varepsilon, r-1}(x). \quad (7.29)$$

The final ledger preserves $(n, r, \mathfrak{p}_{\varepsilon})$, so the predecessor of the left-hand side is unique and ρ^{cal} is well-defined. For $b > a$, set $\rho_{b,a}^{\text{cal}} := \rho_{a+1,a}^{\text{cal}} \circ \dots \circ \rho_{b,b-1}^{\text{cal}}$. This map recovers the complete preceding state, including its signature; it does not purport to invert an overwritten coordinate without consulting the ledger.

Proposition 7.5 (Autonomous realization of the TPK through explicit transitions). *The collection*

$$\begin{aligned} \mathfrak{T}_{\text{TPK}}^{\text{cal}} := & ((\tilde{q}_{n,r})_{n,r}, \pi_{\text{obs}}, \tilde{F}_{\text{obs}}, M_{\text{ph}}, \\ & (\mathcal{A}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{E}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{F}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, \text{Ext}^{\text{APP}, \text{cal}}, \text{Ext}^{\text{fib}, \text{cal}}, \text{Ext}^{\text{enr}, \text{cal}}, \rho^{\text{cal}}, \\ & \text{Sel}^{\text{cal}}, \text{Tra}^{\text{cal}}, \text{Upd}^{\text{cal}}, C_M, \kappa_M, \text{gr}_M, H_{\text{Carr}}, Y_{\text{Carr}}, \\ & \text{Sig}^{\text{cal}}, \pi_{\mathcal{F}}). \end{aligned}$$

is an enriched model of the TPK. In particular, every pair on the right-hand side of (7.23) is an effective TPK edge, not a consequence inferred from a merely necessary condition.

Proof. The labels (n, r) make the carrier copies disjoint and unambiguously fix each source and target. The relation $\text{Ext}^{\text{APP}, \text{cal}}$ transports the complete cell and preserves the two Euclidean identities; the relation $\text{Ext}^{\text{fib}, \text{cal}}$ relates the fibers of the preceding and subsequent complete states: it transports sheet, orientation, prefix, cylinder, boundary, and ledger, and incorporates the updated carry, memory, and signature. Selection does not modify the ledger, and transport concatenates exactly one entry and produces the target presignature (7.14). The update recomputes the signature by means of the typed map $\text{Sig}_{\tilde{q}}^{\text{cal}}$ from the already determined sheet, orientation, carry, route, boundary, and memory. The map Upd^{cal} is unary because the edge contained in $z_{\varepsilon, r}(x)$ determines its cocycles.

The identities are the diagonals, and the word extensions are the compositions in (7.25). Concatenation of prefixes and ledgers is associative; memory satisfies the cocycle law, and carry satisfies (7.9). Therefore, the two parenthesizations of three edges produce the same state. Formula (7.29) defines truncation on complete states, and uniqueness of the final ledger proves that it recovers the source in every field. These are the identity, source, target, composition, memory, carry, and truncation laws of the enriched TPK. In particular, the extension relations are determined as graphs of effective maps rather than as consequences of merely necessary conditions. $\square$

Each $\gamma_{\varepsilon, k_n}$ comprises exactly nine applications of $\text{Upd}_{e_{\varepsilon, n, r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon, n, r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon, n, r}}^{\text{cal}}$. Survival will be proved by means of a constructed tail and will not be used to define G_n .

Lemma 7.6 (Effective internal trisection of one turn). *For every $n \geq 0$, every $x \in G_n$, and every $\varepsilon \in \{-1, 0, +1\}$, the expression $\gamma_{\varepsilon, k_n}(x)$ is defined by nine TPK transitions, with its synchronized relations $\text{Ext}^{\text{APP}, \text{cal}}$ and $\text{Ext}^{\text{fib}, \text{cal}}$, and satisfies*

$$\rho_{k_n+9, k_n}^{\text{cal}} \gamma_{\varepsilon, k_n}(x) = x, \quad (7.30)$$

$$g(\gamma_{\varepsilon, k_n}(x)) = g(x), \quad J_{k_n+9}(\text{mem}_{k_n+9}(\gamma_{\varepsilon, k_n}(x))) = J_{k_n}(\text{mem}_{k_n}(x)) + u, \quad (7.31)$$

$$c(\gamma_{\varepsilon, k_n}(x)) = c(x) + \varepsilon \kappa_{\text{Carr}}. \quad (7.32)$$

The ordered ledger of the nine edges determines ε uniquely. The three images are therefore disjoint. Each endpoint belongs to G_{n+1} and again admits the three maps $\gamma_{-1, k_{n+1}}, \gamma_{0, k_{n+1}}, \gamma_{+1, k_{n+1}}$.

Proof. Formulas (7.1) traverse the nine phases once and return the copy $a_{n+1,1}$ of $a(1)$. At each step, the APP relation transports the complete sextuple; Sel^{cal} fixes the TRIT type from the phase mark r preserved in the state; and Tra^{cal} adds exactly one symbol to the prefix, one segment to the cylinder, and its ordered pair of endpoints to the boundary. By induction on r , the explicit tuples $y_{\varepsilon, r}(x)$ and $z_{\varepsilon, r}(x)$ satisfy the predicates of the indicated relations, and the unary update produces the state at the next level. The identity $\rho_{d_{n,r}+1, d_{n,r}}^{\text{cal}}(x_{\varepsilon, r}(x)) = x_{\varepsilon, r-1}(x)$ is the definition (7.29); its well-definedness follows from uniqueness of the final ledger. The composition of the nine truncation maps recovers x , proving (7.30) without separately inverting the overwritten fields.

The phase $9 \rightarrow 1$ occurs exactly once. By (7.5) and the cocycle law, the linear memory component transports the preceding memory, and its translational component adds exactly u in the direct limit. The compatibility $J_{d+1}J_{d+1, d} = J_d$ proves (7.31); the return concerns the phase, not the enriched state.

By (7.6), eight edges have zero increment, and the edge completing $\mathfrak{p}_\varepsilon$ has the computed increment $\chi(\varepsilon, \varepsilon) \kappa_{\text{Carr}}$. The laws (7.9) and (7.4) give (7.32). The ledger preserves $(\mathfrak{p}_\varepsilon, 2\varepsilon, r_3(2\varepsilon), \chi(\varepsilon, \varepsilon))$; the three values of this ledger are distinct even when the scalar increment is zero. Hence the images are not identified, and the block decoder recovers a unique ε .

The edge definitions depend only on the new phase and concatenate, rather than replace, the prefix, cylinder, boundary, and ledger. They are also invariant under $J_d \text{mem}_d \mapsto J_d \text{mem}_d + nu$ and under translation of the input carry. Consequently, the same construction applies at the endpoint of any turn. For every finite state, the explicit sequence

$$x, \quad \gamma_{0, k_n}(x), \quad \gamma_{0, k_{n+1}} \gamma_{0, k_n}(x), \quad \dots$$

is a compatible infinite tail. This proves the survival of all constructed states and the repeated availability of the three branches, without postulating either property. Consequently, in accordance with the preceding level definitions, we may finally identify

$$H_{k_n}^{\text{cal}} := G_n \quad (7.33)$$

for every $n \geq 0$. □

The preceding closure distinguishes the three clocks involved. Nine emissions return the phase in C_9 and advance memory. Twelve turns, that is, 108 emissions, also return the position of the observable calendar; even then, neither the ledger nor memory is erased. The operator $\mathcal{K}_{\text{ph}}$ acts only afterward, as a recognition of reduced memory from the already constructed turns; it never generates the edges of $\text{Ext}^{\text{enr}, \text{cal}}$.

Let

$$v : \mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad v(0) = 0, \quad v(1) = +1, \quad v(2) = -1$$

be the balanced representative. For $b = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, the block extension is the ordered composition of six turns

$$\Gamma_{b, k}^{[54]} := \gamma_{v(b_6), k+45} \circ \gamma_{v(b_5), k+36} \circ \gamma_{v(b_4), k+27} \circ \gamma_{v(b_3), k+18} \circ \gamma_{v(b_2), k+9} \circ \gamma_{v(b_1), k}. \quad (39d)$$

At block level one always uses $k = k_{6N} = 1 + 54N$; consequently, each factor has the domain constructed by the preceding factor. Fix a canonical APP seed $x_\star \in H_1^{\text{cal}}$ and define recursively

$$\begin{aligned} Y_0 &:= \{x_\star\}, & Y_N &\subseteq H_{1+54N}^{\text{cal}}, \\ Y_{N+1} &:= \{\Gamma_{b,1+54N}^{[54]}(x) : x \in Y_N, b \in \mathbb{F}_3^6\}, \\ \rho_{N+1,N}^{\text{blk}} &:= \rho_{1+54(N+1),1+54N}^{\text{cal}} \upharpoonright Y_{N+1}, \\ \text{Ext}_N^{\text{cal}}(x) &:= \{\Gamma_{b,1+54N}^{[54]}(x) : b \in \mathbb{F}_3^6\}. \end{aligned} \tag{39e}$$

The notation $\rho_{b,a}^{\text{cal}}$ denotes the composition of all deletion maps between levels b and a . By construction, every extension in $\text{Ext}_N^{\text{cal}}(x)$ is a chain of fifty-four TPK edges, not a jump of the Markov quotient.

The reader is defined directly on the ledger, before invoking a representation through Γ_b . The preceding lemma provides a decoder

$$\text{dec}_9 : \{\text{segments generated by nine edges}\} \longrightarrow \{-1, 0, +1\},$$

which reads the TRIT pair and its reduction on the segment. If $\text{Led}_{j,i}^{(9)}(x)$ denotes the i -th nonadic segment of block j in the ledger of x , set

$$\beta_{\text{blk},N}(x) := \left((v^{-1} \text{dec}_9(\text{Led}_{j,i}^{(9)}(x)))_{i=1}^6 \right)_{j=0}^{N-1}. \tag{7.34}$$

For $N = 0$, the value is the empty word. This definition does not presuppose that the decomposition of x as a composition of maps Γ_b is unique.

We now specialize the preceding arithmetic conjugacy to $p = 3$, $d = 6$, and $A = I_6$. If $\pi_{N+1,N} : (\mathbb{Z}/3^{N+1}\mathbb{Z})^6 \rightarrow (\mathbb{Z}/3^N\mathbb{Z})^6$ is the canonical reduction, define

$$\Phi_N := \Psi_N^{-1} \circ \beta_{\text{blk},N} : Y_N \longrightarrow (\mathbb{Z}/3^N\mathbb{Z})^6. \tag{7.35}$$

This definition does not assign a coordinate after observing a branch: the reader $\beta_{\text{blk},N}$ is obtained from the already constructed TPK ledger, and Ψ_N^{-1} is the explicit inverse, fixed beforehand, of the unfiltered arithmetic tower.

Proposition 7.7 (Finite conjugacy and complete covering of the fiber). *For every $N \geq 0$,*

$$\beta_{\text{blk},N} : Y_N \xrightarrow{\cong} (\mathbb{F}_3^6)^N$$

is a bijection compatible with truncations, and for every $x \in Y_N$

$$\begin{aligned} \beta_{\text{blk},N+1}[\text{Ext}_N^{\text{cal}}(x)] &= \{\beta_{\text{blk},N}(x) \frown b : b \in \mathbb{F}_3^6\}, \\ \text{pref}_N \circ \beta_{\text{blk},N+1} &= \beta_{\text{blk},N} \circ \rho_{N+1,N}^{\text{blk}}. \end{aligned} \tag{40_0}$$

The maps Φ_N are bijections and satisfy the intertwining identity

$$\pi_{N+1,N} \circ \Phi_{N+1} = \Phi_N \circ \rho_{N+1,N}^{\text{blk}}. \tag{7.36}$$

Consequently, the tower of generated histories and the Hensel arithmetic tower are isomorphic as finite inverse systems.

Proof. The assertion is immediate for $N = 0$. Assume that it holds at level N . For each word $b \in \mathbb{F}_3^6$, the preceding lemma constructs an extension $\Gamma_{b,1+54N}^{[54]}(x)$. If $b \neq b'$, the first coordinate at which they differ determines a distinct TRIT pair and cocycle in the first divergent nonadic segment of the genealogical ledger. That ledger preserves the ordered edges, not merely the final sum of their carries; hence the two histories remain distinct even if carries from later turns cancel. The direct reading (7.34) then gives

$$\beta_{\text{blk},N+1}(\Gamma_{b,1+54N}^{[54]}(x)) = \beta_{\text{blk},N}(x) \frown b. \tag{7.37}$$

The six successive trisections produce exactly 3^6 distinct extensions. Deleting the last fifty-four steps recovers x , and deleting the final block of the ledger recovers $\beta_{\text{blk},N}(x)$. This simultaneously proves bijectivity and the compatibility of $\beta_{\text{blk},N}$ with truncations.

Proposition 7.4 independently proves that Ψ_N is bijective and that

$$\text{pref}_N \circ \Psi_{N+1} = \Psi_N \circ \pi_{N+1,N}.$$

Composing this equality with the compatibility of β_{blk} and applying Ψ_N^{-1} yields

$$\pi_{N+1,N} \Psi_{N+1}^{-1} \beta_{\text{blk},N+1} = \Psi_N^{-1} \beta_{\text{blk},N} \rho_{N+1,N}^{\text{blk}},$$

which is exactly (7.36). The equality does not arise by defining an ad hoc coordinate on each branch: it results from composing two systems of constructed bijections that are compatible at every depth. $\square$

The inverse limit

$$\Omega_{\Gamma}^{\text{cal}} := \varprojlim_N (Y_N, \rho_{N+1,N}^{\text{blk}}) \quad (40a_0)$$

is the history limit of the TPK model constructed edge by edge in the preceding lemma. Its projection under π_{obs} preserves the configurations of the observable calendar, but the assertion does not depend on identifying this autonomous model with an ambient relation defined only by necessary conditions. The finite bijections pass to the limit and produce the cylinder homeomorphism

$$\Lambda : (\mathbb{F}_3^6)^{\mathbb{N}} \xrightarrow{\cong} \Omega_{\Gamma}^{\text{cal}}, \quad \beta_{\text{blk}} := \Lambda^{-1}. \quad (40b_0)$$

Denote the canonical projection by $\rho_{\infty,N}^{\text{blk}} : \Omega_{\Gamma}^{\text{cal}} \rightarrow Y_N$. Thus, Λ establishes an equivalence compatible with truncations: each ternary tape encodes a TPK history, and every prefix is extended through edges of the same generator.

The subcylinders are now defined as subsets of the limit, rather than as pairs of suffixes. For $x \in Y_N$ and $b \in \mathbb{F}_3^6$, let

$$\mathcal{Z}_N(x) := \{\xi \in \Omega_{\Gamma}^{\text{cal}} : \rho_{\infty,N}^{\text{blk}}(\xi) = x\}, \quad (7.38)$$

$$\mathcal{Z}_{N+1}(x; b) := \{\xi \in \Omega_{\Gamma}^{\text{cal}} : \rho_{\infty,N+1}^{\text{blk}}(\xi) = \Gamma_{b,1+54N}^{[54]}(x)\}. \quad (7.39)$$

Theorem 7.8 (Effective partition into 729 refinement subcylinders). *For every $N \geq 0$ and every $x \in Y_N$, there is a disjoint union*

$$\mathcal{Z}_N(x) = \bigsqcup_{b \in \mathbb{F}_3^6} \mathcal{Z}_{N+1}(x; b). \quad (7.40)$$

The $3^6 = 729$ terms are nonempty clopen refinement subcylinders; each is realized by fifty-four edges of the constructed TPK.

Proof. Let $\xi \in \mathcal{Z}_N(x)$. Its level- $N+1$ coordinate belongs to Y_{N+1} , truncates to x , and, by the bijection $\beta_{\text{blk},N+1}$, has a unique final block $b \in \mathbb{F}_3^6$. Identity (7.37) then implies $\rho_{\infty,N+1}^{\text{blk}}(\xi) = \Gamma_{b,1+54N}^{[54]}(x)$; therefore, ξ belongs to one of the terms on the right-hand side. The reverse inclusion follows immediately by truncation. If two terms intersected, their level- $N+1$ coordinates would be equal and the ledger decoder would give $b = b'$; hence the union is disjoint.

Each endpoint $\Gamma_b^{[54]}(x)$ admits an infinite tail obtained by iterating the zero-block extension; consequently, every term is nonempty. Since Y_{N+1} is finite and discrete, the terms are inverse images of points under a continuous projection of the inverse limit and are therefore clopen. The very definition of $\Gamma_b^{[54]}$ establishes its fifty-four edges. $\square$

Finally, the collection $(\Phi_N)_N$ induces the homeomorphism of limits

$$\Phi_{\infty} : \Omega_{\Gamma}^{\text{cal}} \xrightarrow{\cong} \mathbb{Z}_3^6, \quad \Phi_{\infty}(\xi) := (\Phi_N(\rho_{\infty,N}^{\text{blk}}(\xi)))_{N \geq 0}, \quad (7.41)$$

where $\mathbb{Z}_3^6 = \varprojlim_N (\mathbb{Z}/3^N\mathbb{Z})^6$. Equation (7.36) guarantees that the right-hand side is a compatible class, and the finite bijections prove injectivity, surjectivity, and continuity in both directions. Under Φ_∞ , $\mathcal{Z}_N(x)$ is exactly the congruence class

$$\{z \in \mathbb{Z}_3^6 : z \equiv \Phi_N(x) \pmod{3^N}\},$$

and the sets $\mathcal{Z}_{N+1}(x; b)$ are its 729 refinement classes modulo 3^{N+1} . This is the effective junction between TPK extension, limit cylinders, and the Hensel tower.

For $b = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, set

$$v(b) := \sum_{i=1}^6 [b_i]_3 3^{6-i} \in \{0, \dots, 728\}, \quad (40c_0)$$

where $[b_i]_3 \in \{0, 1, 2\}$ is the positional representative. The 243 visible words in the initial census are the image of the first emission; they are not to be confused with the 3^6 capacity of each extension block. Finally,

$$X_{\infty, \mathbb{Z}}^{\text{cal}} := \mathbb{Z}_{\text{HMT}} \times \Omega_{\Gamma}^{\text{cal}}.$$

For a prefix $s = (m; b_0, \dots, b_{N-1})$, define in the generated arithmetic of $\mathbb{Q}_{\text{HMT}}$

$$a_N(s) := m + \sum_{q < N} \frac{v(b_q)}{729^{q+1}}, \quad \text{Succ}(s) := \{(m; b_0, \dots, b_{N-1}, b) : b \in \mathbb{F}_3^6\}.$$

If $(m; \beta_{\text{blk}, N}(x))$ denotes the prefix obtained by prepending m to the tuple $\beta_{\text{blk}, N}(x)$, (40₀) gives the typed equality

$$\text{Succ}(m; \beta_{\text{blk}, N}(x)) = \{(m; \beta_{\text{blk}, N+1}(y)) : y \in \text{Ext}_N^{\text{cal}}(x)\}.$$

From their first definition, the half-open cylinders are subsets of the HMT completion:

$$I_N^{\text{HMT}}(s) := \{r \in \mathbb{R}_{\text{HMT}} : a_N(s) \leq_{\text{HMT}} r <_{\text{HMT}} a_N(s) + 729^{-N}\}.$$

They satisfy materially

$$I_N^{\text{HMT}}(s) = \bigsqcup_{y \in \text{Succ}(s)} I_{N+1}^{\text{HMT}}(y), \quad |\text{Succ}(s)| = 729, \quad \text{diam}_{\text{HMT}} I_N^{\text{HMT}}(s) = 729^{-N} \longrightarrow 0. \quad (40a)$$

The boundary convention identifies the two representations only after they have been constructed. The partition (40a), nesting, and contraction produce

$$\begin{aligned} P_{\text{ar}} : X_{\infty, \mathbb{Z}}^{\text{cal}} &\twoheadrightarrow \mathbb{R}_{\text{HMT}}, \\ P_{\text{ar}}(m, \xi) &:= \text{the unique element of } \bigcap_N C_N^{\text{cl}}(m; \beta_{\text{blk}, N}(\rho_{\infty, N}^{\text{blk}}(\xi))), \\ C_N^{\text{cl}}(s) &:= \overline{I_N^{\text{HMT}}(s)}. \end{aligned} \quad (40)$$

Therefore,

$$X_{\infty, \mathbb{Z}}^{\text{cal}} / \sim_{P_{\text{ar}}} \cong \mathbb{R}_{\text{HMT}}. \quad (41)$$

The other representation, P_{exc} , preserves incidence, boundary, and memory toward Paley–Witt–Golay–Leech and, from there, toward $V^{\natural}$ –Monster. Both arise from the same antecedent in (35); neither the Archimedean coordinate nor the exceptional recognition selects the seeds. The five constructions sp, sol, gau, coh, det remain consubstantial within $\mathcal{C}_{\text{cont}}^{\text{disc}}$ and are not disaggregated in order to carry out the cardinal step.

7.2.3 Descent of operations to the evaluation quotient

The preceding quotient identifies histories that express the same value, not their complete registers. The following criterion characterizes exactly when an operation on histories induces an operation on this quotient without choosing a privileged history within each fiber; its formulation continues the genealogical construction established in the preceding sections.

Proposition 7.9 (Descent criterion along evaluation fibers). *Let $q : \Omega \rightarrow J$ be the evaluation onto the image of a genealogical chart. Let $\star : D \rightarrow \Omega$ be an operation with $D \subseteq \Omega^2$ saturated by the fibers of $q \times q$. There exists a unique operation $\bar{\star} : (q \times q)(D) \rightarrow J$ such that*

$$q(x \star y) = q(x) \bar{\star} q(y)$$

if and only if, for pairs in the domain,

$$q(x) = q(x'), \quad q(y) = q(y') \implies q(x \star y) = q(x' \star y').$$

Proof. If the visible operation exists, its result depends only on the two values, which proves necessity. For sufficiency, define $s \bar{\star} t = q(x \star y)$, where $q(x) = s$ and $q(y) = t$. Saturation preserves the domain when representatives are changed; the stated condition preserves the result. Surjectivity of $q \times q$ onto the image of the domain proves uniqueness. $\square$

A coordinate does not automatically replace a history as an argument of all its operations: the criterion determines when that substitution is legitimate. In particular, transporting an operation through a bijection is not enough to identify it with a native operation preceding the reading. If a chart Ω is compact and q is continuous and surjective onto a Hausdorff space J , evaluation also induces a homeomorphism $\Omega / \sim_q \rightarrow J$: it is a continuous bijection from a compact space to a Hausdorff space. It is the *quotient*, and not necessarily the space of histories, that is topologically identified with J .

7.2.4 Internal reconstruction of the chart and uniform lift

Within $\mathbb{G}_{\text{HMT}}(h, m_\infty)$, the reconstruction in (35)–(41) is carried out with the same codes and operators. Denote by $\mathbb{R}_{\text{HMT}}^\mathbb{G}$ the internal HMT completion. The uniqueness of the complete Archimedean ordered field provides, **after** its construction, the internal isomorphism

$$\eta : \mathbb{R}_{\text{HMT}}^\mathbb{G} \xrightarrow{\cong} \mathbb{R}^\mathbb{G}. \quad (41_0)$$

For the corresponding internal interval $I_N^\mathbb{G}(s) := I_N^{\text{HMT}}(s) \cap \mathbb{R}_{\text{HMT}}^\mathbb{G}$, the isomorphism the isomorphism preserves endpoints and order:

$$\eta[I_N^\mathbb{G}(s)] = \{r \in \mathbb{R}^\mathbb{G} : a_N(s) \leq r < a_N(s) + 729^{-N}\}. \quad (41_0a)$$

The surjectivity of (40) covers every element of that completion, not only those obtained in a finite execution.

So that “every” is not concealed in the definition of the codomain, we construct the two injections relating the line to its internal power set. Let

$$E(A) = \sum_{n \geq 0} \frac{21_A(n)}{3^{2n+2}} \quad (A \in \mathcal{P}^\mathbb{G}(\omega)). \quad (41a)$$

This representation places 0 or 2 only in the even ternary positions; its image is contained in $[0, 1/4]$. It therefore never reaches the right endpoint of the root cylinder, and two distinct subsets have a different first even position: E is injective and is not subject to boundary ambiguity. Write $E_{\text{HMT}} := \eta^{-1} \circ E$.

Given A , begin with the root cylinder containing $E_{\text{HMT}}(A)$. If $s_N(A)$ is the prefix already chosen, write $s_N(A) = (0; b_0(A), \dots, b_{N-1}(A))$. The partition (40a), transported by η^{-1} , supplies a unique half-open

subcylinder $s_{N+1}(A)$ containing $E_{\text{HMT}}(A)$. By (40₀), that extension is also an admissible TPK extension. By induction,

$$s_{N+1}(A) \upharpoonright N = s_N(A), \quad E_{\text{HMT}}(A) \in I_N^{\text{HMT}}(s_N(A)) \quad (N < \omega). \quad (41b_0)$$

The recursion is a uniform formula with parameters A, h . Since both are internal, Replacement in the model of Lemma 7.3 forms the complete sequence of blocks $(b_q(A))_{q < \omega}$. Without confusing the tape with the sequence of prefixes, define

$$\xi_A := \Lambda((b_q(A))_{q < \omega}) \in \Omega_\Gamma^{\text{cal}, \mathbb{G}}.$$

This yields the **uniform lift of the Cantor embedding**

$$Q : \mathcal{P}^{\mathbb{G}}(\omega) \longrightarrow X_{\infty, \mathbb{Z}}^{\text{cal}, \mathbb{G}}, \quad Q(A) := (0, \xi_A), \quad P_{\text{ar}}(Q(A)) = E_{\text{HMT}}(A). \quad (41b)$$

The map Q is not called an inverse of P_{ar} : their composition expresses the embedding E_{HMT} , not the identity on the entire line. In the opposite direction, let $(q_n)_{n < \omega}$ be the generated enumeration of $\mathbb{Q}_{\text{HMT}}$, and define

$$C(r) := \{n < \omega : \eta(q_n) < r\} \quad (r \in \mathbb{R}^{\mathbb{G}}).$$

The rational cut belongs to $\mathcal{P}^{\mathbb{G}}(\omega)$, and C is injective by density. The injections E and C , followed by the internal Cantor–Bernstein theorem, prove

$$P_{\text{ar}}[X_{\infty, \mathbb{Z}}^{\text{cal}, \mathbb{G}}] = \mathbb{R}_{\text{HMT}}^{\mathbb{G}}, \quad \mathbb{R}_{\text{HMT}}^{\mathbb{G}} \cong \mathbb{R}^{\mathbb{G}} \approx \mathcal{P}^{\mathbb{G}}(\omega). \quad (41c)$$

Here an arbitrary real is used only as a universally quantified variable to prove surjectivity of an already constructed tree. It selects neither APP, TRIT, TPK, the constants, nor the transitions of that tree.

7.3 Enumeration by genealogical codes and HMT condensation

Rule (37) supplies derivation names from

$$(\ulcorner \varphi \urcorner, \sigma(p_0), \dots, \sigma(p_{k-1})). \quad (42)$$

For any object a of the hierarchy, let $\beta(a)$ be its first stage and $\sigma(a)$ its least derivation tree in the recursive order of Lemma 7.3. The term $\sigma(a)$ is not required to be a natural number when the stage is uncountable. The relation

$$a <_{\mathbb{G}} b \iff (\beta(a), \sigma(a)) <_{\text{lex}} (\beta(b), \sigma(b)) \quad (42a)$$

is the global genealogical well-order $<_{\mathbb{G}}$ induced by the constructor. Repeated names are eliminated by taking the least one; neither a real nor an enumeration of the line is inserted as data.

Lemma 7.10 (Internal countability of levels below $\omega_1^{\mathbb{G}}$). *For every $\beta < \omega_1^{\mathbb{G}}$, the level $\mathbb{G}_\beta(h, m_\infty)$ is countable within $\mathbb{G}_{\text{HMT}}(h, m_\infty)$, and possesses a canonical surjection*

$$e_\beta : \omega \twoheadrightarrow \mathbb{G}_\beta(h, m_\infty). \quad (43)$$

Proof. The initial level has a countable code. If e_β enumerates level β , the formula codes and finite tuples of parameters in (42) form a countable set and enumerate the successor. If $\lambda < \omega_1^{\mathbb{G}}$ is a limit, there is internally a surjection $s : \omega \twoheadrightarrow \lambda$; take the $<_{\mathbb{G}}$ -least one, in the order already constructed in Lemma 7.3, and diagonalize the $e_{s(n)}$. Finally, among all surjections so obtained, define e_β as the $<_{\mathbb{G}}$ -least. The recursion $\beta \mapsto e_\beta$ is a uniformly definable class of $\mathbb{G}$; every e_β , and every restriction of the recursion to an ordinal, belongs to $\mathbb{G}$. It uses only $h, u_{h,m}$ and the global well-order. $\square$

Lemma 7.11 (Condensation of internal reals below $\omega_1^\mathfrak{G}$). *Every real in $\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ appears at a level whose index is smaller than $\omega_1^\mathfrak{G}$.*

Proof. Let $r \subseteq \omega$ be a real in $\mathfrak{G}$, and let $B := \mathfrak{G}_0(h, m_\infty)$. This base is transitive and internally countable. With the codes of natural numbers and finite pairs already fixed, $\text{Ord} \cap B = \omega + 1$. By induction,

$$\text{Ord} \cap \mathfrak{G}_\alpha = (\omega + 1) + \alpha.$$

The successor adds the ordinal formed by the ordinals of the preceding level; it cannot add a larger one, because that would not be a subset of that level. Limits are unions. Choose a limit Θ , with $\Theta > \omega^2$, such that $r \in \mathfrak{G}_\Theta$. Thus $\text{Ord} \cap \mathfrak{G}_\Theta = \Theta$, and every level $\mathfrak{G}_\xi$, with $\xi < \Theta$, belongs to $\mathfrak{G}_\Theta$. Internally form the expanded structure

$$\mathcal{A}_\Theta = \langle \mathfrak{G}_\Theta, \in, u_{h,m}, H_\Theta, <_\mathfrak{G} \upharpoonright \mathfrak{G}_\Theta \rangle$$

with the level relation as a predicate of its signature, $H_\Theta = \{ \langle \xi, x \rangle : \xi < \Theta \text{ \& } x \in \mathfrak{G}_\xi \}$, and its definable Skolem functions. Let

$$Y = \text{Hull}^{\mathcal{A}_\Theta}(B \cup \{B, r\}) < \mathcal{A}_\Theta.$$

Closure under the countable collection of Skolem functions shows within $\mathfrak{G}$ that Y is countable. Let $c : Y \rightarrow M$ be its Mostowski collapse, and let

$$\bar{\Theta} := \text{otp}(Y \cap \Theta) < \omega_1^\mathfrak{G}.$$

Since $B \subseteq Y$ is transitive, induction on membership shows that c fixes every element of B , as well as B itself. In particular, it fixes $\omega, u_{h,m}, h, m_\infty$. Since $r \subseteq \omega \subseteq Y$, it also follows that $c(r) = r$.

We prove by induction on $\xi \in Y \cap \Theta$ that

$$c^n(Y \cap \mathfrak{G}_\xi(h, m_\infty)) = \mathfrak{G}_{c(\xi)}(h, m_\infty). \quad (43a)$$

For $\xi \in Y \cap \Theta$, the predicate H_Θ defines the set $\mathfrak{G}_\xi$ from ξ , so $\mathfrak{G}_\xi \in Y$. Relativizing each formula to that set gives $Y \cap \mathfrak{G}_\xi < \mathfrak{G}_\xi$, in the language into which Lemma 7.1 translates the HMT definitions.

The base case of (43a) has already been proved because the collapse fixes B . At a successor, if $\xi + 1 \in Y$, then also $\xi \in Y$ and $c(\xi + 1) = c(\xi) + 1$. For $a \in Y \cap \mathfrak{G}_{\xi+1}$, fix a finite formula defining it over $\mathfrak{G}_\xi$. The assertion

$$a = \{z \in \mathfrak{G}_\xi : \varphi^{\mathfrak{G}_\xi}(z, \bar{p})\}$$

is an ordinary formula over $\mathfrak{G}_\Theta$, with parameters $a, \mathfrak{G}_\xi$. Elementarity supplies the parameters $\bar{p}$ within $Y \cap \mathfrak{G}_\xi$. The induction hypothesis and the elementarity of that level show that $c(a)$ is the subset defined by the same formula and the collapsed parameters over $\mathfrak{G}_{c(\xi)}$. Conversely, given a formula and parameters at the collapsed level, lift the parameters to the original level. The subset defined there exists, is unique, and belongs to Y by elementarity. Its collapse is the required subset. This proves both inclusions at the successor stage without introducing a uniform truth predicate.

For a limit $\lambda \in Y \cap \Theta$, elementarity supplies, for every $x \in Y \cap \mathfrak{G}_\lambda$, a witness $\xi \in Y \cap \lambda$ with $H_\Theta(\xi, x)$. Therefore,

$$Y \cap \mathfrak{G}_\lambda = \bigcup_{\xi \in Y \cap \lambda} (Y \cap \mathfrak{G}_\xi), \quad c^n(Y \cap \lambda) = c(\lambda).$$

Continuity of the hierarchy and the induction hypothesis give (43a) at the limit. Cofinality of $Y \cap \lambda$ in the ambient ordinal λ is not required: its collapsed indices range over $c(\lambda)$.

Finally, every $x \in Y$ has a witness $\xi \in Y \cap \Theta$ with $H_\Theta(\xi, x)$, and $\bar{\Theta}$ is a limit. Taking unions,

$$M = \bigcup_{\eta < \bar{\Theta}} \mathfrak{G}_\eta(h, m_\infty) = \mathfrak{G}_{\bar{\Theta}}(h, m_\infty).$$

Since $c(r) = r \in M$, it follows that $r \in \mathfrak{G}_{\bar{\Theta}}$ with $\bar{\Theta} < \omega_1^\mathfrak{G}$. □

The lemma does not say that a finite-length prefix enumerates the line. It says that every generated hereditary object has a first transfinite level and a first genealogical code within that level. This is precisely the information lost by the extensional reduct when it preserves only the expressed value.

7.4 Injection of the generated line into the first uncountable ordinal of the genealogical closure

Let $\iota(r) \subseteq \omega$ be the rational cut of $r \in \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$ relative to the enumeration of $\mathbb{Q}_{\text{HMT}}$ previously constructed from APP integers and quotients. The map $r \mapsto \iota(r)$ is definable from r and that enumeration, so $\iota(r) \in \mathfrak{G}$; Lemma 7.11 places it at a stage with countable index. Define

$$\begin{aligned}\beta(r) &:= \min\{\beta : \iota(r) \in \mathfrak{G}_{\beta+1}(h, m_\infty)\}, \\ n(r) &:= \min\{n : e_{\beta(r)+1}(n) = \iota(r)\}, \\ j_{\text{HMT}}(r) &:= \omega \cdot \beta(r) + n(r).\end{aligned}\tag{44}$$

Definition (44) is uniform in the parameters h, m_∞ . Since $\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$ is a set of $\mathfrak{G}$, Separation fixes the domain of the definition, and Replacement in $\mathfrak{G}$ forms its graph:

$$\{\langle r, \omega \cdot \beta(r) + n(r) \rangle : r \in \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}\} \in \mathfrak{G}.$$

Therefore, j_{HMT} is an internal function of $\mathfrak{G}$.

By Lemmas 7.10 and 7.11, $j_{\text{HMT}}(r) < \omega_1^{\mathfrak{G}}$ for every generated real. If $j_{\text{HMT}}(r) = j_{\text{HMT}}(s)$, ordinal division by ω recovers the same pair (β, n) ; (43) recovers the same rational cut and, by completeness, $r = s$. Hence

$$j_{\text{HMT}} : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \hookrightarrow \omega_1^{\mathfrak{G}}\tag{45}$$

is an injection constructed from the rank and name of the HMT genealogy. Its existence uses precisely the semantic closure (37), the well-ordering of names from Lemma 7.3, and the condensation (43a); it does not receive the Continuum Hypothesis in order to choose the stage. The subsequent set-theoretic identification describes this same machinery in different notation without reversing its causal provenance.

7.5 Injection of the first uncountable ordinal into the line and cardinal equivalence

For every $\beta < \omega_1^{\mathfrak{G}}$, Lemma 7.10 allows a countable well-order of type β to be encoded in a subset $w_\beta \subseteq \omega$. To avoid confusing the empty order with the one-point order, both the domain $D \subseteq \omega$ and its strict order relation $R \subseteq D^2$ are encoded. Once a bijective pairing $\langle \cdot, \cdot \rangle : \omega^2 \rightarrow \omega$ is fixed, the code is

$$w(D, R) = \{2n : n \in D\} \cup \{2\langle a, b \rangle + 1 : (a, b) \in R\}.$$

The even bits recover the domain, and the odd bits recover the relation. Take the $<_{\mathfrak{G}}$ -least real encoding such a well-order. Since the property “encodes a well-order of type β ” and the global well-order are internal, Replacement forms the function $\beta \mapsto w_\beta$ within $\mathfrak{G}$. Distinct order types produce distinct codes. The ternary map, already used in (41a), is defined by

$$E(w) = \sum_{n \geq 0} \frac{2\mathbf{1}_w(n)}{3^{2n+2}}\tag{46}$$

and is injective and avoids the endpoint of the root cylinder. The lift (41b) ensures that $E_{\text{HMT}}(w_\beta)$ belongs to the line expressed by an HMT history. Thus

$$k_{\text{HMT}}(\beta) := E_{\text{HMT}}(w_\beta), \quad k_{\text{HMT}} : \omega_1^{\mathfrak{G}} \hookrightarrow \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}.\tag{47}$$

The injections (45) and (47), constructed without using the equality sought, give by Cantor–Bernstein

$$|\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}| = \aleph_1^{\mathfrak{G}}.\tag{48}$$

Since (41c) identifies the complete HMT line with $\mathcal{P}^{\mathfrak{G}}(\omega)$, (48) rewrites exactly as

$$\mathfrak{G}_{\text{HMT}}(h, m_\infty) \models 2^{\aleph_0} = \aleph_1. \quad (49)$$

The equality does not arise from 729, from an informal product of cardinality and ordinality, from the digits of a constant, or from a numerical bracket. It arises from the covering of the line by the inverse limit, the condensation of the genealogical closure, and the two explicit injections.

7.6 Full quantifier over all internal subsets

Let $A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}_{\text{HMT}}^{\mathfrak{G}})$. If A is countable, the first alternative holds. If it is not, $j_{\text{HMT}}[A]$ is an uncountable subset of $\omega_1^{\mathfrak{G}}$. It cannot be bounded by a countable ordinal, since every initial segment of ω_1 is countable. Its increasing enumeration has order type $\omega_1^{\mathfrak{G}}$, and the inverse of j_{HMT} on its image produces an injection $\omega_1^{\mathfrak{G}} \hookrightarrow A$. The inclusion $A \subseteq \mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$, (48), and Cantor–Bernstein give

$$\forall A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}), \quad |A| \leq \aleph_0 \quad \text{or} \quad |A| = |\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}|. \quad (50)$$

Here the full power set of the integral continuum generated by HMT is quantified. The bijective transport of the folding with memory carries (50) to any of its representations without losing parts; it does not replace the injections (45)–(47).

7.7 Representation of the genealogical closure as $L[h, m_\infty]$

Only after (49)–(50) is the construction in (36) assigned its name in the usual set-theoretic notation. Let $L[u_{h,m}]$ be the constructible hierarchy relative to the real already produced in (33a). Since the HMT signature is countable and all its operators are graph relations encoded by h , Lemma 7.1 eliminates its symbols uniformly. This yields the equality of classes

$$\mathfrak{G}_{\text{HMT}}(h, m_\infty) = L[u_{h,m}] = L[h, m_\infty]. \quad (51)$$

To prove the first inclusion, strengthen the induction: for every ordinal β , both $\mathfrak{G}_\beta$ and the sequence $\langle \mathfrak{G}_\xi : \xi \leq \beta \rangle$ belong to $L[u_{h,m}]$. The base is the transitive closure of a set constructed from ω and $u_{h,m}$; the parameters h, m_∞ are recovered from $u_{h,m}$. At a successor, the preceding level is an *internal set*, not merely a subclass of $L[u_{h,m}]$. Its satisfaction relation and the uniform elimination (37c) make it possible to form internally the set of its definable subsets. At a limit, transfinite recursion, Replacement, and Union in $L[u_{h,m}]$ form the preceding sequence and its union. Transitivity then gives $\mathfrak{G} \subseteq L[u_{h,m}]$.

Conversely, Lemma 7.3 supplies an internal transitive model of ZF, containing all ordinals and $u_{h,m}$ as an element. The recursion constructing $L[u_{h,m}]$ is carried out in that model: at every stage, satisfaction over the preceding level and relative definability are absolute between the two transitive domains. By induction, its levels coincide with those of the ambient construction and belong to $\mathfrak{G}$. Thus $L[u_{h,m}] \subseteq \mathfrak{G}$. Finally, (33a) makes $u_{h,m}$ and the pair (h, m_∞) primitive-recursively interdefinable, from which the second equality follows.

Equation (51) explicitly identifies the closure with the constructible hierarchy relative to the generated parameter. This representation does not alter causal provenance: $L[u]$ neither selects the genealogical ledger nor defines the extensions or the line; the decision of the Continuum Hypothesis follows from the exhaustivity rule and the maps proved between (36) and (50). The Taken together, the composition of (33a), (36), (37d), and (49)–(51) and Theorem 7.12 give a complete and uniform expression of the realized universe, semantic coincidence, and cardinal closure; none of these representations replaces APP–TRIT–TPK generation.

7.8 Theorem giving an affirmative resolution of the Continuum Hypothesis

Theorem 7.12 (Decision of the continuum in HMT). *For every realized history m_∞ produced by the joint action $\text{APP} \rightarrow \text{TRIT} \rightarrow \text{TPK} \rightarrow X_\infty^{\text{enr}} \rightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}$, the HMT genealogical extension (36) generates the internal reconstruction of the integral line (40), $\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$, constructs the injections (45) and (47), and satisfies $2^{\aleph_0} = \aleph_1$ over its entire internal power set. Therefore, HMT proves the Continuum Hypothesis affirmatively for the integral continuum that it defines and generates. The notation $L[h, m_\infty]$ is a subsequent representation of that hierarchy, not the premise deciding the result.*

7.9 Falsification criteria for the cardinal closure

The following list identifies failures that would invalidate a step of the preceding closure. Construction, semantic, and cardinality controls are distinguished:

1. there exists a level N or a surviving state without a compatible successor in (34);
2. $u_{h,m}$ does not hereditarily encode h, m_∞ , or some graph relation of Σ_{HMT} does not admit the uniform elimination (37c);
3. an object admitted by the HMT semantic rules lacks a term in (37a), or a term in (37a) expresses an object outside those rules, breaking (37d);
4. $\mathfrak{G}$ does not satisfy one of the axioms used, its definable well-order fails, or a purported enumeration e_β is not internal and uniform;
5. the subcylinders in (40a) do not form an exhaustive disjoint partition, their cylinders cease to contract, or the recursion (41b₀) does not produce an internal compatible history;
6. the collapse in Lemma 7.11 does not commute with a successor or limit stage, or there exists an internal real that appears in no level with index smaller than $\omega_1^{\mathfrak{G}}$;
7. the selection $\beta \mapsto w_\beta$ is not internal or uniform, or does not preserve order type;
8. two distinct reals have the same genealogical pair (β, n) in (44), two distinct ordinals produce the same code in (47), or one of the injections $E, C, j_{\text{HMT}}, k_{\text{HMT}}$ fails;
9. there exists an uncountable internal subset of the line whose range under j_{HMT} is bounded in $\omega_1^{\mathfrak{G}}$;
10. some generative stage preceding the covering (40a) uses the Continuum Hypothesis, $\mathbb{R}$, a target constant, the Committee on Data of the International Science Council (CODATA), the Particle Data Group (PDG), or a subsequent physical identification as a causal antecedent to define a seed, a relation $\mathcal{L}_\tau$, Ext, a coefficient, admissibility, or an extension of the tree. The subsequent lift (41b₀) does not constitute such an inversion: there an arbitrary already constructed real is a universal variable and selects, within the previously proved exhaustive partition, the unique subcylinder containing it.

No execution restricted to a preset number of levels replaces the universal quantifier of the inverse system; an effective failure of compatibility at some depth would falsify it. No subsequent set-theoretic notation proves (49) by itself. The closure jointly uses Lemma 7.3, the line–power-set identification in (41c), Lemmas 7.10–7.11, and the two injections constructed from the genealogy.

8 Phase, dodecaphase register, and exceptional realization of the common state

The theorem in Section 7 decides the cardinal question within the complete HMT state. The constructions that follow are not a separable ontological appendix: they make explicit correlated coordinates of the same enriched state. The dual projection preserves complementary information: Archimedean value in one branch, and incidence, boundary, and genealogy in the other.

The holonomy Γ_9 composes nine updates U_t on the enriched state. The reduced kernel $\mathcal{K}_{\text{ph}}$ is a subsequent representation over emissions already constructed that preserves reduced phase and memory:

$$U_t \longrightarrow \Gamma_9 \longrightarrow \mathcal{K}_{\text{ph}}, \quad g(k+9) = g(k), \quad q_{\text{mem},k+9} = q_{\text{mem},k} + 1, \quad \hat{x}_{k+9} \neq \hat{x}_k.$$

The phase returns, the memory advances, and the state is not reset. By contrast, the integral dodecaphase seal is constructed through the signed reader and the Hadamard transform:

$$\mathcal{S}_{12} (= R_{\text{sgn}}) : \mathcal{X}_{\text{TPK}}^{\text{term, enr}} \longrightarrow \mathcal{U}_{12}^{\text{sgn}} := \text{im } \mathcal{S}_{12}, \quad U_{\text{term}} \in \mathcal{U}_{12}^{\text{sgn}}, \quad K := \frac{1}{4} \mathcal{H}_{12} U \in \mathbb{Z}^{12}.$$

The coordinate U_{term} is obtained from the enriched state by means of the oriented register and the channel readers:

$$\mathcal{S}_{12}(x_{\text{term}}) = \Pi_H(\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}}))) = U_{\text{term}}.$$

Section 11 develops this evaluation, the reversible production of K , and the transverse return to the same twenty-five channels. The serialization of C_{term} consumed by the Python controls is a verification entry point, not a mathematical generator or a restriction on the printed proof. The vector K does not by itself produce either branch. Together with the correlated representations P, E, Φ , it feeds the integral closure

$$\begin{aligned} Z_m &:= P_m + E_m - \Phi_m - K_m, & R_{N+1}(Z) &:= \sum_{r \geq 0} Z_{N+1+r} 1000^{-r-1}, \\ c_{N+1}^\infty &:= [R_{N+1}(Z)], & s_m &= Z_m + c_{m+1}^\infty, & A_m &= s_m \bmod 1000, & c_m &= \frac{s_m - A_m}{1000}, \\ (P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}; c_{N+1}^\infty)_{N \geq 1} &\xrightarrow{\tau_{\alpha, \infty}} (\alpha_N)_N \mapsto \alpha_{\text{HMT}} := \varprojlim_N \alpha_N. \end{aligned}$$

The condition $c_{13} = 0$ belongs exclusively to the isolated execution of twelve blocks and expresses the rational $\alpha_{12}^{(0)}$. It is not the boundary of the infinite arrow: there, each horizon receives the quotient determined by its signed tail; in particular, the first circuit of the complete section has $c_{13}^\infty = -1$. The exceptional representation preserves a distinct codomain:

$$\begin{aligned} B_K &= \beta_{\text{arq}}(K) = \{i : K_i \geq 729\}, & s^{(0)} &= P + E - \Phi - K, \\ H_\alpha^- &= \eta_{\text{arq}}(s^{(0)}) = \eta_{\text{exc}}(B_K, P_\pi, H_e, H_\varphi, H_{\text{APP}}), \\ \tilde{J}_* &= (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}) \mapsto o_* = 1 \mapsto v_\alpha = 4\rho_1 + \rho_2 + \dots + \rho_{12}, \\ N_{\alpha,0} &= \{x \in N(C_W) : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, & N_\alpha &= N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3} \cong \Lambda_{24}, \\ C_W &\mapsto N(C_W) := \bigcup_{c \in C_W} (A_2^{12} + \phi(c)) \xrightarrow{\text{Nbr}_{v_\alpha}} N_\alpha \cong \Lambda_{24} \longrightarrow V^\natural \xrightarrow{\text{Aut}} \mathbb{M}. \end{aligned}$$

The scalar α_{HMT} , the face B_K , the hexad H_α^- , the vector v_α , and the lattice N_α belong to different codomains. They share provenance; they are not interchangeable objects.

Proposition 8.1 (Causal unity of the correlated representations). *The objects*

$$\mathcal{K}_{\text{ph}}, \quad K, \quad \pi_{\text{HMT}}, \varphi_{\text{HMT}}, e_{\text{HMT}}, \alpha_{\text{HMT}}, \quad H_\alpha^-, \quad v_\alpha, \quad N_\alpha, \quad V^\natural$$

are neither external data nor retroactive selectors of the cardinal inference. All are produced as coordinates, representations, or realizations of the two projections of the single enriched state. The cardinal comparison

does not query their conventional values as oracles, but it does operate on the complete HMT structure that preserves their genealogy, regions, and memory.

Proof. The cardinal proof uses the inverse system of histories, the integral real line, the semantic closure, the ordinal classifier, and the two injections within the same APP–TRIT–TPK state. No conventional digit determines these maps. In the internal constructive order, $\mathcal{K}_{\text{ph}}$ factors the memory of Γ_9 ; the seal K is obtained reversibly from the terminal coordinate produced by $\mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}$; the fourfold relation is expressed from the coinductive sections; and the flag, the vector v_α , the Leech neighbor, and $V^\natural$ are obtained through the exceptional projection. The following sections provide the complete constructions and their proofs. $\square$

In particular, the following type separations are preserved:

$$\mathcal{K}_{\text{ph}} \neq K, \quad \mathcal{H}_{12} \neq H_\alpha^-, \quad \alpha_{\text{HMT}} \neq v_\alpha \neq N_\alpha.$$

Nor is there a canonical arrow $W_{24} \rightarrow \Lambda_{24}$: the Witt design and the ternary Leech neighbor are distinct prolongations of the common code. The Monster group acts on $V^\natural$, not on decimal digits, TPK routes, or codes of α .

9 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

9.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (9.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 9.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (9.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 9.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (9.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (9.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (9.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 9.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (9.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (9.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (9.5) and its residual bound. $\square$

9.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (9.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (9.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 9.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (9.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 9.2. For the second, apply (9.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (9.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (9.8) determines its compatibility with every subsequent extension exactly.

Example 9.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (9.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

9.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (9.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 9.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 9.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

9.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (9.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 9.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (9.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (9.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (9.13)$$

Lemma 9.8 (Integer criterion for intersection). *The cylinders in (9.10) have a nonempty intersection if and only if (9.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (9.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (9.13). $\square$

Definition 9.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 9.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (9.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (9.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 9.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 9.2 places v_c simultaneously in both intervals in (9.10); the intersection criterion gives compatibility. The update identities are (9.5) in each base. Recovery follows from (9.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 9.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

9.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 9.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (9.16)$$

Theorem 9.14 (Uniqueness of the prolongation compatible with the reader). *Condition (9.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (9.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (9.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

9.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (9.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (9.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (9.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (9.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 9.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (9.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (9.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (9.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (9.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (9.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (9.23). $\square$

Example 9.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 9.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (9.24)$$

Proposition 9.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (9.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

9.7 Composition theorem for the block, cylinder, and signature

Theorem 9.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (9.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (9.14) and (9.15);*
2. *its six trits produce the signature $S(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (9.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 9.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 9.11 to be applied. The deterministic application of (9.6) produces the bands used by (9.17)–(9.20). The recovery identities are those of Theorem 9.15. Finally, Theorem 9.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

9.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{9.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{9.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (9.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

10 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

10.1 Regional bands, encoding, and integer signatures

10.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (10.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (10.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (10.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

10.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (10.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (10.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (10.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (10.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 10.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

10.2 Conversion clock and first zero advance

10.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (10.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 10.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (10.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 10.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

10.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (10.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

10.3 Oriented construction of the twenty-four-trit descriptor

10.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

10.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are $(5, 7, 7, 7, 7, 7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (10.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (10.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

10.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (10.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^X : the two objects share a length, but their producers and their places in the composition differ.

Proposition 10.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (10.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (10.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

10.4 Functional panel and terminal repertoire

10.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (10.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

10.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (10.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its $8!$ orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (10.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

10.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 6: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 6,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 10.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 7, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 7: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

10.5 Comparative readers, affine reader, and temporal incidence

10.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (10.14)$$

Lemma 10.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (10.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

10.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (10.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

10.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (10.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

10.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (10.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (10.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 10.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (10.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

10.6 Enumeration of cylinders and decimal candidates

10.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (10.19)$$

10.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 10.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (10.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 10.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 10.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (10.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

10.7 Exceptional incidence and heterotypy condition

10.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (10.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (10.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (10.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

10.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (10.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (10.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (10.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

10.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 &\rightarrow 4, & 1 &\rightarrow 5, & 2 &\rightarrow 6, \\ 3 &\rightarrow 7, & 4 &\rightarrow 8, & 5 &\rightarrow 9, \\ 6 &\rightarrow 10, & 7 &\rightarrow 11, & 8 &\rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

10.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{10.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 10.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{10.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (10.22); the destination bands T_8, T_{12} and residues $R_{4,8}, R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (10.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (10.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

10.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

$$\begin{array}{cccc} 020022 & 222111 & 211201 & 021101 \\ 002001 & 020111 & 200110 & 121012 \\ 010122 & 001001 & 221110 & 222110 \\ 121020 & & & \end{array}$$

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

10.8 Regional composition and continuity of the realizations

10.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{10.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (10.13).

The register is then formed by the twelve Euclidean divisions in (10.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

10.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

10.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

10.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

10.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (10.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean's native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library's customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms

actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

10.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 6, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

10.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (10.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 10.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1) - 1}{h} \right\rfloor. \quad (10.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (10.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 10.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (10.32)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (10.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(10.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (10.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D / Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1) / Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (10.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D / Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (10.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

10.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (10.36)$$

The words are ordered as in (10.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 10.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (10.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (10.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h). \end{aligned} \quad (10.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (10.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (10.35) and the profile (10.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

10.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_\partial(B) = (1, 2, 2, 1, 2, 0), \quad (10.39)$$

where w_{col} counts the nonzero elements in each column and q_∂ sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 10.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango, pesos}} = 18, \quad N_{\text{rango, pesos, sumas}} = 2. \quad (10.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (10.39) give (10.40). Substituting the two final indices into (10.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are (0, 2, 0) and (2, 2, 1); only the latter belongs to this line. This recovers the last step of the selector in Section 10.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080}x_\chi]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

10.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

11 Construction of the dodecaphase register and its rational phase coordinate

The generation of the regional coordinates does not exhaust the information in the APP–TRIT–TPK state. An Archimedean reading determines a value through compatible cylinders; another reading preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphase register belongs to this second reading before entering the closure of α . It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with path memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block word from the catalog, a list of twelve oriented events, a signed integral vector, and a base-one-thousand dodecaphase register are distinct objects. Comparing them requires the mappings that relate them. Equality of length does not supply those mappings, and the loss of signs is not a mere simplification of notation.

This chapter maintains the causal cut fixed in the preceding chapters: the nine APP positions of the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving remainder, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite calendar. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow \mathcal{K}_{\text{ph}}$ reversed: the last mapping is a reduced-memory reading, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence among three coordinatizations of a record previously produced by APP–TRIT–TPK. The Hadamard transformation relates the signed record to the integral record; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12/4}} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the subset of the integrality sublattice $\mathcal{L}_H$ that constitutes the domain of this chart; the second, on the image of the extractor. Once the base 1000, the 12 positions, and the phase origin are fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean division recovers the complete record. The proofs of §11.7, §11.8, and §11.9 establish these equivalences on their respective domains.

Two compositions subsequently act on that record. The positional composition combines its components with the closure, propagation, and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and obtains

$$K \longmapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete record makes it possible to reconstruct both compositions. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (14.12) links this integral extraction to the incidence of the regional codes. The negative support of its balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional prolongation. Thus, the function of K is defined through an invertible transformation and precise subsequent mappings; the additional depth and transport information remains in the state from which the record arises.

11.1 Cyclic extension and memory grading

The observable calendar contains one hundred eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the cyclic translation law. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphase sector and the interior nonadic position. Neither of them by itself preserves the total number of turns.

Proposition 11.1 (The carry of the calendar). *The mappings*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section that chooses the representatives $0, \dots, 8$, the additivity defect is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . The reduction is surjective, and multiplication by nine is injective on the indicated domain. If the sequence split, the central group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. The contradiction proves that the sequence is nonsplit. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the displayed carry and verifies the identity of the section. $\square$

The result explains a difference that will reappear in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts the nonadic turns, the calendar preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and the vertical memory. The closure of a finite chart does not turn that history into a memoryless cycle.

11.2 The oriented record and the integral 1 + 5 + 6 chart

For a history produced by TPK, let τ_m be the subroute on E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with path memory, not the set of phase labels. The projection to the calendar forgets precisely some of the data that the signed reader requires.

An initial chart of the record can be written through oriented events. The construction developed in this article fixes the set of event codes $\{2, 4, 6, 8\}$ and the sectorial orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , one obtains

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event census and the orientation separate. It does not identify a cardinal direction with a TRIT sign. Nor is the vector e , whose components are signed local censuses, yet the vector U that will appear below.

The opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

define the integral chart

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The $1 + 5 + 6$ decomposition is not an arbitrary distribution of coordinates: one coordinate preserves the charge, five compare the pairs relative to a reference pair, and six preserve the opposition of the half-coronas.

Lemma 11.2 (Inversion of the integral chart). *An integral tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which yields the division by six. Once the p_i have been reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the stated solution. This solution is integral exactly under the parity congruence. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize the suppression of the rest of the ledger. The complete record has a larger domain than this census chart. The extraction of the channels that feed U uses that complete record and is not deduced from the list (e_i) through an identification of names.

11.3 Positional and incidence readings of the terminal state

Denote by X_{term} the state produced by the composition of selection, transport, and updating up to the terminal cut. The operator expression is

$$X_{\text{term}} = (\mathcal{U}_{108} \circ \dots \circ \mathcal{U}_1)(X_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readings are

$$X_{\text{term}} \xrightarrow{(\pi_{\text{term}}, \mathcal{S}_{12})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary reading; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector developed in this article operates on catalogs of sizes 196, 197, and 196. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column margin $q_\partial = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; therefore $B_{\text{term}} = B_1$. This verification proves the final step of the selector from the stated census. It does not turn the column margin into a consequence of the row charge, nor does it replace the preceding enumeration with inspection of those two matrices.

The signed reader is the composition

$$\mathcal{S}_{12} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (11.1)$$

Its evaluation is developed below on the complete enriched state, without introducing K , α , or conventional metrology as selectors. The raw CT108 projection, which forgets the signed coordinate, is not used as a substitute domain for this reader.

11.4 Signed dodecaphase chart of the full TPK state

The dodecaphase chart and a possible factorization through the raw 108-step projection (CT108) are objects of different types. Let $X \in \text{TPKFullState}$ be a complete enriched state, and let $\rho_{108}(X) \in \text{CT108RawProjection}$ be the object obtained by forgetting orientation, sheet, carry, boundary signature, memory, and the dodecaphase ledger. In the canonical coordinates of the full state, write

$$X = (X_{\text{APP}}, X_{\text{TRIT}}, X_{\text{route}}, X_{\text{mem}}, C_{\text{term}}, U_{12}^{\text{sgn}}, X_{\text{boundary}}, \dots).$$

The active reader is, by definition, the projection of that already generated and preserved coordinate:

$$\mathcal{S}_{12} := \text{pr}_{U_{12}^{\text{sgn}}} : \text{TPKFullState} \longrightarrow \text{SignedDodecaphaseState}.$$

On the compatible sublattice of the full state, this projection coincides with the composite reader

$$\mathcal{S}_{12} \equiv R_{\text{sgn}} := \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (11.2)$$

Therefore, $\mathcal{S}_{12}$ reads a preserved coordinate of the same state; it receives neither a conventional constant nor a metrological table. Its domain is not replaced by CT108RawProjection, since that replacement would additionally require a factorization $\mathcal{S}_{12} = \Xi \circ \rho_{108}$. Such a factorization is neither presupposed nor needed here; it would require proof that the signed reading is constant on the fibers of ρ_{108} . The internal theorem operates directly on the full TPK state.

To fix the content of a possible incidence realization, let $\mathcal{L}$ be a ledger of visits and oriented traces belonging to X . Each visit preserves the route, APP position, sheet, instant, multiplicity, and boundary incidence. The arithmetic evaluation produces the integer and its decomposition

$$n_v = r_v + 9q_v, \quad 1 \leq r_v \leq 9.$$

The integer n_v is calculated from the sum or product sheet; its remainder is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits of remainder 9.

In the temporal window $E_m = \{9(m-1) + 1, \dots, 9m\}$, the domains are distinguished

$$\mathcal{V}_m = \{v : t(v) \in E_m\}, \quad \mathcal{T}_m = \{\theta : \text{origin}(\theta) \in E_m\}.$$

The presentation by counts considers

$$A_m = \sum_{v \in \mathcal{V}_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (11.3)$$

$$C_m = - \sum_{\theta \in \mathcal{T}_m} u(\theta) \sigma(\theta) = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (11.4)$$

$$V_m = \sum_{v \in \mathcal{V}_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \varepsilon_{\partial}(v). \quad (11.5)$$

Here $w(v)$ is the incidence multiplicity, and $\varepsilon_{\partial}(v) = 1$ on the corona, -1 in the interior. The traces counted by $\nu^{\pm}$ must be enumerated by the corresponding ledger rule. The integral sum requires finite support or an alternative construction that determines its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; their relation requires the reader's unit of length.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative and preserve the quotients

$$x = [x]_{10} + 10 \lfloor x/10 \rfloor.$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (11.6)$$

By composition one obtains the mapping

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (11.7)$$

The indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; therefore, the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is preserved in §11.8, where the complete integral reconstruction is proved. This formula produces its coordinates from the counts. It uses neither K , U , α , nor the expected transverse coordinates as arguments.

11.4.1 Terminal representation and exact closure of its charts

The canonical terminal state is

$$X_{\text{term}} = (\mathcal{U}_{108} \circ \dots \circ \mathcal{U}_1)(X_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t,$$

where each $\mathcal{U}_t$ acts on the full TPK state and not on its raw projection. Among the coordinates produced and preserved before the signed representation is

$$C_{\text{term}} = \text{pr}_C(X_{\text{term}}) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

No subsequent table is introduced here: C_{term} is the transverse coordinate that $\mathcal{E}_{108}^{90,120}$ produces within the evolution of the full state. On the compatible sublattice of TPK states, the chart satisfies the operator identity

$$\mathcal{S}_{12} = \text{pr}_U = \Pi_H \circ \text{pr}_C. \quad (11.8)$$

The identity is formulated on the full TPK state and does not presuppose the descent of $\mathcal{S}_{12}$ through ρ_{108} .

The subsequent evaluation from the twenty-five expressed fields is displayed without a gap. For $1 \leq r \leq 3$, let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90},$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2.$$

Integral substitution of the preceding twenty-five fields gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479).$$

Applying

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2})$$

one obtains, component by component,

$$U^{(1)} = (2378, -452, -668, -322),$$

$$U^{(2)} = (1406, 998, -204, -28),$$

$$U^{(3)} = (2479, -551, -371, -997).$$

Columnwise interleaving therefore evaluates the signed chart:

$$U_{\text{term}} := \mathcal{S}_{12}(X_{\text{term}}) = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (11.9)$$

Theorem 11.3 (Subsequent exact closure among U , K , and the channels). *Once the signed chart (11.9) has been expressed by projection, the orbital Hadamard transform expresses, at a subsequent stage,*

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601),$$

and the transverse extractor (D_3, D_4, Q) produces

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned} \quad (11.10)$$

The composition is an exact bijection among the compatible sublattice of signed states, the seal K , and its transverse coordinate $C_{\text{term}} = (b^{90}, b^{120}, Q_{\text{TPK}})$. In particular,

$$C_{\text{term}} = (D_3K, D_4K, Q(K)). \quad (11.11)$$

Proof. Grouping U_{term} into its three step-three orbits yields

$$\begin{aligned} u^{(0)} &= (2378, -452, -668, -322), & \frac{1}{4}H_4u^{(0)} &= (234, 729, 621, 794), \\ u^{(1)} &= (1406, 998, -204, -28), & \frac{1}{4}H_4u^{(1)} &= (543, 659, 58, 146), \\ u^{(2)} &= (2479, -551, -371, -997), & \frac{1}{4}H_4u^{(2)} &= (140, 824, 914, 601). \end{aligned}$$

The divisions are exact in $\mathbb{Z}$. Columnwise interleaving expresses K . Finally, the twelve cyclic step-three and step-four subtractions and the total sum give, component by component, (11.10). Proposition 11.13 proves the integral inverse, and Theorem 11.9 proves naturality under refinement. No operation receives α , CODATA, a metrological table, or a conventional constant. $\square$

Remark 11.4 (Complete operator form and focal executable control). The terminal arrow is not postulated from a table: it is the typed composition

$$X_{\text{term}} \xrightarrow{\mathcal{R}_{12}} \mathcal{L}(X_{\text{term}}) \xrightarrow{\mathcal{E}_{108}^{90,120}} C_{\text{term}} \xrightarrow{\Pi_H} U_{\text{term}} \xrightarrow{\mathcal{H}_{12}^{-1}} K.$$

The preceding definitions traverse the twelve windows that partition the one hundred eight steps and express each channel coordinate as a finite sum of its visits and traces, with orientation, sheet, carry, boundary, and memory. This is the mathematical realization of the production of C_{term} ; a row-by-row tabular expansion would be another serialization of the same sum, not an additional premise of the theorem. The focal executable exactly reevaluates the finite chart $C_{\text{term}} \leftrightarrow U_{\text{term}} \leftrightarrow K$ as a supplementary reproducible control. The proof of the continuum hypothesis remains formulated over APP-TRIT-TPK, the full enriched state, its correlated representations, and the joint discrete structure.

Neither $\mathcal{S}_{12}$, nor U , nor K , nor α are external inputs or conventional selectors of the discrete construction of the continuum. They are typed coordinates and representations of the same APP-TRIT-TPK state. The coordinate α_{HMT} is a derived and correlated output of that single generation; its representation order does not break the continuity of the object or weaken its derivation.

The incidence formula (11.7) describes an additional realization when a complete ledger $\mathcal{L}(X_{\text{term}})$ is unfolded. The preceding theorem does not claim that the signed coordinate can be reconstructed after that ledger has been forgotten through ρ_{108} ; nor does it require such a claim. This separation avoids confusing the internal closure of the enriched state with an inverse problem posed on a projection that has removed precisely the orientation and memory information.

The mapping Π_H , executed within the chart, produces from C_{term} the signed coordinate and its three step-three orbits:

$$\begin{aligned} U_{\text{term}}^{(1)} &= \Pi_H^{(1)}(C_{\text{term}}) = (2378, -452, -668, -322), \\ U_{\text{term}}^{(2)} &= \Pi_H^{(2)}(C_{\text{term}}) = (1406, 998, -204, -28), \\ U_{\text{term}}^{(3)} &= \Pi_H^{(3)}(C_{\text{term}}) = (2479, -551, -371, -997). \end{aligned}$$

Let $\mathcal{P}_{\text{orb}} : \mathbb{Z}^{12} \rightarrow (\mathbb{Z}^4)^3$ be the orbital regrouping

$$\mathcal{P}_{\text{orb}}(U) = ((U_1, U_4, U_7, U_{10}), (U_2, U_5, U_8, U_{11}), (U_3, U_6, U_9, U_{12})),$$

and let $\text{Int}_{3,4} = \mathcal{P}_{\text{orb}}^{-1}$ be the columnwise interleaving of three tetrads. We also define

$$\text{Dig}_{10}^{(3)}(z) = \left(\left\lfloor \frac{z}{100} \right\rfloor, \left\lfloor \frac{z}{10} \right\rfloor \bmod 10, z \bmod 10 \right), \quad 0 \leq z \leq 999.$$

The terminal modular chart is the effective mapping on the compatible channel state

$$\mathcal{G}_{\text{term}}(C) := \left[(\text{Dig}_{10}^{(3)})^{12} \circ \text{Int}_{3,4} \circ \left(\frac{1}{4}H_4 \oplus \frac{1}{4}H_4 \oplus \frac{1}{4}H_4 \right) \circ \mathcal{P}_{\text{orb}} \right] (\Pi_H(C)). \quad (11.12)$$

Its domain is the subset of the integrality sublattice of compatible coordinates in $\mathbb{Z}^{25}$ on which Π_H is integral, the three orbital images under H_4 are divisible by four, and the twelve resulting coordinates belong to $\{0, \dots, 999\}$. It receives neither U_{term} , K , α , a decimal table, nor a metrological value; the interleaving, the phase origin, and the orientation are part of the type of the channel coordinate.

The verification coordinate is serialized in `metadata/semilla-registro-terminal.json`. That file contains C_{term} , the phase origin, and the orientation; it contains neither U_{term} , K , nor the twelve residual rows. The focal program applies $\mathcal{G}_{\text{term}}$, expresses U_{term} , K , and the residues, and returns through (D_3, D_4, Q) to the same C_{term} . Starting there permits an exact focal verification of the subsequent finite chart. It does not redefine the domain of the signed chart (11.9), does not replace the preceding proof, and does not attribute to the auxiliary the production of the coordinate it receives.

Proposition 11.5 (Modular calculation of the twelve terminal rows). *The evaluation of $\mathcal{G}_{\text{term}}$ on the produced coordinate C_{term} first obtains U_{term} and then the phase tetrads*

$$(234, 729, 621, 794), \quad (543, 659, 058, 146), \quad (140, 824, 914, 601),$$

and, after interleaving, produces exactly the twelve rows of the following table. It is the algebraic verification subsequent to Theorem 11.3; it does not replace the representation of the signed chart of the full TPK state.

Proof. The formulas for Π_H give the three signed blocks written above. The subsequent product by $H_4/4$ gives, without rounding,

$$\begin{aligned}\frac{1}{4}H_4(2378, -452, -668, -322)^T &= (234, 729, 621, 794)^T, \\ \frac{1}{4}H_4(1406, 998, -204, -28)^T &= (543, 659, 58, 146)^T, \\ \frac{1}{4}H_4(2479, -551, -371, -997)^T &= (140, 824, 914, 601)^T.\end{aligned}$$

The twelve divisions are exact in $\mathbb{Z}$. Columnwise interleaving gives

$$\begin{aligned}(234, 543, 140, 729, 659, 824, \\ 621, 058, 914, 794, 146, 601),\end{aligned}$$

and $\text{Dig}_{10}^{(3)}$ uniquely provides its three decimal residues. The auxiliary verifier included in the package executes these same operations; its negative controls alter the channel state or its phase origin. $\square$

Denote by

$$\mathcal{F}_{\text{term}}^{(10)} := ((a_m, c_m, v_m))_{m=1}^{12}$$

the digit chart expressed by $\text{Dig}_{10}^{(3)}(K_m)$. When a complete incidence realization is available, $(a_m, c_m, v_m) = ([A_m]_{10}, [C_m]_{10}, [V_m]_{10})$.

This equality is not postulated in order to reconstruct the ledger after it has been forgotten. Each row preserves the window index and is therefore not a bag of interchangeable digits.

m	a_m	c_m	v_m	K_m
1	2	3	4	234
2	5	4	3	543
3	1	4	0	140
4	7	2	9	729
5	6	5	9	659
6	8	2	4	824
7	6	2	1	621
8	0	5	8	058
9	9	1	4	914
10	7	9	4	794
11	1	4	6	146
12	6	0	1	601

Consequently,

$$z^{\text{term}} = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (11.13)$$

Proposition 11.6 (Terminal return of the computed collection to the channels). *The decimal recombination of $\mathcal{F}_{\text{term}}^{(10)}$, followed by the transverse extractor, produces, without consulting α or a metrological table,*

$$\begin{aligned}D_3 z^{\text{term}} &= (-495, -116, -684, 108, 601, -90, & -173, -88, 313, 560, -397, 461), \\ D_4 z^{\text{term}} &= (-425, -281, -481, 671, -255, 30, & 475, -543, 680, 251, 6, -128), \\ Q(z^{\text{term}}) &= 6263.\end{aligned}$$

These are exactly the twenty-five coordinates $(b^{90}, b^{120}, Q_{\text{TPK}})$ used by the harmonic reader.

Proof. The decimal formation of each row gives (11.13). Cyclically subtracting the entry $m + 3$ and the entry $m + 4$ produces, component by component, the two vectors displayed; the sum of the twelve entries is 6263. The calculation is repeated in the package's autonomous verifier. The terminal collection is a computed representation of the channel state. The transverse evaluation returns exactly to the seed C_{term} ; the harmonic reading then recovers U_{term} , and the inverse Hadamard transform recovers the same K . This proves the agreement of both coordinatizations without using alpha as a selector. $\square$

The chart $\mathcal{F}_{\text{term}}^{(10)}$ is sufficient and necessary for this transverse evaluation. The table is a decimal certificate of the expressed output, not a visit-by-visit reedition of the oriented ledger. The enriched state additionally preserves Euclidean quotients, visits, traces, and provenance; those fields are not declared recoverable from the thirty-six residues, nor are they necessary for the cardinal theorem of this article.

Proposition 11.7 (Sufficient and necessary arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$, ordered by the counts (A, C, V) . For the mapping defined by (11.6)–(11.7),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sectorial residues exactly determine the output of 25 coordinates. The preservation of the quotients and provenance constitutes additional information in the record.

Proof. Equality modulo 10 produces the same blocks and, therefore, the same transverse coordinates. Conversely, if the transverse coordinates coincide, $D_3(z - z') = D_4(z - z') = 0$. The steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of the total charges makes that constant vanish. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. The three residues of each window are thus recovered. The assertion is an algebraic identity for every N, N' , not an inference obtained from finite tests. $\square$

11.5 Conjugation of the counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the order of the windows, preserves the arithmetic class and the boundary class of the visits, and interchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (11.14)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula preserves exactly this difference. Negating the channel is not equivalent to negating the complete decimal block.

Applying this identity to a concrete TPK involution requires verifying its action on the incidences. For an occupancy reader evaluated before the advance, the reversal of a route $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

The endpoint terms are incorporated before reduction modulo 10. They vanish on closed routes; in open windows they may alter the residues. This correction, already present in the bidirectional development of transport, specifies what the linkage between the route and the counts must preserve.

Proposition 11.8 (Necessity of the nonperiodic information in the record). *Suppose that the observable of a fixed collection of routes satisfies $o(t + 54) = o(t)$, that multiplicity is preserved under that return, and that the same local reader acts on the windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. The translation $t \mapsto t + 54$ carries E_m bijectively to E_{m+6} and preserves each counted incidence and its multiplicity. The equality is transmitted to the sum and to decimal reduction. $\square$

Therefore, a presentation with 12 distinct blocks requires the reader to preserve some information that does not factor through that observable of period 54: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition delimits an inadmissible reduction of the state; it does not assert that the complete dynamics has period 54, nor does it by itself determine its terminal reader.

Theorem 11.3 expresses the signed chart of the full TPK state, applies the orbital Hadamard transform, and obtains through the transverse extractor

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are internal outputs of the terminal chart, not parameters fitted through alpha. Formulas (11.16)–(11.17) recover from them

$$U_{\text{term}} = S_{12}(X_{\text{term}}) = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (11.15)$$

Proposition 11.6 also verifies the transverse return to the same twenty-five channels. The arithmetic preserves the distinction among the terminal matrix, the incidence collection, the signed vector, and the dodecaphase register.

11.6 Construction of the signed vector from the channel record

The harmonic reader Π_H can be written without using either K or alpha. To avoid confusing its orbital sums with the alpha blocks, denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (11.16)$$

Division by three belongs to the reader of the three orbits. Its integrality is verified on the compatible outputs of the record; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (11.17)$$

Integral substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns recovers

$$U = U_{\text{term}} = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (11.18)$$

The presence of negative values and components greater than 999 shows that U is not a base-one-thousand digit word. Nor is it $U_6 \| U_6$, a catalog concatenation of period six. The difference concerns domain and preserved information, not merely values.

The complete representation and its return control are, respectively,

$$\begin{aligned} X_{\text{term}} &\xrightarrow{\mathcal{R}_{12}} \mathcal{R}_{12}(X_{\text{term}}) \xrightarrow{\mathcal{E}_{108}^{90,120}} C_{\text{term}} \xrightarrow{\Pi_H} U_{\text{term}} \xrightarrow{H_4^{-1}} K, \\ K &\longrightarrow \mathcal{F}_{\text{term}}^{(10)} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U_{\text{term}}. \end{aligned}$$

The Hadamard inversion develops $U_{\text{term}} \rightarrow K$, while the incidence evaluation and harmonic reading develop the return $\mathcal{F}_{\text{term}}^{(10)} \rightarrow U_{\text{term}}$. Their equality is a reversible concordance square; it does not turn a received table into the generator of the signed state. The executable auxiliary serializes C_{term} in order to isolate and falsify this finite chart. The initial equality is evaluated in the preceding passage through Π_H on the coordinate C_{term} already produced in the full state; the auxiliary neither redefines the APP-TRIT-TPK generator nor factors the chart through CT108RawProjection.

Theorem 11.9 (Naturality of the signed reader). *For $N' \geq N \geq 108$, if*

$$\tau_{N',N} : \mathcal{X}_{\text{TPK}}^{[N'],\text{enr}} \longrightarrow \mathcal{X}_{\text{TPK}}^{[N],\text{enr}}$$

is the truncation that preserves the first N updates and their ledger, then

$$\mathcal{S}_{12,N} \circ \tau_{N',N} = \mathcal{S}_{12,N'}.$$

Proof. Each updater adds the next incidence without rewriting the one hundred eight terminal windows already generated. Therefore $\mathcal{R}_{12}$, $\mathcal{E}_{108}^{90,120}$, and Π_H read the same coordinates after truncating any subsequent prolongation. The equality is one of mappings on produced states; it does not result from preserving the output as a first coordinate of the input. $\square$

11.7 Orbitwise Hadamard transform and integral reconstruction

In the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation into four sectors of each orbit is what, once origin and orientation have been fixed, admits the ninety-degree angular reading. The transformation does not act on three contiguous tetrads in natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders according to the three stated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 11.10 (Hadamard inversion and integrality). *One has*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integral elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly the divisibility by four of the four coordinates of H_4u . $\square$

Definition 11.11 (Integral dodecaphase transformation). The subset of the integrality sublattice formed by the signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The mapping $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of Abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines exactly the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves that the two compositions are inverse. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 11.12 (Canonical integral record). *The inversion $K^{(r)} = H_4 U^{(r)}/4$ of the blocks in (11.17) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In natural order, the dodecaphase register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (11.19)$$

It is the unique rational preimage of U , and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4 U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4 U^{(2)} &= (2172, 2636, 232, 584), \\ H_4 U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (11.19). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in terms of lattices. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even and one has modulus two; the minors of order three are multiples of four and one has modulus four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integral vector admits an integral preimage; the vector obtained satisfies the precise congruences required by the inversion. The factor 1/4 is forced by the matrix and does not have the status of a free coefficient.

11.8 Cyclic differences and uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to those separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 11.13 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven unit factors. In particular, its outputs determine a unique dodecaphase register.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this final freedom because $Q(\mathbf{1}) = 12$.

For the integral assertion, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter vanish in the incidence rows and equal twelve in row Q . All maximal minors are divisible by twelve, and the tree produces one of modulus exactly twelve. The final factor is therefore twelve. $\square$

Verification on (11.19) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (11.16) and (11.17) in turn reconstruct U from those differences. A triangle of exact representations of the same object is thus closed: transverse record, signed vector, and dodecaphase register. Equality of results does not turn their codomains into a single codomain; it proves that the stated mappings recover the relevant information.

11.9 Reading of one turn and periodic prolongation of the dodecaphase register

Once K has been constructed, the base-one-thousand chart permits two distinct operations. The first reads a single turn; the second periodically prolongs the dodecaphase register. Let $B = 1000$ and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (11.20)$$

The leading zeros of a block, such as that of 058, are part of its three-digit representation. They do not change its integral value, but they do permit the twelve blocks to be concatenated unambiguously.

Definition 11.14 (Finite and periodic readings of the dodecaphase register). The reading of one turn is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 11.15 (Exact value and period of the prolonged dodecaphase register). *One has*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12}\kappa_{\text{per}}. \quad (11.21)$$

The infinite word of the dodecaphase register has minimum period twelve in base one thousand and minimum period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series yields $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; therefore the minimum block period is twelve. The decimal expansion is canonical, since the dodecaphase register has no tail of 999 blocks. If its minimum decimal period were d , it would divide 36. Grouping by threes would produce a block period $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under that operation. Hence the minimum decimal period is 36. $\square$

Both readings are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not permit one to be substituted for the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

11.10 Reversibility of the rational constant and phase change

The rational constant κ_{per} completely determines the record K when the base, length, and reading origin are preserved. Its numerical function and its structural function are related by an exact inversion.

Proposition 11.16 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1)\kappa_{\text{per}}.$$

The integral expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros of each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000\kappa(K) - K_1. \quad (11.22)$$

Proof. The first identity inverts the definition of the periodic evaluation. Iterated Euclidean division by 1000, with twelve positions fixed, recovers the vector. On the other hand,

$$I(\rho K) = 1000I(K) - K_1(1000^{12} - 1).$$

Division by $1000^{12} - 1$ gives (11.22). $\square$

The recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations D_3K, D_4K, Q . The rational evaluation preserves the signed record and the cyclic differences that depend on it. The total depth and the transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphase register.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

The phase acts exactly on the constant: its value is fixed in the canonical reading, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integral components. The signed support associated with α also uses its regional records and the boundary normalization. The relation between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented record, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved jointly.

11.11 Periodicity and cokernel of the transport operator

The periodicity just proved belongs to the dodecaphase register. It does not imply periodicity of the regional coordinates, of the carries of the closure, or of alpha. It is useful to specify this separation through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with orientation fixed, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it is not an authorization to modify the dodecaphase register already selected by the record.

Proposition 11.17 (Quotient of the periodic carry). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ is equivalent to imposing $t = b$, and leaves the relation $b^{12} - 1 = 0$. Evaluation of the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain distinct: one is a rational evaluation and the other a quotient of a carry module. The finite closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

These results fix the internal inputs of the dodecaphase closure of α . The dodecaphase register has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet decides the arithmetic type of the output of the complete closure.

12 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase register. The operation combines their blocks with fixed orientation and transports the quotients of the positional normalization. The resulting parameter expresses the compatibility of that oriented difference with the prolongation of the cylinders. Its value and the incidence data used in its construction remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with path memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. The correlated generation of the four coordinates shares an origin and coinductive compatibility, without thereby requiring that its four components be evaluated in a single step.

The two representations act in distinct operatorial codomains. The first carries out an integral closure by Euclidean division in base 1000, preserves the tail, and produces the complete section α_{HMT} . The analytic readout has two strata that must not be conflated. Its ninth-order jet, constructed by the vacancy resolvent $E_{21/22}$, provides a finite characterization, isolates a simple root, and determines an exact involution on that fiber. The subsequent analytic completion is a correlated representation of the same state: it computes all its coefficients from the regional and terminal coordinates already expressed, proves uniform convergence, and constructs rational intervals compatible with the integral cylinders. It is not presented as an independent selector of alpha. The final section proves the equality of the two representations and explicitly states their common dependence.

The nonuniqueness of a tail chosen from the bare jet prevents the completion from being inferred from the nine isolated coefficients. The complete rule displayed below removes this freedom once the enriched state has been fixed; it neither introduces a conventional constant nor alters the causal order of the dodecaphase generation of alpha. These results remain within the complete HMT state on which the continuum is constructed and the cardinality proof is formulated; their conventional digits are not used as retrospective selectors.

12.1 First route: positional determination by integral closure

12.1.1 Domain and orientation of the integral closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase register by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase register is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum that incorporates the incoming quotient. The normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (12.1)$$

Thus, $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted merely because the table is printed from left to right.

Definition 12.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (12.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no choice of residual sign: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of the closure are not anomalies; they are the integer information required to maintain the oriented equality.

12.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions that determine the coordinate P preserve their distinct antecedents. Sharing this readout does not identify them as regions or as histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, the incoming carry, the total before normalization, the resulting block, and the outgoing carry are shown. The table is a reproducible integer calculation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 8: Initial closure with zero right boundary. Dependence propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the readout of the window is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (12.2)$$

The superscript records the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readouts becomes visible when the carry of the remote tail is transported.

12.1.3 Telescoping and reversibility on a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (12.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate yields a false equality. Only an evaluation that preserves its endpoints permits it to telescope.

Proposition 12.2 (Integer identity at arbitrary depth). *For $t_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$t_N(P) + t_N(E) - t_N(\Phi) - t_N(K) - t_N(A) = B^N c_1 - c_{N+1}. \quad (12.4)$$

Proof. Multiplying (12.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1})B^{N-j} = B^N c_1 - c_{N+1}.$$

Every interior carry appears twice with opposite sign; only the two endpoints survive. $\square$

For Table 8, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant that appears is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize deletion of either condition upon passing to the complete section.

The local inversion is also exact:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , the blocks A , and the complete carries, it recovers the dodecaphase register. Concatenation also recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal prescription that begins with alpha and subsequently manufactures the record. The direction of construction remains signed record, dodecaphase register, closure, and output.

12.1.4 Prolongation and canonical normalization of the complete section

The prolongation preserves the periodic dodecaphase register and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so that $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 12.3 (Existence and uniqueness of the complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (12.5)$$

converges absolutely. In the domain of the resulting blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no tail eventually constant at 999, and a unique bounded integer sequence (c_j) satisfying (12.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying the equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Moreover, $R_j(A) \in [0, 1)$, hence $-3 < c_j < 2$, which places all carries in $\mathcal{C}$. The shift identities for the tails give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique and, once fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes the normalization of the subsequent reader explicit: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it yields favorable digits, nor is compatibility at arbitrary depth deduced from having observed a long run.

The initial comparison shows why this precaution matters. The stabilized prolongation yields $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first cycle. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of the block 999 does not violate the canonical convention: what is excluded is a tail that is eventually constant at 999. The window with zero boundary ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

12.1.5 Operative definition and complete additive equation

Definition 12.4 (Alpha coordinate of the closure). The integral route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (12.6)$$

where (A_j, c_j) is the compatible canonical normalization of (12.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}}. \quad (12.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base one thousand arises from the cubic chart and period twelve from the dodecaphase register. The signs are those of the oriented channel and of the closure, and the remote boundary is the one imposed by the compatible section. These are the effective origins of the coefficients in the equation.

At this level, alpha may be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determined algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, the signed record, the orientation, and the carry law, the scalar equality would not be a sufficient presentation of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the sign of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None of them is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}$, which denotes the periodic prolongation also denoted by K_j in (12.1), the balance prior to quotient transport is

$$Z_j := P_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (12.1) denotes the sum that already includes the incoming contribution, and therefore

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_α^- uses the signs of the balance Z_j in the initial window; digit determination additionally uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphase register arise from the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on readout depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (12.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window and that of its prolongation, both expressed in the same coordinates.

Proposition 12.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the records in the domain under consideration, one has*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (12.8)$$

The quotient of the complete prolongation is $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^\infty)} < B^{-N},$$

so that the resulting window is the canonical prefix of order N .

Proof. Telescoping (12.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, every incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, one has $c_1 = 0$. Substituting this equality into the decomposition of the series proves (12.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For the complete canonical normalization, $c_{N+1}^\infty = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. Integrality of the quotient implies $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^\infty)} = B^{-N} R_{N+1}(A)$, which proves membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed on the scale of the next position. Computation over the five boundaries preserves this information as a finite collection of possibilities until the blocks to the left stabilize. The quantitative contraction arises from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain and status of the analytic representation. The analytic composition displayed in §12.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation produces exactly $T_{7,9}$; degree 9 is the last component of that domain. The identity $N^3 = 0$ explains why the resolvent is expressed by three summands. The local resolvent does not by itself determine coefficients of higher degree. Section 12.3 proves this nonuniqueness of the bare jet and separates two operations: the dodecaphase representation produces the coordinate α_{HMT} from the state records; subsequently, the analytic representation consumes the same precoordinate and generates, through an explicit recurrence, all coefficients of its analytic image. No conventional root is consulted, nor is the analytic representation promoted to an independent generator. The compatibility squares and nested cylinders prove equality of both representations at every depth. Local nilpotence, dodecaphase representation, and analytic representation thus perform distinct functions: the first certifies the finite result of order nine; the second determines α_{HMT} ; the third represents that same coordinate at arbitrary depth and verifies its compatibility.

12.1.6 Constructive determination and the role of the dodecaphase register

The preceding definition makes precise what it means to determine alpha from the structure. The domain of the operation contains four records of known provenance: the three regional sequences and the integral phase record. The normalization also incorporates a boundary condition compatible with prolongation. Once these data have been fixed, Euclidean division determines each remainder and its quotient. Proposition 12.3 extends that determination to the complete system. Independence from a target physical value is therefore a property of the order of construction: the value compared at the end does not participate in selecting any of the arguments of this map.

This causal independence must be distinguished from algebraic independence among numbers. The former is established by identifying the domain, the operations, and the provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through records produced by APP-TRIT-TPK and subsequently makes it possible to verify the relations it satisfies. This order makes it possible to use an already generated coordinate in a later operation without turning it into an external parameter.

The record K has a more precise role than that of a numerical correction added to a formula. Its twelve positions arise from the reversible transformation of the signed record. The equation $U = \mathcal{H}_{12}K$ preserves the ability to return to that record, whereas the transverse differences and the total sum permit another reconstruction. In the periodic evaluation, the ordering of positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation are therefore part of the definition of the number appearing in (12.7).

The normalization of alpha composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented balance of blocks is formed; this balance is then represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and permits the subsequent extraction of a support. The second permits the formation of the prefixes of the real coordinate. Both readouts remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel pairwise when a prefix is evaluated, while their endpoints remain in the equality. That cancellation is a property of the weighted sum; it neither removes the quotients from the state nor permits them to be suppressed before the sum is taken. The right boundary of a window receives the contribution of the continuation. For this reason, evaluation of an isolated window and restriction of a prolonged section are distinct operations. The previously proved change of the final block from 663 to 662 provides an explicit realization of this difference without altering the blocks already stabilized to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible prolongations; normalization transmits the integer information at their boundary from positions of lower weight to positions of higher weight. The equation preserves the contribution of both directions

through their indices and boundary conditions. Reversibility on the fixed fiber is verified by reconstructing K from the other records and the complete quotients. That inversion certifies the consistency of the original operation; the genealogical direction remains the one that first produces the record and then the alpha coordinate.

The correlated analytic representation has complementary content. In it, the immediate object is a vacancy balance, and the coordinate already produced by the state is represented as a root in an isolating domain. The dodecaphase representation provides a canonical normalization of records; the analytic representation studies the vanishing of an operator at that same coordinate. The joint presentation preserves the difference between codomains without asserting causal independence. Their relations with the common state are expressed through the refinement maps and the comparison of their complete representations, as developed in the corresponding subsection. Agreement at a finite approximation constitutes a localized control; equality at arbitrary depth requires the nested cylinders and the coinductive recurrence proved there.

Finally, this organization explains the position of alpha in the article. Its equation appears after regional generation and the construction of K because it uses both results. The electron section subsequently uses this coordinate as a factor in its composition. The exceptional prolongation receives the records and their supports, in which incidence information is preserved. These two continuations take up the same provenance and develop distinct mathematical roles: evaluation of a central section and construction of a boundary configuration. Developing each one makes it possible to trace the structure through the equation instead of restricting the comparison to its scalar value.

12.2 Analytic readout: finite witness and correlated representation

12.2.1 Transport invariants and graded resolvent

The finite analytic witness receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (12.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification readout; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full and with its exact logical force. Recent developments consolidate a marked signature of the state. The synchronization matrix and associated count are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

Its determinant yields

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy readout in the base-one-thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the evaluation of the truncation. They are not obtained from its root or from a metrological approximation to it.

The lower branch also coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the record from the preceding chapter gives $r_{\text{dod}} = 11$, and the phase semicorona provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{E}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (12.10)$$

Substitution recovers the coefficients of P_6 and of its ninth-order prolongation. The pair 45/46 crosses the separation between them: the upper branch occupies degree six, and the lower branch reappears in degrees eight and nine.

The factorization of (12.10) is formulated for a state that preserves origin, channel order, and orientation. Naturality with respect to morphisms preserving these data is not equivalent to uniqueness of every conceivable reader after those data have been forgotten. In the marked domain, the preceding formula fixes each coefficient through $(\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$; therefore, two readers that preserve the signature and apply the same graded evaluation agree term by term through degree nine. This is the uniqueness used below, and it requires no additional documentary cross-reference.

12.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits a particularly explicit finite operatorial construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N denotes a nilpotent operator here, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 12.6 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

The evaluation $e_n \mapsto x^n$ produces

$$T_{7,9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (12.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45, and 495 are not decimal adjustments: they arise from the semicrown, the APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only the symmetry of the cycle and replacing this operator with another symmetric operator does not necessarily preserve the coefficients.

12.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$P_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = P_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has an existence-and-uniqueness result of its own.

To isolate its root by rational arithmetic, let

$$A_n(t) = \sum_{j=0}^n \frac{(-1)^j t^{2j+1}}{2j+1}, \quad L_n(t) = 2 \sum_{j=0}^n \frac{t^{2j+1}}{2j+1} \quad (0 < t < 1).$$

The alternating-series criterion and the geometric sum of the tail give

$$A_{2n+1}(t) < \arctan t < A_{2n}(t), \quad (12.12)$$

$$L_n(t) < \log \frac{1+t}{1-t} < L_n(t) + \frac{2t^{2n+3}}{(2n+3)(1-t^2)}. \quad (12.13)$$

We apply (12.12) to Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and (12.13) to

$$\log(10/9) = \log \frac{1+1/19}{1-1/19}, \quad \log 10 = \log(10/9) + 2 \log \frac{1+1/2}{1-1/2}.$$

Write

$$\delta := \frac{\log(10/9)}{\log 10}.$$

With truncation orders 80, 20, 40, 120, applied respectively to the four preceding series, one obtains rational intervals for π and δ , both of width less than 10^{-70} . No value of alpha is consulted in these bounds.

Proposition 12.7 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and this root is simple. It also has the rational isolating interval*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (12.14)$$

Proof. The bounds on the internal coordinates $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ are sufficient for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms is greater than 0.0626, so the value of the operator is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. For the left endpoint l and right endpoint u of (12.14), rational substitution of the bounds (12.12)–(12.13) into the polynomial gives, with directed rounding,

$$-4.6 \cdot 10^{-20} < E_{21/22}^{(9)}(l) < -4.5 \cdot 10^{-20} < 0, \quad 0 < 1.6 \cdot 10^{-20} < E_{21/22}^{(9)}(u) < 1.8 \cdot 10^{-20}.$$

The monotonicity already proved therefore places the unique root between the two endpoints. The interval is verified after the operator has been constructed and does not enter the selection of its coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. The root of a truncation may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity makes it possible to control the motion of the root under perturbations; it does not eliminate those perturbations.

12.3 Compatibility between the dodecaphase and analytic representations of alpha

Compatibility is established at arbitrary depth. The reversible dodecaphase representation and the analytic representation $E_{21/22}$ of the alpha coordinate are two internal representations with distinct codomains, produced from the same enriched APP-TRIT-TPK state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but it does not by itself determine the tail. The completion constructed subsequently is not deduced from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the record K are prior outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}, r \geq 36$, let $\hat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states that prolong the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves remainder, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, record, memory, survival, provenance, and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$\mathcal{U}_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6,r}^{\alpha, \rightarrow} = \text{graph}(\mathcal{U}_r^\alpha).$$

The functionality of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \hat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\hat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6,r}^{\alpha, \rightarrow}). \quad (12.15)$$

12.3.1 The dodecaphase closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha, N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (12.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. If $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be restriction to the first N blocks.

Theorem 12.8 (Complete dodecaphase output and projective compatibility). *The collection $(\mathsf{T}_{\alpha, N})_{N \geq 1}$ is compatible:*

$$\rho_{N', N} \circ \mathsf{T}_{\alpha, N'} = \mathsf{T}_{\alpha, N} \quad (N' \geq N). \quad (12.16)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (12.17)$$

are nested, have diameter at most 1000^{-N} , and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (12.18)$$

Consequently, the first route expresses a unique infinite section, which we denote by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 12.3 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (12.3) and its telescoping (12.4) preserve the remote boundary; Proposition 12.5 proves that restricting a prolonged execution recovers exactly the preceding prefix. This gives (12.16). Membership of the complete sum in each cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (12.7) identifies that point with the expression in (12.18). All its coefficients and signs are fixed before the representation of alpha. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36,12}(\alpha_{A,36}) = \alpha_{A,12} \quad (12.19)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

12.3.2 Exact ninth-order analytic characterization

Let

$$\mathbb{K} := \mathbb{Q}\left(\pi_{\text{HMT}}, \log_{10} \frac{10}{9}\right), \quad E_9(x) := E_{21/22}^{(9)}(x) = P_9(x) - \delta.$$

Equations (12.10) and (12.11) construct $P_9 = P_6 + T_{7,9}$ from the signature of the state and from the nilpotent resolvent; Proposition 12.7 proves that E_9 has a unique simple root $\xi_9 \in I_\alpha$, isolated by the rational interval (12.14).

Finite bidirectionality can be written without introducing an analytic tail. Define

$$B_9(x) := \delta + \frac{7}{4}x^2 - \frac{x^4}{20} + \frac{2x^5}{21} + \frac{x^6}{46} + \frac{x^7}{120} - \frac{x^8}{45} - \frac{2x^9}{495}, \quad (12.20)$$

$$q_x(T) := xT^2 - B_9(x)T + x^3, \quad x \in I_\alpha = (0, 10^{-2}), \quad (12.21)$$

$$J_x : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \quad J_x(T) := \frac{x^2}{T}. \quad (12.22)$$

Proposition 12.9 (Finite quadratic equivalence and involution). *For every $x \in I_\alpha$ and every $T \in \mathbb{R}^\times$,*

$$\frac{q_x(2\pi_{\text{HMT}})}{2\pi_{\text{HMT}}} = P_9(x) - \delta = E_9(x), \quad (12.23)$$

$$J_x(J_x(T)) = T, \quad T^2 q_x(J_x(T)) = x^2 q_x(T). \quad (12.24)$$

Let $Z_x = \{T \in \mathbb{R}^\times : q_x(T) = 0\}$. Consequently, $J_x|_{Z_x} : Z_x \rightarrow Z_x$ is involutive. When $E_9(x) = 0$, it exchanges the roots $2\pi_{\text{HMT}}$ and $x^2/(2\pi_{\text{HMT}})$. This is an exact involution on the scalar fiber of the ninth-order jet; it does not act on the enriched state.

Proof. Dividing $q_x(T)$ by T and substituting $T = 2\pi_{\text{HMT}}$ reproduces the definition of $P_9(x) - \delta$ term by term. The first identity in (12.24) is immediate. For the second,

$$T^2 q_x(x^2/T) = x^5 - B_9(x)x^2T + x^3T^2 = x^2 q_x(T).$$

Thus, J_x preserves the set of roots and, by Vieta's formula, exchanges the two roots whose product is x^2 . $\square$

The finite decimal witness therefore retains its exact force:

$$\text{Tr}_9(\xi_9^{-1}) = \text{Tr}_9\left((\alpha_{12}^{(0)})^{-1}\right) = 137.035999084. \quad (12.25)$$

Equality (12.25) certifies the finite characterization and the preceding involution. It does not turn the root of a truncated polynomial into the root of every compatible prolongation.

12.3.3 The jet does not by itself determine an analytic prolongation

Let

$$j_9 : \mathbb{K}[[x]] \longrightarrow \mathbb{K}[x]/(x^{10})$$

be the truncation, and for $h \in \mathbb{K}[x]$ set

$$E_h(x) := E_9(x) + x^{10}h(x). \quad (12.26)$$

All these polynomials satisfy $j_9(E_h) = j_9(E_9)$.

Theorem 12.10 (Nonuniqueness of the root from the jet). *There is no map*

$$r_9 : \mathbb{K}[x]/(x^{10}) \longrightarrow I_\alpha$$

that assigns, from the jet alone, the positive root of every polynomial prolongation of E_9 having a unique root in I_α . In particular, E_0 and $E_{1/729}$ have the same jet and distinct positive roots.

Proof. The bound proved in the proof of 12.7 gives $E'_9(x) > 6.24$ on $[0, 10^{-2}]$. For $t \geq 0$, write $E_t(x) = E_9(x) + tx^{10}$. Then

$$E'_t(x) = E'_9(x) + 10tx^9 > 6.24,$$

whereas $E_t(0) = -\delta < 0$ and $E_t(10^{-2}) > E_9(10^{-2}) > 0$. Each E_t therefore has a unique positive root $\eta(t) \in I_\alpha$. For $t = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0;$$

since $E_{1/729}$ is strictly increasing and changes sign exactly once, its root satisfies $\eta(1/729) < \xi_9 = \eta(0)$. Nevertheless, $j_9(E_{1/729}) = j_9(E_0)$. A map that factored the root through j_9 would have to assign the same point to both polynomials, a contradiction. $\square$

The variation is not merely existential. The implicit function theorem applies because $\partial_x E_t(\eta(t)) > 6.24$, and gives

$$\eta'(t) = -\frac{\eta(t)^{10}}{E'_9(\eta(t)) + 10t\eta(t)^9} < 0. \quad (12.27)$$

This proves that coefficients beyond degree nine modify the root even though they leave the entire jet constructed through that degree invariant. This conclusion does not deny the tail produced by the enriched state: it proves only that the root does not factor through the bare truncation j_9 . Choosing a tail h from an external conventional root would be circular. The completion that follows consults neither that root nor a metrological table: it directly evaluates the regional and terminal coordinates of the state. Because this evaluation agrees algebraically with the integral representation, the result is a correlated representation, not an assertion of independence between two selectors.

12.3.4 Analytic representation of the alpha coordinate

The control of the preceding jet requires the tail to be exhibited: merely naming a complete reader is not sufficient. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the precoordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (12.28)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$, and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. No conventional α is introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational readout of the terminal record. The integral closure applies to (12.28) the normalization of (12.1); the chart that follows expresses a compatible analytic representation on the same precoordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it contains no edge $\alpha_A \rightarrow \lambda$, although the equality proved below permits the two representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and set $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. Define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - qa(s)^3)}{a(s)^{10}}. \quad (12.29)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (12.29) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique prolongation in the geometric class defined below. It does not constitute a second independent selection of $a(s)$, but rather a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$, set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, one has

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (12.30)$$

Thus, the coefficient seed contains no freedom once the state and the class of q -geometric prolongations have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet by itself selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (12.31)$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (12.32)$$

This is the rule that computes every coefficient beyond degree nine. No free parameter remains in each prolongation.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (12.33)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (12.34)$$

To formulate the projective system without implicit types, fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{J}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{enr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (12.35)$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (12.36)$$

Proposition 12.11 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (12.37)$$

The projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (12.32) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the preceding polynomial literally. Compatibility for $M' \geq M$ follows by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

12.3.5 Uniform convergence, simple root, and rational intervals

The regional cylinders and the periodic record provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational number P_c as though it were the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, one obtains

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (12.38)$$

The natural interval extension of (12.29) then transports A_{1000} , the cylinder of π_{HMT} , and the directed rational bound on δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\begin{aligned} \frac{7297352569283800997285105472380662}{10^{36}} &< a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \\ -8.711 \times 10^{-6} &< \lambda(s) < -8.709 \times 10^{-6}. \end{aligned} \quad (12.39)$$

These inequalities use certified prefixes with their rational tail bound; they do not receive an alpha table. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (12.40)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1-x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (12.41)$$

The tail of the derivatives beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (12.42)$$

Theorem 12.12 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$, together with its derivatives, converges uniformly to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. One has*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (12.43)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (12.40); term-by-term differentiation and the sum of the two geometric series give (12.41) and (12.42). The proof for the jet establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; therefore, the complete perturbation leaves the derivative above 6.239. Finally, substituting (12.29) into (12.34), the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; hence its unique root is simple and coincides with that precoordinate. $\square$

To make the identification effective at arbitrary depth, directed evaluations at the endpoints first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

By continuity there is a root in the interval and, by the second inequality, it is unique. Only after this existence, monotonicity, and the invariant that the bracket contains the root have been fixed is the mean-value-theorem rescue applied. Start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$, and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (12.44)$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half is retained. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (12.45)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (12.44), one has $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, which proves (12.45). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm need not recognize that equality in order to retain it.

If the available enclosure does not satisfy (12.44), the source cylinders are first refined and the evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches, the new interval retains the root and has diameter at most $w_j/2$. For every finite target depth, directed refinement makes it possible to satisfy the width bound without making an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (12.46)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four levels in base one thousand and checks that the complete outer closure A_{1000} , not merely its lower endpoint, lies inside every expressed cylinder. Its negative tests reject an enclosure that is too wide, a disordered enclosure, a point that is not the midpoint, a domain with no sign change, and a degenerate bracket. The coinductive prolongation permits one thousand to be replaced by any required depth.

Concretely, for $S_N = 1000^N$, the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (12.47)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left cell an endpoint lying exactly on its right boundary.

Let $C_N^A(s)$ be the integral cylinder in (12.17). Recursively choose $m(N) > m(N-1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (12.48)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N need not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (12.49)$$

Bound (12.40) permits a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the complete

function and the membership already proved are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

In the positive domain under consideration, where $\lambda(s) < 0$, one obtains at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Therefore, certification of the weak signs in (12.49) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the supremum that may be excluded from the cylinder.

Proposition 12.13 (Compatibility squares and common cylinders). *The restrictions of words and analytic representations commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N'}} & \mathcal{W}_{N'} \\ \parallel & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N}} & \mathcal{W}_N \end{array} \quad \begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M. \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and materially satisfies $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of the integral normalization; the second is Proposition 12.11. The inclusion follows from Definition (12.48), and nonemptiness and the signs follow from Theorem 12.12. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

We now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (12.50)$$

Theorem 12.14 (Projective equality of the two correlated representations). *For every N , the integral and analytic routes express the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (\text{A7}_0)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (\text{A8}_0)$$

Proof. The integral normalization expresses exactly $a(s)$. Theorem 12.12 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Therefore, their prefixes agree for every N ; the nesting and the diameters tending to zero give equality of the limits. Metrological recognition occurs subsequently. $\square$

The analytic function constitutes a correlated representation of the same state, not a second derivation causally independent of the integral closure. This precision does not weaken the proved equality: it identifies its exact mechanism, prevents attribution to the jet of a tail that it does not determine, and prevents promotion of a conventional value to a generator.

13 Coinductive depth of the fourfold relation

The regional readers and the dodecaphase closure express, respectively, π, φ, e and α from the common enriched state. We reserve N for operator depth and n for decimal length: the two indices are not identified, nor is a uniform readout cost presupposed. For each exact realization χ , its integer part is preserved separately, and $p_{\chi,n}$ denotes the length- n prefix of its fractional part in the canonical decimal expansion that is not eventually equal to nine. Define

$$\rho_{n+1,n}^{\text{dec}}(d_1 \dots d_n d_{n+1}) = d_1 \dots d_n, \quad p_{\chi,0} := \emptyset.$$

The uniqueness of $p_{\pi,n}, p_{\varphi,n}, p_{e,n}$ follows from the uniqueness of the exact character already generated and of its canonical decimal cylinder. That of $p_{\alpha,n}$ follows from the integral dodecaphase closure, its tail identity, and the projective compatibility proved in Theorem 12.8. The ninth-order analytic resolvent provides an additional finite characterization, while the complete analytic reader computes the remaining coefficients by $b_{10+3n} = \lambda/729^n$, proves uniform convergence, and constructs rational intervals contained in the integral cylinders. It thereby identifies that same section at arbitrary depth by Theorem 12.14. The finite jet is used neither as a generator nor as a substitute for the coinductive prolongation.

Theorem 13.1 (Projective compatibility of π, φ, e, α at arbitrary depth). *The coinductive action of the common state determines*

$$\begin{aligned} \forall \chi \in \{\pi, \varphi, e, \alpha\} \forall n \geq 1 \quad \exists! p_{\chi,n}, \\ \rho_{n+1,n}^{\text{dec}}(p_{\chi,n+1}) = p_{\chi,n}, \quad p_{\chi,\infty} := (p_{\chi,n})_{n \geq 0} \in \varprojlim_{n \geq 0} \{0, \dots, 9\}^n. \end{aligned} \quad (\text{D1})$$

In particular,

$$(p_{\pi,\infty}, p_{\varphi,\infty}, p_{e,\infty}, p_{\alpha,\infty}) \quad (\text{D2})$$

is a correlated coinductive representation of the unique enriched state.

Proof. Result 6.15 fixes the exact realizations of the three regional readers. Theorem 12.8 constructs the complete section α_{HMT} and proves the compatibility of all its prefixes. The canonical decimal expansion of each of these four realizations has a unique prefix of every length; truncating the length- $n+1$ prefix gives the length- n prefix. Applying these readers to the common state gives the stated correlated collection. The existence and compatibility of this exact representation neither identify n with N nor add a cost bound to the readout procedures already established. $\square$

The internal constructive order preserves the readers of π, φ, e , the signed state, the integral dodecaphase seal K , and the dodecaphase wheel that expresses α ; no conventional constant is promoted to a generator. A run to one thousand, ten thousand, or one million digits verifies the corresponding finite computational instance of (D1); it does not constitute a bound on the theorem. The theorem is the compatible collection for every n and the infinite section it determines.

This depth is not used to infer the Continuum Hypothesis. The four representations certify the coinductive action of the same generator; the cardinal closure was already obtained through the covering, condensation, and the two injections. Its verification criterion is twofold: compatibility of all truncations in (D1) and internal provenance of the data of the dodecaphase wheel that expresses α_{HMT} .

14 Exceptional prolongation: from dodecaphase incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point makes it possible to type the bifurcation without writing an informal quotient α/K . The terminal state generates the signed record U , the integral seal $K = \frac{1}{4}\mathcal{H}_{12}U$, and the regional representations P, E, Φ . The arithmetic branch composes these objects through $T_{\alpha,\infty}$ and expresses α_{HMT} ; the incidence branch applies the support and hexad readers to the same seal:

$$(x_{\text{term}}, U, K; P, E, \Phi) \longrightarrow \begin{cases} \alpha_{\text{HMT}} \in \mathbb{R}_{\text{HMT}}, \\ (B_K, H_{\alpha}^-, o_*, v_{\alpha}) \longrightarrow N_{\alpha}. \end{cases}$$

Thus, an isolated real number determines neither a code, a lattice, nor a group: both branches transport typed components of the same enriched state. The ordered record K can be recovered from its rational constant κ under the already fixed conventions of base, length, and origin; projection onto a support preserves only the selected incidences. For its part, $\mathcal{K}_{\text{ph}}$ denotes a different, reduced-memory reading, subsequent to the holonomy Γ_{g} .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

The argument begins with the compatibility of prefixes and frames, constructs the ternary code, and determines the flag through K and the signed record of α . The marked incidence then selects the lattice origin and the Leech neighbor. The final step realizes the vertex algebra and its automorphisms. At each stage, the structure being preserved is specified: linear combinations, supports, discriminant classes, bilinear form, or graded product. The explanations of the role of K , of the normalization, and of the norm-54 origin accompany their respective constructions in Sections 14.3 and 14.9.

14.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (14.1)$$

Words are written as row vectors. If $A^f = fA f^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(w f^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (14.2)$$

The enriched codomain preserves f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_w f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (14.3)$$

Proposition 14.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (14.3) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

14.2 Finite realization of the elliptic TRIT regime and construction of the ternary code C_W

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (14.4)$$

This quadratic equality realizes the elliptic regime without identifying the field $\mathbb{F}_3$ as an Archimedean realization of the imaginary unit.

The two are distinct realizations of a common operator relation, with their corresponding domains.

Definition 14.2 (Code of the dodecaphase chart). The ternary code associated with (14.4) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 14.3 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (14.5)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (14.5); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs 1, -1 occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [7]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

14.3 Dodecaphasic flag associated with K and α

Articulation of the regional coordinates and closure incidence. The role of this prolongation is understood through a distinction among three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the structural thesis of the article: a numerical realization has an operatorial provenance, and certain data from that provenance also admit a combinatorial and lattice realization. The equations make it possible to follow both paths, together with their domains and the information retained by each.

The starting point is the terminal state with its compatible prolongations. It preserves the remainders and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scaling regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve

windows determines the signed vector U . The subsequent emergence of K arises from the integral transformation proved in §11.7:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (14.6)$$

The congruence defining $\mathcal{L}_H$ guarantees the integrality of the inverse. Thus, K constitutes a reversible coordinatization of the signed record in the relevant sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve the transverse relations; the sum $Q(K)$ determines the uniform component. This structure explains why the record enters both the numerical composition and the selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable once base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves the membership of each position in a class defined by a threshold and groups records having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

In the frame inherited from the dodecaphase register, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), \\ w_\pi &= 010211, \\ w_\varphi &= 121200, \end{aligned} \quad \begin{aligned} w_e &= 201101, \\ w_{\text{APP}} &= 022020. \end{aligned} \quad (14.7)$$

Applying γ_{A_W} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (14.8)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (14.2). In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (14.9)$$

Its negative support is

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}. \quad (14.10)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Normalization of alpha and its signed configuration. The coordinate of α is determined along the positional route through the composition of the regional blocks and the dodecaphase register.

Before positional division, the oriented combination has components $Z_j = B_j + E_j - \Phi_j - K_j^{\text{per}}$, and the balance with incoming quotient is $s_j = Z_j + c_{j+1}$. The normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (14.11)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of the compatible prefixes determines the scalar coordinate of the closure. The signed configuration, in turn, preserves which positions of the pre-normalization combination exhibit a negative defect. In the initial window, this configuration is the vector Z_0 in (14.9), and its negative support is H_α^- .

Two contents of the same operation are thereby defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records prior to normalization. Its incidence readout remains available because the state preserves that balance together with the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

Proposition 14.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (14.10) is the unique hexad H satisfying*

$$B_K \subset H \subset P_\pi, \quad (|H \cap H_e|, |H \cap H_\varphi|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_\alpha^-$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_π . For the second, the intersection with H_φ has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (14.9). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{I}_* = (B_K \subset H_\alpha^-), \quad \tilde{\mathcal{I}}_* = (\mathcal{I}_*; P_\pi, H_e, H_\varphi, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 14.5 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{I}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary encoding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \longmapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and appends its transform under A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. Thus, quotienting by sign produces the hexads, while the complete code remains available for operations that require its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (14.12)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the encoded propagation and closure supports. The threshold belongs to the declared block chart and remains explicit in the definition of the reader. The full record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. Membership in the support P_π , together with the intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The proofs that follow bring together the two determinations: the one obtained from the sign of Z_0 and the one obtained from incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (14.13)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π recover their internal differentiation here. They share w_π as their visible ternary code, but preserve distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to the lifts of the four hexads in its star. The negative region is distinguished by H_α^- , and the chart order individuates the three positive regions. This correspondence preserves the regional roles and the distribution $432 + 4 \cdot 144 = 1008$, while also realizing the incidence partition $1 + 4$.

14.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (14.14)$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 14.6 (Isometry of the dodecaphase screen). *We have*

$$I^\top I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 14.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

14.5 The direction selected by the dodecaphase register

The record K has a geometric role that complements its rational evaluation and its support readout. Once the representation of the twelve positions has been constructed, its components determine a direction in the space of zero-mean variations. The reduction $12 \rightarrow 11$ removes the uniform mode; the subsequent reduction $11 \rightarrow 10$ removes the direction selected by the record itself. The two operations therefore have different antecedents.

To specify the chart used completely, consider the group A_5 of even permutations of five objects and the subgroup $C_5 = \langle (1\ 2\ 3\ 4\ 5) \rangle$. The twelve left cosets $r_p C_5$ are ordered by the following representatives, written as image lists:

p	r_p	p	r_p
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

The action by left multiplication defines an orthogonal representation $\rho : A_5 \rightarrow O(\mathbb{R}^{12})$: $\rho(a)e_p = e_q$ whenever $ar_p C_5 = r_q C_5$. This representation chart acts on the record already obtained; the group characters are not used to choose its twelve values.

Let χ_3 denote the values of the three-dimensional character

class	1	2^2	3	5_a	5_b
χ_3	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$.

The class 5_a contains the displayed generator of C_5 , and 5_b contains its square. This convention distinguishes the two conjugate three-dimensional sectors of the representation. Define

$$B_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad B_{11} = I_{12} - \frac{1}{12} J_{12}. \quad (14.15)$$

Both matrices are determined by finite sums in $\mathbb{Q}(\sqrt{5})$. The appearance of this quadratic field belongs to the realization of the representation, subsequent to the regional generation of φ .

Lemma 14.8 (Projection of the record onto the three-dimensional sector). *One has*

$$P_3 = P_3^\top = P_3^2, \quad P_3 \mathbf{1} = 0, \quad \text{tr } P_3 = 3, \quad P_3 P_{11} = P_{11} P_3 = P_3.$$

For the record (14.7), the vector

$$u_K = P_3 P_{11} K \quad (14.16)$$

is nonzero. Its exact norm is

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0. \quad (14.17)$$

Proof. The character sum in (14.15) is the isotypic projector. The real character and the substitution $a \mapsto a^{-1}$ give the symmetry. The multiplicity of the sector in the action on A_5/C_5 is the dimension of its vectors fixed by C_5 ,

$$\frac{1}{5} \left(3 + 2 \frac{1 + \sqrt{5}}{2} + 2 \frac{1 - \sqrt{5}}{2} \right) = 1.$$

Therefore its rank is three. Its orthogonality to the trivial character gives $P_3 \mathbf{1} = 0$ and the two compositions with P_{11} . These identities may also be verified by multiplying the matrices in the finite sum.

Finally, $\sum K_p = 6263$, and substituting the twelve components into the same sum gives

$$\|u_K\|^2 = K^\top P_3 K = \frac{1}{20} \sum_{a \in A_5} \chi_3(a^{-1}) K^\top \rho(a) K = \frac{6638585 + 2275584\sqrt{5}}{20}.$$

The evaluation of the sixty terms can be checked within the argument itself. For $a \in A_5$, set $q(a) = K^\top \rho(a) K$. Enumerating the even permutations by cycle type produces the following multisets; in the last three classes, every printed value has multiplicity two:

class	C	{q(a) : a ∈ C}, with multiplicities
1	1	4251257
2 ²	15	2539906, 2552978, 2664658, 2740810 2784738, 2892172, 2911830, 3133488 3239012, 3263782, 3404106, 3501032 3554410, 3675796, 3809354
3	20	2835944, 2857732, 2943734, 3003237 3232885, 3317239, 3354996 3410055, 3455630, 3638920 (each twice)
5 _a	12	3396405, 3515761, 3555976 3587516, 3592629, 3765948 (each twice)
5 _b	12	2829137, 2917179, 3001518 3202919, 3232872, 3955026 (each twice).

The table is obtained without approximation: for each of the sixty image lists a , one locates $ar_p C_5 = r_q C_5$ and sums $K_p K_q$. Their class sums are, respectively,

$$S_1 = 4251257, \quad S_{2^2} = 46668072, \quad S_3 = 64100744, \quad S_{5_a} = 42828470, \quad S_{5_b} = 38277302.$$

Since the character vanishes on the class of order three,

$$\begin{aligned} K^\top P_3 K &= \frac{1}{20} \left(3S_1 - S_{2^2} + \frac{1 + \sqrt{5}}{2} S_{5_a} + \frac{1 - \sqrt{5}}{2} S_{5_b} \right) \\ &= \frac{1327717}{4} + \frac{568896}{5} \sqrt{5} = \frac{6638585 + 2275584\sqrt{5}}{20}. \end{aligned}$$

The two positive coefficients prove nonvanishing, and the table permits the finite evaluation to be repeated directly. $\square$

Definition 14.9 (Dodecaphase direction and transverse complement). The direction selected by K in the declared chart is $\ell_K = \mathbb{R}u_K$. Its transverse complement within $V_{11} = \mathbf{1}^\perp$ is

$$V_{10}(K) = V_{11} \cap u_K^\perp.$$

Proposition 14.10 (Orthogonal reduction determined by K). *The operator*

$$P_{10}(K) = P_{11} - \frac{u_K u_K^\top}{\|u_K\|^2} \quad (14.18)$$

is the rank-ten orthogonal projector onto $V_{10}(K)$. One has

$$V_{11} = \ell_K \oplus V_{10}(K), \quad P_{10}P_{11} = P_{10}, \quad P_{10}u_K = 0.$$

Proof. Let $E_K = u_K u_K^\top / \|u_K\|^2$. The proved nonvanishing guarantees that it is defined. It is symmetric, $E_K^2 = E_K$, has rank one, and, since $u_K \in V_{11}$, $P_{11}E_K = E_K P_{11} = E_K$. Hence

$$(P_{11} - E_K)^2 = P_{11} - 2E_K + E_K = P_{11} - E_K.$$

Its image is $V_{11} \cap u_K^\perp$, and its trace is $11 - 1 = 10$. The decomposition and the remaining identities follow by projection onto the two orthogonal summands. $\square$

This result specifies a role of K not exhausted by its designation as a record: in addition to reversibly representing oriented data and selecting the tetrad B_K , it determines a linear polarization. The direction depends on the complete record through $P_3 K$, not merely on the four points of its support. The causal order is therefore record, representation, direction, and dimensional reduction.

The transformation has a precise covariance law. If the chart is simultaneously changed by an orthogonal permutation matrix S , with $K' = SK$ and $P_3' = SP_3 S^{-1}$, then

$$u_{K'} = S u_K, \quad P_{10}'(K') = S P_{10}(K) S^{-1}.$$

On the other hand, replacing u_K by au_K , with $a \neq 0$, does not change E_K . The direction retains a polarization and does not by itself permit recovery of either the amplitude of u_K or the complete record. The reversibility of $\kappa \leftrightarrow K$ remains in its proper domain.

The geometric realization established here is a reduction of inner-product spaces. Its subsequent use as a compact direction requires transporting this line to the corresponding physical realization. Section 14.15 presents the algebraic map that keeps this reduction and the lattice screen coordinated.

14.6 The Witt star and the generation of M_{12}

Definition 14.11 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

Proposition 14.12 (The collection of stars). *The $\binom{[12]}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

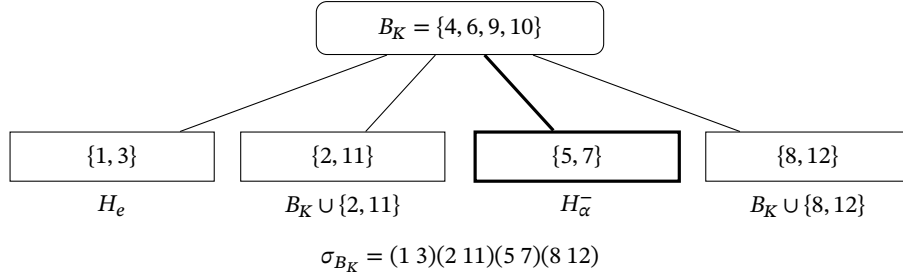


Figure 8: The concrete star of the dodecaphase flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proof. The preservation statement reduces to the explicit definition (14.2). Indeed, the 3^6 words $c(w) = (w, wA)$ are traversed, the 132 weight-six supports are retained, and, for each $B \in \binom{[12]}{4}$, the four pairs $H \setminus B$ with $B \subset H$ are formed. These pairs partition $[12] \setminus B$; replacing each coordinate by its mate maps the 132 supports back onto themselves. This verification is purely finite and proves $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$ for all 495 tetrads.

To make the generation visible, six sufficient stars are written below in cycle notation; the second column displays the tetrad fixed by each:

i	B_i	$g_i = \sigma_{B_i}$
1	$\{1, 2, 3, 4\}$	$(5\ 12)(6\ 11)(7\ 9)(8\ 10)$
2	$\{1, 2, 3, 5\}$	$(4\ 12)(6\ 10)(7\ 11)(8\ 9)$
3	$\{1, 2, 3, 6\}$	$(4\ 11)(5\ 10)(7\ 8)(9\ 12)$
4	$\{1, 2, 4, 5\}$	$(3\ 12)(6\ 9)(7\ 8)(10\ 11)$
5	$\{1, 3, 4, 5\}$	$(2\ 12)(6\ 8)(7\ 10)(9\ 11)$
6	$\{2, 3, 4, 5\}$	$(1\ 12)(6\ 7)(8\ 11)(9\ 10)$

(M12.1)

Let $G_i = \langle g_1, \dots, g_i \rangle$. The finite closure can be repeated without any implicit data through

$$G_i^{(0)} = \{1\}, \quad G_i^{(n+1)} = G_i^{(n)} \cup \{g_j h : 1 \leq j \leq i, h \in G_i^{(n)}\}, \quad G_i = \bigcup_{n \geq 0} G_i^{(n)}. \quad (\text{M12.2})$$

Stabilization of this recurrence gives

i	1	2	3	4	5	6
$ G_i $	2	6	18	360	7920	95040.

(M12.3)

A second way to verify the last order, useful for detecting composition errors, consists in successively fixing the points 1, 2, 3, 4, 5. For $G = G_6$ one obtains

fixed points	0	1	2	3	4	5
$ G_{(1, \dots, k)} $	95040	7920	720	72	8	1
size of the next orbit	12	11	10	9	8	—

(M12.4)

The entries in (M12.3) and (M12.4) are obtained solely by applying the composition of image lists in (M12.1); by the orbit–stabilizer theorem, their product is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$.

Since every g_i preserves $\mathcal{H}$, one has $G \leq \text{Aut}(\mathcal{H})$. By the uniqueness of the hexad containing a five-element set and the incidence enumeration in (14.2), the pointwise stabilizer of a five-element set is trivial; consequently, $|\text{Aut}(\mathcal{H})| \leq 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$. Equality of orders proves $G = \text{Aut}(\mathcal{H})$; in particular, all 495 stars belong to G . The automorphism group of the resulting $S(5, 6, 12)$ system is, by definition and recognition of its natural action, the Mathieu group M_{12} . $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is

the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 14.13 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (14.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

14.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 14.14 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The three positive regions preserve the tail $169|272|272$; the negative region changes it to $418|674|521$; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_\alpha^-)$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

14.7.1 Pentadic representation of the microfiber and its incidence

The arrangement of the five regions makes it possible to visualize simultaneously the unity of the ternary quotient and the differentiation preserved by the catalog. The relevant map is the restriction

$$\Psi|_{\mathcal{R}_\pi^{\text{micro}}} : \mathcal{R}_\pi^{\text{micro}} \longrightarrow \{010211\}, \quad \Psi(U) = (-U) \bmod 3.$$

Its five preimages are the vectors U_6 in the preceding table. Figure 9 recovers their radial organization while preserving the chart order, signatures, and multiplicities of the catalog constructed in the preceding sections.

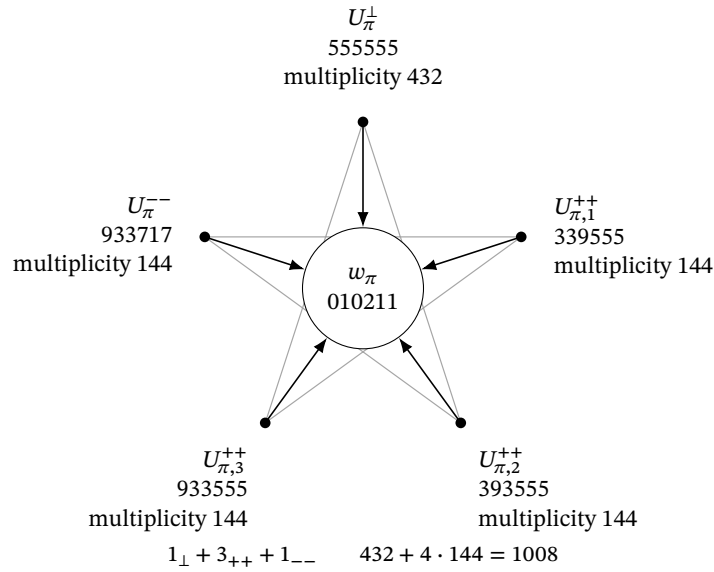


Figure 9: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and its multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five chart positions. Their realization in the five Witt octads is carried out through the regional alignment and the $4 + 1$ decomposition in the text.

Projection to the center identifies the five ternary images; the record of the outer positions retains the regional information that makes the incidence realization possible. The distribution $1_\perp + 3_{++} + 1_{--}$ specifies one transverse region, three positive regions, and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the remaining four: the transverse multiplicity is threefold, even though the five images under Ψ coincide.

The link with the Witt star is formulated through the already constructed flag $B_K \subset H_{\alpha}^{-}$, alignment ℓ , and lift ι_D . The five regional positions are realized in the five octads over $\ell(B_K)$:

$$\begin{aligned} U_{\pi}^{\perp} &\longmapsto O_{\ell(B_K)}^{\perp}, \\ U_{\pi}^{--} &\longmapsto \iota_D(H_{\alpha}^{-}), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \{\{1, 3\}, \{2, 11\}, \{8, 12\}\}. \end{aligned}$$

The assignment of the three positive positions to the three pairs is carried out in an ordered chart. The transverse row and the negative row are distinguished by their regional predicates and by the flag.

The two figures are thereby articulated: Figure 8 shows the four hexads of the small Witt design over B_K ; Figure 9 shows the five regions that, under the alignment, occupy the four residual octads and the transverse octad of the binary design. The $4 + 1$ law, proved in Proposition 14.14, specifies the change of incidence. In this composition, K determines the marked tetrad, and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions preserve the structure required for this transport to the exceptional realization.

14.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (14.19)$$

Theorem 14.15 (Ternary gluing). *The object (14.19) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (14.5) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

14.9 Incidence selection of the rootless unimodular neighbor

From the incidence origin to the lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its role is to distinguish one component among the twelve involved in the lattice realization. The complete ternary code acts as a gluing code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights, and distance of the code translate, respectively, into integrality and determinant properties, parity, and norm bounds for the gluing classes. Here the symbolic information of the code performs a role that the isolated support no longer performs.

The A_2 metric and the positive radial chamber are fixed in the lattice construction. Within that chamber, the congruence required for an even neighbor of index three has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered the closure and now individuates a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adjoining the class $v_\alpha/3$ produces the neighbor N_α in (14.21). This modification preserves the rank and recovers unimodularity and parity; selection by congruence eliminates the previous roots, and the local bounds exclude roots in the new classes. Identification with the Leech lattice occurs after these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as gluing data.

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (14.20)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (14.21)$$

Theorem 14.16 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod{3}$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$

belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (14.20); the scalar value of the constant is not converted into a length of the lattice.

14.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^\natural = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (14.22)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 14.17 (Prolongation to the Moonshine module). *The algebra (14.22), constructed on the neighbor (14.21), is a realization of the Moonshine algebra $V^\natural$. It has central charge 24, a zero weight-one component, and an automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^\natural} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i \tau}. \quad (14.23)$$

Proof. Theorem 14.16 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^\natural$ and establish its character and its automorphisms. The composition of the flag, the neighbor, and the vertex operator algebra (VOA) construction functor gives the realization indicated here. It is not deduced from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof in this article of the Frenkel–Lepowsky–Meurman (FLM) construction [5]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\mathrm{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^\natural$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

Grading, unit, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, the central extension of the lattice, the vertex product, and the sector twisted by the lift of negation. The result is the algebra $V_{\mathrm{HMT}}^\natural$ in (14.22), whose automorphism group is isomorphic to the Monster. Its domain of action is thus fixed by an algebraic structure with a grading and a product. Borchers’s Moonshine theorem subsequently determines the graded traces with insertion of those automorphisms [5, 6].

At this level the unit acquires a precise realization: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, the refinement maps preserve the constant function by precomposition. Each preservation statement therefore has its own object and map. The development of the proofs makes this continuity of construction visible, from the arithmetic operations and memory through incidence, the metric, and the vertex products.

This articulation places K and the closure of α within the main argument. K reversibly preserves an integral coordinate of the history; the normalization of α composes the regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The proofs that follow develop these maps, distinguishing reversible equivalences, information quotients, and realizations that add a metric or algebraic structure. A provable relation among structure, equation, and value is thus expressed while preserving the attribution of each construction and the scope of its theorems.

14.11 Ternary orientation of the Leech neighbor

The order-two presentation constructed above uses negation of the Leech lattice. The same lattice, obtained from the incidence marked by K and the dodecaphase support of α , admits a second presentation compatible with the ternary character of the construction. To make this continuity explicit, the gluing is retained, its order-three isometry is constructed, and preservation of the already selected neighbor is verified. This step requires no alteration of K , its rational reading, or the preceding definition of α . The lattice operations and the hypotheses of the recognition theorems used are specified below.

We work in the basis of simple roots of each A_2 factor, with

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (14.24)$$

Multiplication gives

$$C^3 = I_2, \quad C^2 + C + I_2 = 0, \quad C^\top B_2 C = B_2, \quad R C R = C^{-1}, \quad R^\top B_2 R = B_2.$$

These identities fix both the metric and the orientation of the action. In the discriminant $A_2^*/A_2 \cong \mathbb{F}_3$, we take the class of $\varpi = (2, 1)^\top/3$ as generator. We have $C\varpi - \varpi = (-1, 0)^\top$, so C acts trivially on the discriminant; the reflection R reverses its sign.

The Paley–Witt matrix already constructed produces a full-support element by the following operation:

$$\begin{aligned} q_* &= (1, 1, 1, 1, 1, 1), & q_* A_W &= (2, 1, 1, 1, 1, 1), \\ c_* &= (q_*, q_* A_W) = (1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1) \in C_W. \end{aligned} \quad (14.25)$$

Every coordinate of c_* is nonzero. For the twelve positions, define the frames and the isometry

$$R_b(c_*) = \begin{cases} R, & (c_*)_b = 1, \\ I_2, & (c_*)_b = 2, \end{cases} \quad g_c = \bigoplus_{b=1}^{12} R_b(c_*) C R_b(c_*)^{-1}. \quad (14.26)$$

The symbol $R_b(c_*)$ here denotes an invertible frame of an A_2 factor, not the three-dimensional isotypic projector P_3 .

Proposition 14.18 (Order-three isometry on the marked neighbor). *The operator g_c is an order-three isometry with no nonzero fixed vectors. It preserves the Niemeier lattice $N = N(C_W)$ and the neighbor N_α constructed in (14.21).*

Proof. Each block of g_c is C or C^{-1} ; its minimal polynomial is $X^2 + X + 1$. This proves its order and the absence of fixed vectors. Both blocks act trivially on the discriminant, so they preserve every gluing class and therefore the lattice N .

Write $v = v_\alpha = 4\rho_{o_*} + \sum_{b \neq o_*} \rho_b$, with the frames and incidence origin already fixed. Its radial coefficients are $a_{o_*} = 4$ and $a_b = 1$ for $b \neq o_*$. All satisfy $a_b \equiv 1 \pmod{3}$. In one factor,

$$(C - I_2)\rho = (-2, -1)^\top, \quad (C^{-1} - I_2)\rho = (-1, -2)^\top.$$

After division by three, their discriminant classes are 2 and 1, respectively. Consequently, the vectors

$$y = \frac{g_c v - v}{3}, \quad z = \frac{g_c^{-1} v - v}{3} \quad (14.27)$$

have discriminant words c_* and $-c_*$, both in C_W . Hence $y, z \in N$. The local product is

$$\left\langle \frac{(C^{\pm 1} - I_2)a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2,$$

and summing gives $\langle y, v \rangle = \langle z, v \rangle = -(16 + 11) = -27$. Thus, $y, z \in N_0 = \{x \in N : \langle x, v \rangle \equiv 0 \pmod{3}\}$. For $x \in N_0$,

$$\langle g_c x, v \rangle = \langle x, g_c^{-1} v \rangle = \langle x, v \rangle + 3\langle x, z \rangle \equiv 0 \pmod{3}.$$

The argument applied to g_c^{-1} proves $g_c N_0 = N_0$. Finally, $g_c(v/3) = v/3 + y$ preserves the class generating $N_\alpha = N_0 + \mathbb{Z}(v/3)$. We obtain $g_c N_\alpha = N_\alpha$. $\square$

The proof preserves more than the lattice dimension: it uses the gluing code, its orientation, the flag origin, and the congruence of the radial vector. These data explain why the second presentation remains linked to the dodecaphase register from which the construction began.

14.12 Ternary trace, twisted weight, and order-three extension

Let $\Lambda = N_\alpha$, and let $\hat{g}_c$ be the standard lift of g_c to the lattice algebra $V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda]$, with the central extension and cocycle already specified. The odd order allows this lift to be chosen of order three. Each complex A_2 factor contributes the eigenvalues ζ_3 and ζ_3^2 , where $\zeta_3^2 + \zeta_3 + 1 = 0$.

Proposition 14.19 (Graded trace and type of the lift). *For $j = 1, 2$,*

$$\mathrm{Tr}_{V_\Lambda}(\hat{g}_c^j q^{L_0-1}) = t_3(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12}. \quad (14.28)$$

The lowest weight of the nontrivial twisted modules is $4/3$, and the lift is of type zero.

Proof. Only the zero vector contributes to the lattice part of the trace, because g_c and g_c^2 have no nonzero fixed vectors. At each oscillator degree, the trace of symmetric powers is the inverse of the determinant $I - g_c^j q^n$. Each A_2 factor contributes $1 + q^n + q^{2n}$, proving (14.28).

The twisted-vacuum weight formula for an order-three lattice automorphism, applied to the two twelve-dimensional eigenspaces, gives

$$\rho_{\hat{g}_c} = \frac{1}{4 \cdot 3^2} (1 \cdot 2 \cdot 12 + 2 \cdot 1 \cdot 12) = \frac{4}{3}.$$

The type is the class of $3^2 \rho_{\hat{g}_c} = 12$ modulo three, which is zero. The weight formula is a result of twisted-module theory; the multiplicities and isometry supplied to it have been constructed here. $\square$

Theorem 14.20 (Second presentation of the Moonshine algebra). *The type-zero cyclic extension*

$$V_{(3)} = \mathrm{Orb}_{\mathbb{Z}/3}(V_\Lambda, \hat{g}_c) \quad (14.29)$$

is isomorphic to the vertex operator algebra $V^\natural$.

Proof. Theorem 14.16 provides a positive, even, unimodular, rootless lattice of rank twenty-four. Proposition 14.18 constructs an order-three fixed-point-free isometry on that same lattice; Proposition 14.19 verifies the lift and its type. The Chen–Lam–Shimakura theorem on the order-three orbifold of the Leech lattice algebra then applies [8, Theorem 1.1]. This identification is also covered by the uniform treatment of Abe–Lam–Yamada [9, Theorem 4.4 and Sections 2–4]. This application follows construction of the lattice data; equality of characters does not replace the hypotheses of the theorem used. $\square$

To specify the dual symmetry, write $V_{(3)} = W_0 \oplus W_1 \oplus W_2$, where W_i is the integral sector selected in the $\hat{g}_c^i$ -twisted module. The product satisfies $Y(W_i, z)W_j \subset W_{i+j \bmod 3}((z))$. Hence

$$g^\vee|_{W_i} = \zeta_3^i I \quad (14.30)$$

defines an order-three automorphism. Indeed, the factor on a product is $\zeta_3^{i+j} = \zeta_3^i \zeta_3^j$, and the vacuum and conformal vector belong to W_0 . The classical identification of this automorphism in the Monster is class $3B$, with series $T_{3B} = t_3 + 12$ [10, 6]. This normalization can be checked in the construction. The lattice character is $\mathrm{ch} V_\Lambda = J + 24$: its initial pole is q^{-1} , its weight-one space has dimension 24, and the weight-zero modular character is determined by these data. If F is the character of the fixed subspace and H that of each integral twisted sector, projection onto invariants and conjugation of the two sectors give

$$F = \frac{J + 24 + 2t_3}{3}, \quad F + 2H = J, \quad \mathrm{Tr} g^\vee q^{L_0-1} = F - H = t_3 + 12.$$

Its domain is the extended algebra, whereas $\hat{g}_c$ acts on the preceding lattice algebra.

14.13 Equivalence of the order-two and order-three presentations

Denote by

$$V_{(2)} = (V_{\Lambda})^+ \oplus (V_{\Lambda}^T)^+$$

the FLM presentation in (14.22). Both presentations preserve the same Leech neighbor as their antecedent. Their extensions employ different automorphisms and vertex products determined by the respective lifting data.

Theorem 14.21 (Graded equivalence and transport of automorphisms). *Given vertex operator algebra isomorphisms*

$$\phi_2 : V_{(2)} \xrightarrow{\cong} V^{\natural}, \quad \phi_3 : V_{(3)} \xrightarrow{\cong} V^{\natural},$$

the composition

$$\Phi_{23} = \phi_2^{-1} \phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)} \quad (14.31)$$

preserves the vacuum, conformal vector, grading, and vertex product. Conjugation by Φ_{23} identifies the automorphism groups and preserves their graded traces.

Proof. The composition of the two isomorphisms preserves the stated structures. In particular,

$$\Phi_{23} Y_{(3)}(a, z)b = Y_{(2)}(\Phi_{23}a, z)\Phi_{23}b, \quad \Phi_{23} L_0^{(3)} = L_0^{(2)} \Phi_{23}.$$

If h is an automorphism of $V_{(3)}$, then $h' = \Phi_{23} h \Phi_{23}^{-1}$ is an automorphism of $V_{(2)}$. Equality of traces on each finite-dimensional homogeneous component gives

$$\mathrm{Tr}_{V_{(3)}} h q^{L_0-1} = \mathrm{Tr}_{V_{(2)}} h' q^{L_0-1}.$$

□

The choice of ϕ_2 and ϕ_3 forms part of the statement; Φ_{23} is not presented as a canonical identification independent of these choices. Nor does it transform an order-three automorphism into the order-two involution: conjugation preserves order. Its content is that two different extensions realize the same graded structure, with explicit transport of their products, representations, and traces.

14.14 Eta functions, Fricke involution, and vacuum normalization

For $\mathrm{Im} \tau > 0$, the traces of the two lattice automorphisms admit the expressions

$$\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n), \quad t_2 = \left(\frac{\eta(\tau)}{\eta(2\tau)} \right)^{24}, \quad t_3 = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12}. \quad (14.32)$$

They follow directly from the factorizations $1 - q^{2n} = (1 - q^n)(1 + q^n)$ and $1 - q^{3n} = (1 - q^n)(1 + q^n + q^{2n})$. The order in q of the level-three quotient, before raising it to the twelfth power, is

$$\frac{1}{24} - \frac{3}{24} = -\frac{1}{12}; \quad 12 \left(-\frac{1}{12} \right) = -1.$$

Here $-1/12$ denotes the exact difference between the initial exponents of two eta functions. The operation is specified and does not require interpreting a divergent sum as a convergent one. Agreement with other normalized defects in the corpus preserves their respective realizations; it does not identify their operators through equality of the scalar.

Proposition 14.22 (Fricke involution on the order-two trace). *Let $w_2\tau = -1/(2\tau)$. Then*

$$t_2(w_2\tau) = \frac{2^{12}}{t_2(\tau)}, \quad T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2} \quad (14.33)$$

and $T_{2A}(w_2\tau) = T_{2A}(\tau)$.

Proof. The transformation law $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$ gives

$$\frac{\eta(-1/(2\tau))}{\eta(-1/\tau)} = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}.$$

The twenty-fourth power yields the first identity. Substitution interchanges the two variable terms of T_{2A} and fixes the constant term. Recognition of this expression as the McKay–Thompson series of class $2A$ corresponds to the level-two symmetrization of Conway–Norton [10, §§4 and 6, Table 3] and the Moonshine theorem [6]. $\square$

The modular uniformizations of levels two and three, with the preceding normalizations, are

$$j = \frac{(t_2 + 256)^3}{t_2^2}, \quad (14.34)$$

$$J = j - 744 = t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \quad (14.35)$$

They are used as classical modular identities on the traces already constructed, not as rules for selecting the code or the neighbor. The second can also be written $j = (t_3 + 27)(t_3 + 243)^3/t_3^3$. The direct product expansion of (14.28) begins with

$$t_3 = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5).$$

In particular, $[q]J = 54 + 10 \cdot 3^9 = 196884$; this is a consequence of the traces and uniformization. The construction of $V^\natural$ retains its lattice and orbifold proof, independently of this check of the first coefficient.

The three operations occurring in this extension have different domains. The dual symmetry $g^\vee$ acts on sectors of the vertex algebra; Fricke acts on the modular parameter and its function; radius duality interchanges momentum and winding in the compact lattice sector. Their reciprocity formulas can be coordinated through the corresponding maps, but the name “duality” does not replace those maps. This distinction allows the construction to continue toward dimensional screens while preserving incidence, grading, and the origin of each operation.

14.15 Exceptional incidence and coordination of the screens

The direction selected by K and the Leech lattice arise from distinct representations of the constructed incidence. The former determines the orthogonal reduction $12 \rightarrow 11 \rightarrow 10$; the latter enters the isotropic reduction $26 \rightarrow 24$. The coordination preserves the preceding space, its relations, and the information transmitted by each projection.

The sectoral representation on T-duality and dimensional screens jointly develops this coordination. Its first block brings together the involution of integer coordinates, unitary conjugation with its domains, and the change of residual section with quotient memory. The second constructs the morphism $\mathcal{R}_M^{\text{alg}}$ and proves the isomorphism of the quotient presentations. The graph preserves the coordinate in V_{11} together with its projection onto V_{10} , while the isotropic quotient recovers the Leech lattice. The respective kernels and codomains are thus preserved.

The action ellipse is constructed before duality is applied and fixes its dimensionless radius together with the reciprocal radius. The compact sector then links exceptional incidence to a determined quadratic form. The fields, action, and operators of the subsequent physical realization are presented in their proper domain, preserving this algebraic antecedent and its internal proofs.

14.16 Graded Moonshine traces and representation of the Monster

For $g \in \text{Aut}(V_{\text{HMT}}^{\mathfrak{h}})$, the trace with insertion is

$$T_g(\tau) = \text{Tr}_{V_{\text{HMT}}^{\mathfrak{h}}} g q^{L_0 - 1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [6, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\mathfrak{h}} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (14.22), and a choice of isomorphism with $V^{\mathfrak{h}}$ transports its automorphism group. This does not construct an action of the Monster on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^{\mathfrak{h}}}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_\alpha \longrightarrow V_{\text{HMT}}^{\mathfrak{h}}. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of this section is therefore an assembled derivation: the finite and lattice proofs make the passage from dodecaphase incidence to the hypotheses of the recognition theorems effective. Every arrow used is defined in the development, and its conclusion is obtained within the stated domains. Experimental identification of physical realizations remains separate from this internal proof and from the subsequent comparison of constants.

15 Conclusions: affirmative resolution of the Continuum Hypothesis

The main result is the absence of intermediate cardinals between the natural numbers and the real line within the HMT genealogical closure. For each realized history, the construction of that closure and the two cardinal injections establish

$$\mathbb{G}_{\text{HMT}}(h, m_\infty) \models 2^{\aleph_0} = \aleph_1. \quad (15.1)$$

The equality is not presupposed at any generative stage. Exhaustive cylinder coverage establishes the equinumerosity

$$\mathbb{R}_{\text{HMT}}^{\mathbb{G}} \approx \mathcal{P}^{\mathbb{G}}(\omega).$$

Genealogical condensation assigns to every internal real an ordinal strictly below $\omega_1^{\mathbb{G}}$ and constructs

$$j_{\text{HMT}} : \mathbb{R}_{\text{HMT}}^{\mathbb{G}} \hookrightarrow \omega_1^{\mathbb{G}}.$$

The coding of countable well-orders and its ternary lift construct the injection in the opposite direction

$$k_{\text{HMT}} : \omega_1^{\mathbb{G}} \hookrightarrow \mathbb{R}_{\text{HMT}}^{\mathbb{G}}.$$

The Cantor–Bernstein theorem yields the cardinal equality in (15.1); it does not supply that equality as an input.

The quantifier of the result ranges over every subset of the real line that belongs to the complete genealogical closure. For each $A \in \mathcal{P}^{\mathbb{G}}(\mathbb{R}_{\text{HMT}}^{\mathbb{G}})$, the proved cofinality imposes the dichotomy

$$|A| \leq \aleph_0 \quad \text{or} \quad |A| = |\mathbb{R}_{\text{HMT}}^{\mathbb{G}}|.$$

Within the integral continuum generated by HMT, there is no cardinal strictly intermediate between $\aleph_0$ and the cardinal of the real line. Consequently, Triadic Modular Holography gives an affirmative resolution of the Continuum Hypothesis in the complete genealogical closure defined in this work.

The structural significance of this equality lies in its connection to an explicit construction of the real line. The composition

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_\infty^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}$$

generates a single compatible system of histories endowed with value, sheet, orientation, residue, quotient, carry, boundary, route, memory, and survival. Its Archimedean projection produces the real line $\mathbb{R}_{\text{HMT}}^{\mathbb{G}}$; its incidence projection preserves the boundary information that the evaluation of the Archimedean coordinate does not retain. The real line is an output of the construction, while the state retains the operations and relations from which that output arises.

The finiteness of each level does not restrict this scope. The fiber $\mathbb{F}_3^6$, consisting of $3^6 = 729$ refinement subcylinders, describes local branching; the coinductive quantifier acts at every prescribed depth. Executions to one thousand, ten thousand, or one million digits are finite witnesses of a law parametrized by depth, not bounds on the construction. The nonadic return recovers the phase while carry and memory advance, so the observable closure does not reset the enriched state.

The emitter development specifies three levels of preservation. The third-digit check congruence accompanies the two recoverable signatures and determines an alphabet of 42 reachable blocks. Six witness emissions construct a regional incidence cycle with explicit seed provenance. Comparing two words with equal frequencies and different temporal correlation identifies the ordering information preserved by the archive. Their cardinalities, graphs, and functionals belong to defined finite domains; the proof of the principal cardinal assertion separately retains its quantifiers, closure, and two constructed injections.

The five regional closures attain the minimum of eight in the finite problem of covering the four prescribed edges. The graph retains emission labels, and the quotient retains their incidences between phase classes. Oriented reversal makes the reader ν_{120} odd and preserves the even parity of ν_{270} ; these identities make explicit the response of memory to a change of orientation.

The same genealogy produces the correlated results of the dual realization. They include the nonadic holonomy Γ_9 , its subsequent reduced-memory reading $\mathcal{K}_{\text{ph}}$, the holographic arithmetic–geometric coupling constant K , the fourfold relation $(\pi, \varphi, e, \alpha)$, and the Paley–Witt–Golay–Leech– $V^{\mathfrak{h}}$ –Monster corridor. They are distinct codomains and readers of the same state; none is introduced from conventional values to force the cardinal comparison.

In particular, K is the canonical integral representation of the oriented dodecaphase register. Its production follows the chain

$$X_{\text{term}} \longrightarrow \mathcal{R}_{12}(X_{\text{term}}) \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U_{\text{sgn}}, \quad U_{\text{sgn}} \xrightarrow{\mathcal{K}_{12/4}} K, \quad K \xrightarrow{\mathcal{K}_{12}} U_{\text{sgn}}.$$

The Hadamard equivalence expresses a reversible coordinatization of the generated register; it does not turn U_{sgn} , K , or its rational reading into external data. The positional reading of K subsequently enters the closure of α ; its incidence and geometric readings preserve support, direction, and dimensional reduction. This function must not be confused with $\mathcal{K}_{\text{ph}}$, which belongs to another type and is obtained by memory reduction after the holonomy.

The identification

$$\mathcal{G}_{\text{HMT}}(h, m_{\infty}) = L[u_{h,m}] = L[h, m_{\infty}]$$

is the final recognition of a hierarchy already constructed. It makes its representation by relative constructibility explicit, but does not generate the real line, select the injections, or enter the cylinder reduction. The causal order remains APP–TRIT–TPK, enriched state, joint discrete structure of the continuum, Archimedean projection, and subsequent identification.

Thus jointly proved are the construction of the HMT continuum, the equinumerosity of its real line with the power set of ω , the two cardinal injections, the absence of intermediate cardinals in the complete closure, and the equality $2^{\aleph_0} = \aleph_1$. This is the affirmative resolution of the Continuum Hypothesis established by Theorem 7.12.

References

- [1] R. P. Feynman, “Space-Time Approach to Non-Relativistic Quantum Mechanics”, *Reviews of Modern Physics* **20**, 367–387 (1948). doi:10.1103/RevModPhys.20.367.
- [2] B. Musto and J. Vicary, “Quantum Latin Squares and Unitary Error Bases”, *Quantum Information and Computation* **16**, 1318–1332 (2016). doi:10.26421/QIC16.15-16-4.
- [3] G. De las Cuevas, T. Drescher and T. Netzer, “Quantum Magic Squares: Dilations and Their Limitations”, *Journal of Mathematical Physics* **61**, 111704 (2020). doi:10.1063/5.0022344.
- [4] G. De las Cuevas, T. Netzer and I. Valentiner-Branth, “Magic Squares: Latin, Semiclassical, and Quantum”, *Journal of Mathematical Physics* **64**, 022201 (2023). doi:10.1063/5.0127393.
- [5] I. B. Frenkel, J. Lepowsky and A. Meurman, *Vertex Operator Algebras and the Monster*, Pure and Applied Mathematics, vol. 134, Academic Press, 1988.
- [6] R. E. Borcherds, “Monstrous Moonshine and Monstrous Lie Superalgebras”, *Inventiones Mathematicae* **109**, 405–444 (1992). doi:10.1007/BF01232032.
- [7] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der mathematischen Wissenschaften 290, Springer, 1999. doi:10.1007/978-1-4757-6568-7.
- [8] H.-Y. Chen, C. H. Lam and H. Shimakura, “ $\mathbb{Z}_3$ -orbifold construction of the Moonshine vertex operator algebra and some maximal 3-local subgroups of the Monster”, preprint, arXiv:1606.05961 (2016), version 2 (2017). arXiv:1606.05961.
- [9] T. Abe, C. H. Lam and H. Yamada, “A remark on $\mathbb{Z}_p$ -orbifold constructions of the Moonshine vertex operator algebra”, preprint, arXiv:1705.09022 (2017), version 4. arXiv:1705.09022.
- [10] J. H. Conway and S. P. Norton, “Monstrous Moonshine”, *Bulletin of the London Mathematical Society* **11**, 308–339 (1979). doi:10.1112/blms/11.3.308.
- [11] G. Cantor, “Ein Beitrag zur Mannigfaltigkeitslehre”, *Journal für die reine und angewandte Mathematik* **84**, 242–258 (1878). doi:10.1515/crelle-1878-18788413.
- [12] D. Hilbert, “Mathematical Problems”, *Bulletin of the American Mathematical Society* **8**, 437–479 (1902). doi:10.1090/S0002-9904-1902-00923-3.
- [13] K. Gödel, “The Consistency of the Axiom of Choice and of the Generalized Continuum-Hypothesis”, *Proceedings of the National Academy of Sciences* **24**, 556–557 (1938). doi:10.1073/pnas.24.12.556.
- [14] P. J. Cohen, “The Independence of the Continuum Hypothesis”, *Proceedings of the National Academy of Sciences* **50**, 1143–1148 (1963). doi:10.1073/pnas.50.6.1143.
- [15] P. J. Cohen, “The Independence of the Continuum Hypothesis, II”, *Proceedings of the National Academy of Sciences* **51**, 105–110 (1964). doi:10.1073/pnas.51.1.105.
- [16] T. Jech, *Set Theory*, 3rd ed., Springer Monographs in Mathematics, Springer, 2003; Chapter 13, “Constructible Sets”, pp. 175–200. doi:10.1007/3-540-44761-X.